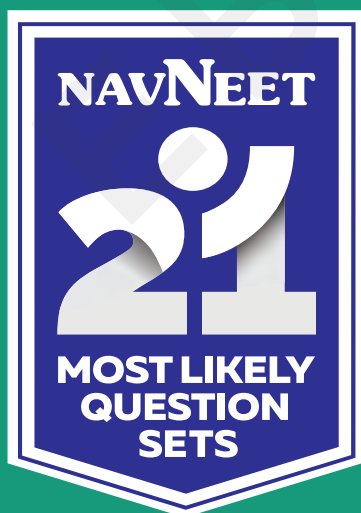


For
2026
Examinations



SCIENCE &
TECHNOLOGY (PART 1)

Board's
Activity Sheet
of July 2025
Included

Solution on
QR Code

STD. X



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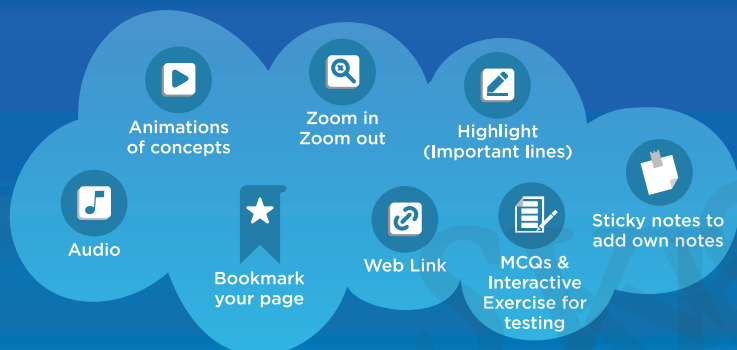


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2026

**An excellent book for the
Preparation of the Board's
Examination**

**SCIENCE & TECHNOLOGY
(PART – 1)
STANDARD X**

This book contains ...

1. Format of the Activity Sheet at the beginning of the book
2. Study Tips for Students
3. **Section 1** : Board's July 2024 Activity Sheet with full Solution
4. **Section 2** : Question Sets for Question-wise study and Revision :
(a) Guidance on how to attempt the question/activity
(b) Solved questions
5. **Section 3** : Assignment – Questions for Practice
6. **Section 4** : Board's July 2025 Activity Sheet for Practice
(with Answers on QR Code)

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2626
CTP
(23-9-25)

Activity Sheet Paper Pattern

Time : 2 Hours]

[Max. Marks : 40

Important points and instructions :

- (1) All questions are compulsory.
- (2) Use of a calculator is not allowed.
- (3) Draw scientifically correct, labelled diagrams wherever necessary.
- (4) Start each main question on a new page.
- (5) The numbers to the right of the questions indicate full marks.
- (6) For each MCQ [Q. 1. (A)], evaluation will be done for the first attempt only.
- (7) For each MCQ, write only the alphabet corresponding to the correct answer, e.g. (i) (A); (ii) (C) ...

		Marks	Marks with option
Q.1(A)	5 Multiple Choice Questions carrying 1 mark each	5	5
Q.1(B)	5 Questions carrying 1 mark each (No fill in the blanks questions) Each question will be of different type.	5	5
Q.2(A)	3 Scientific reason questions carrying 2 marks each (Attempt any 2) (Minimum 1 question from chemistry and 1 from physics)	4	6
Q.2(B)	5 Questions carrying 2 marks each (Attempt any 3) (Minimum 2 questions from chemistry and 2 from physics)	6	10
Q.3	8 Questions carrying 3 marks each (Attempt any 5) (4 questions from chemistry and 4 from physics)	15	24
Q.4	2 Questions carrying 5 marks (Attempt any 1) (1 question from chemistry and 1 from physics)	5	10
Total		40	60

Study Tips

Dear Students,

The SSC examination is an important milestone in your life. If you take appropriate steps in revising Science and Technology (Part 1), you can secure maximum marks.

For this, what you need is 'Smart Study'. Read all the ten chapters of Science and Technology (Part – 2) carefully with full concentration. In Navneet Digest, various types of important questions and their model answers are given. You must have used the Digest throughout the year. Now prepare for the final examination using '**Navneet 21 Most Likely Question Sets**'.

Let us understand the pattern of the activity sheet.

- In **Q. 1.** there will be two sub-questions (A) and (B). These are objective questions, carrying a total of **10 marks**. In **Q. 1. (A)**, there will be 5 multiple choice questions (1 mark each). **Q. 1. (B)** will contain 5 different objective question types (1 mark each) such as 'Find the odd one out', 'Find the correlation', 'Match the columns/pairs', 'Write whether the given statement is true or false', 'Write the name/molecular formula/electronic configuration', or any other type of question. Revise these questions (Sets 1 to 6) everyday to secure full marks.
- **Q. 2. (A)** has three 'Give scientific reasons' type of question carrying **2 marks each**. You have to attempt any two out of these. **Set 7** covers these questions.
- **Q. 2. (B)** will contain 5 questions carrying **2 marks each**. You have to attempt any three of these. Here the variety of questions asked will be : 'Solve the numerical problems', 'Write the law, Definition/principle and explain with an example', 'Write a short note', 'Questions on chemical reactions', 'Complete the flow chart', 'Distinguish between', 'Write properties/characteristics/uses/advantages/effects', Give examples. It may contain any other type of question.

To solve a numerical problem, you must remember all formulae given in chapters 1, 4, 5, 6, 7 and 10 from the textbook. To attempt a question on chemical reactions, you must remember all chemical reactions. Some of these questions may be difficult to answer.

- **Q. 3** will contain 8 questions carrying **3 marks** each. You have to attempt any five out of these. 'Give explanation using the given statements', 'Suggest remedies/measures', 'Explanation of diagrams', 'Complete the table/chart', 'Explain with the help of examples', 'Solve the numerical problem', 'Complete the diagram', 'Answer questions based on figures', 'Write answers with explanation', 'Write the law, theory and explain the same', 'Complete the paragraph'. It may contain any other type of question.
- **Q. 4** will contain two questions carrying **5 marks each**. You have to attempt any one out of these. 'Draw a figure and write explanation', 'To correct the incorrect diagram and write explanation', 'Classify with detailed explanation', 'Read the given paragraph and answer the questions', 'Complete the given incomplete table/chart and write explanation', 'Answer the questions in detail', 'Draw a concept diagram based on the given item and explain'. It may contain any other type of question.

To score maximum marks in the final examination you must study regularly. Also try to solve the practice activity sheet in the stipulated time. Read the entire activity sheet carefully for the first 5 minutes to decide the order in which the questions should be attempted.

Guidelines for Time Management

Question	Sub-question	Time to complete the answers
Read the entire activity sheet carefully for first 5 minutes		
Q. 1 (A) and (B)	10 objective questions	20 minutes
Q. 2 (A) and (B)	5 questions of 2 marks each (5 minutes each)	25 minutes
Q. 3	5 questions of 3 marks each (8 minutes each)	40 minutes
Q. 4	1 question of 5 marks	15 minutes
Use the last 15 minutes to check the answer paper.		

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Section 1**BOARD'S ACTIVITY SHEET : JULY 2024**
(With Full Solution and Marking Scheme)**Time : 2 Hours]****[Marks : 40****Note :** (i) All questions are compulsory.

(ii) Use of a calculator is not allowed.

(iii) The numbers to the right of the questions indicate full marks.

(iv) In case of MCQs (Q. No. 1(A)), only the first attempt will be evaluated and will be given credit.

(v) For each MCQ, the correct alternative (A), (B), (C) or (D) with subquestion number is to be written as an answer.

e.g. (i) (A), (ii) (B), (iii) (C)

(vi) Scientifically and technically, labelled diagrams should be drawn wherever necessary.

Q. 1. (A) Choose the correct alternative and write the correct option : 5**Note :**

■ This part will contain five multiple choice questions, carrying 1 mark each.

(i) In modern periodic table, the non-metals are found in

(July '23)

(A) s-block (B) p-block (C) d-block (D) f-block

(ii) $\text{CuSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$.

This is type of chemical reaction.

(A) displacement (B) double displacement
(C) combination (D) decomposition

(iii) The unit of electric power is the

(A) joule (B) volt (C) watt (D) coulomb

(iv) We can see the sun even when it is a little below the horizon because of

(March '24)(A) reflection of light (B) refraction of light
(C) dispersion of light (D) absorption of light

(v) is one of the combustible components of L.P.G.

(March '22)

(A) Ethane (B) Propane (C) Methane (D) Ethene

ANSWERS

(i) (B); (ii) (A); (iii) (C); (iv) (B); (v) (B). (1 mark each)

Q. 1. (B) Answer the following questions :

5

Note :

- This question will contain five types of objective questions 1 mark each.
- All questions will be of different types.
- Different types of questions :
 - (1) Find the odd one out.
 - (2) Correlate.
 - (3) Match the columns/make pairs.
 - (4) True or False.
 - (5) Give the names/molecular formula.
 - (6) Diagram-based question.
- Any other type of question may be asked.

(i) Write the correlation :

Brass : Copper and Zinc : : Bronze :

(ii) Write True or False :

Benzene is an aromatic compound.

(iii) Find the odd one out :

Myopia; Presbyopia; Hypermetropia; Retina

(iv) What is the angle of refraction when the angle of incidence is 0° ?

(v) Name *two* appliances which work on the principle of magnetic effect of electric current.

ANSWERS

(i) Copper and Tin (1 mark)

(ii) True (1 mark)

(iii) Retina (1 mark)

(iv) The angle of refraction is 0° . (1 mark)

(v) Electric motor, Electric fan (Any two appliances). ($\frac{1}{2}$ mark each)

Note :

- This question will contain 3 subquestions of 2 marks each. Attempt any 2 out of 3.
- Minimum 1 question from chemistry and 1 from physics will be asked.
- Here we have provided answers to all the subquestions.

- (i) Sodium is always kept in kerosene. (Sept. '21)
- (ii) A simple microscope is used for watch repairs.
- (iii) It is necessary to manage the space debris.

ANSWERS

- (i) (1) Sodium is a very reactive metal. It reacts vigorously with oxygen in the air/water even at room temperature and evolves large amount of heat, therefore catches fire. (1 mark)
- (2) It does not react with kerosine and sinks in it. Hence, to protect sodium and to prevent accidental fires it is always kept in kerosine. (1 mark)
-
- (ii) (1) When an object is placed within the focal length of a magnifying glass or simple microscope (convex lens), its larger and erect image is obtained on the same side of the lens as that of the object. (1 mark)
- (2) By adjusting the distance between the object and the lens, the image can be obtained at the minimum distance of distinct vision. Thus, a watch repairer can see the minute parts of a watch more clearly with the aid of a magnifying glass (a simple microscope) than with the naked eye, without any stress on the eye. Hence, watch repairers use a magnifying glass (a simple microscope) while repairing the watches. (1 mark)
-
- (iii) (1) In space, nonessential objects such as the parts of launchers and satellites, revolving around the earth are called the debris in space. (1 mark)
- (2) The debris can be harmful to the artificial satellites. They can collide with the satellites or spacecrafts and damage them.

In that case, the satellite/spacecraft will not be able to function properly.

Hence, it is necessary to manage the debris.

(1 mark)

Q. 2. (B) Answer the following questions : (Any three)

6

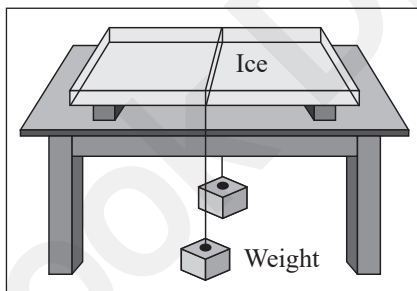
Note :

- This question will contain 5 subquestions carrying 2 marks each. Attempt any 3 out of 5.
- Minimum 2 questions from chemistry and 2 from physics will be asked.
- Different types of questions :
 - (a) Numerical problems.
 - (b) Laws/Definitions/Principles.
 - (c) Short notes.
 - (d) Chemical reactions.
 - (e) Flow chart.
 - (f) Distinguish between.
 - (g) Properties/characteristics/uses/advantages/effects.
 - (h) Examples.
- Any other type of question may be asked.
- Here, we have provided answers to all the subquestions.

(i) The absolute refractive index of water is 1.33. What is the velocity of light in water?

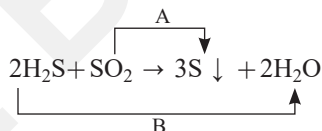
(Velocity of light in vacuum = 3×10^8 m/s)

(ii) Observe the figure and answer the questions given below :



- (a) Name the phenomenon shown in the figure.
(b) Define the above phenomenon.

(iii)



- (a) Identify the above chemical reaction.
(b) Name the type of reaction in A and B.

(iv) Rearrange the Columns 2 and 3 so as to match with the Column 1 :

Column 1	Column 2	Column 3
(1) Triad	Concentrated mass and +ve charge	Thomson
(2) Octave	Average of the atomic masses of first and third elements	Newlands
	Properties of the eighth element are similar to first	Dobereiner

(v) Define the following terms :

- (a) Escape velocity (b) Free fall

ANSWERS

(i) Data :

$$v_1 = 3 \times 10^8 \text{ m/s}, \quad n = 1.33 \quad v_2 = ?$$

Solution :

$$n = \frac{v_1}{v_2} \quad (1/2 \text{ mark})$$

$$\therefore 1.33 = \frac{3 \times 10^8}{v_2} \quad (1/2 \text{ mark})$$

$$\therefore v_2 = \frac{3 \times 10^8}{1.33} \quad (1/2 \text{ mark})$$

$$v_2 = 2.25 \times 10^8 \text{ m/s (Velocity of light in water)} \quad (1/2 \text{ mark})$$

(ii) (a) Regelation. (1 mark)

(b) The phenomenon in which ice converts into liquid due to applied pressure and then reconverts to ice, once the pressure is removed is called regelation. (1 mark)

(iii) (a) The above chemical reaction is Redox reaction. (1 mark)

(b) Type of reaction A is Reduction reaction. (1/2 mark)

Type of reaction B is Oxidation reaction.. (1/2 mark)

(iv)	Column 1	Column 2	Column 3
(1) Triad	Average of the atomic masses of first and the third elements. (½ mark)	Dobereiner (½ mark)	
(2) Octave	Properties of the eighth element are similar to first. (½ mark)	Newlands (½ mark)	

(v) (a) Escape velocity : The minimum velocity required by an object to overcome the downward pull by the earth and escape the earth forever is called escape velocity. (1 mark)

(b) Free fall : Whenever an object moves under the influence of gravity alone, its path is called free fall. (1 mark)

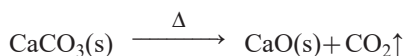
Q. 3. Answer the following questions : (Any five)

15

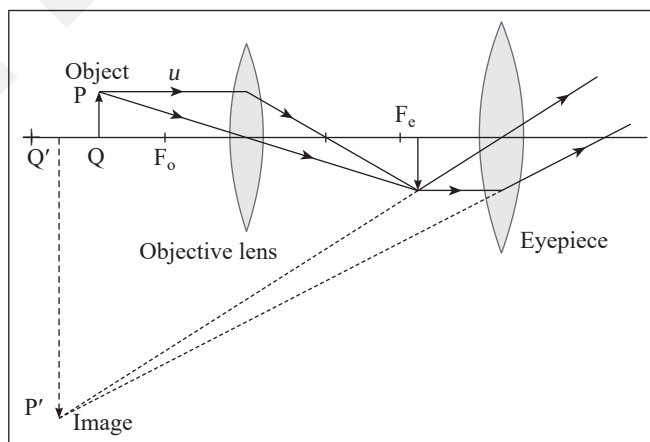
Note :

- This question will contain 8 subquestions carrying 3 marks each. Attempt any 5 out of 8.
- 4 questions from chemistry and 4 from physics will be asked.
- Different types of questions :
 - (a) Give explanation using the given statements.
 - (b) Suggest remedies/measures.
 - (c) Explanation of a diagram.
 - (d) Complete the table/chart.
 - (e) Explain with the help of examples.
 - (f) Solve numerical problems.
 - (g) Complete the diagram.
 - (h) Answer questions based on figures.
 - (i) Write answers with explanation.
 - (j) Write the law, theory and explain.
 - (k) Complete the paragraph.
- Any other type of questions may be asked.
- Here, we have provided answers to all the subquestions.

(i) What information will you get from the given reaction?



- (ii) Observe the following ray diagram and answer the questions given below :



- Identify the instrument shown in the diagram.
 - What is the combined effect of two lenses?
 - Write the use of the above instrument.
- (iii) Read the following paragraph and answer the questions given below :

Rainbow is a beautiful natural phenomenon. It is the combined effect of dispersion, refraction and total internal reflection of light. It can be seen mainly after rainfall. Small droplets of water act as small prisms. When light rays from the sun enter these droplets they get refracted and dispersed. Then there is internal reflection and after that once again the light gets refracted while coming out of the droplet. All these three processes together produce the rainbow.

- Which *three* major phenomena are responsible for rainbow formation?
 - How does refraction take place?
 - Draw a diagrammatic representation of rainbow production.
- (iv) From the elements of second period of the modern periodic table given below, answer the following questions :

3 Li Lithium 6.941	4 Be Beryllium 9.012	5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.908	10 Ne Neon 20.180
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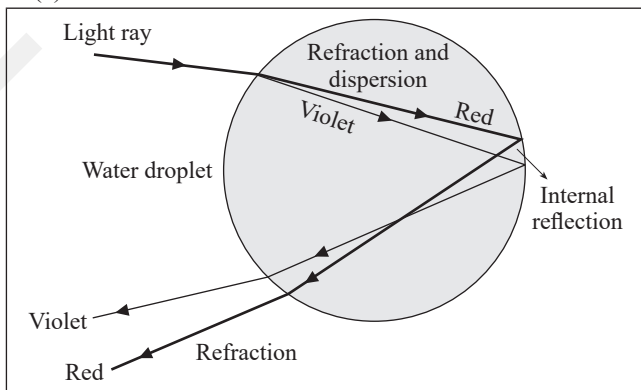
- (a) Name the elements in which both the shells are completely filled with electrons.
- (b) Name the element which has same number of electrons in both the shells.
- (c) Which is the most electropositive element in this period?
- (v) Explain on which factors does the value of acceleration due to gravity g depend.
- (vi) Explain the following terms :
(a) Metallurgy (b) Ores (c) Gangue.
- (vii) Give the names of the following satellite series developed by India :
(a) Satellite series actively working in the field of telecommunication, television and meteorological services.
(b) Satellite series specially working in the field of education.
(c) Satellite series to exactly locate the position of any place on the earth's surface.
- (viii) (a) What is meant by anomalous behaviour of water?
(b) Explain the role of anomalous behaviour of water in preserving aquatic life in cold climate.

ANSWERS

- (i) (1) Type of chemical reaction is Decomposition reaction. **(1 mark)**
(2) When lime stone is heated, calcium oxide and carbon dioxide are formed. When this carbon dioxide gas is passed through freshly prepared lime water, insoluble calcium carbonate and water are formed. In this reaction, lime water turns milky. **(1 mark)**
(3) It is an Exothermic reaction. **(1 mark)**
-
- (ii) (a) Compound microscope. **(1 mark)**
(b) Higher magnification can be obtained by the combined effect of the two lenses. **(1 mark)**
(c) It is used to observe blood corpuscles, plant and animal cells, microorganisms like bacteria, etc. **(1 mark)**
-
- (iii) (a) The three major phenomena responsible for rainbow formation are dispersion, internal reflection and refraction. **(1 mark)**

- (b) When a ray of light from the sun enters a water droplet, it gets refracted as the medium changes from air to water. **(1 mark)**

(c)



(1 mark)

- (iv) (a) The element in which both the shells are completely filled with electrons is Neon (Ne). **(1 mark)**

- (b) The element which has same number of electrons in both the shells is Beryllium (Be). **(1 mark)**

- (c) Lithium is the most electropositive element in this period.

(1 mark)

- (v) (1) The value of the acceleration due to gravity, g , changes from place to place on the earth. It also varies with the altitude and depth below the earth's surface. The factors affecting the value of g are the shape of the earth, altitude and depth below the earth's surface. The earth is not perfectly spherical. It is somewhat flat at the poles and bulging at the equator. At the surface of the earth, the value of g is maximum (9.832 m/s^2) at the poles as the polar radius is minimum, while it is minimum (9.78 m/s^2) at the equator as the equatorial radius is maximum. **(1 mark)**

- (2) As the height (h) above the earth's surface increases, the value of g decreases. It varies as $\frac{1}{(R+h)^2}$, where R is the radius of the earth. **(1 mark)**

- (3) In the interior of the earth, on the average, the value of g is less than that at the earth's surface. As the depth below the

earth's surface increases, the value of g decreases and finally it becomes zero at the centre of the earth. (1 mark)

(vi) (a) Metallurgy : The process used for extraction of metals in their pure form from their ores, then metals are further purified by different methods of purification. All the process is called metallurgy. (1 mark)

(b) Ores : The minerals from which metals are extracted profitably and conveniently are called ores.

Examples : Bauxite ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$), Cinnabar (HgS). (1 mark)

(c) Gangue : Ores contain metal compounds with some of the impurities like soil, sand, rocky material, etc. These impurities are called gangue. (1 mark)

(vii) (a) INSAT and GSAT satellite series are actively working in the field of telecommunication, television and meteorological services. (1 mark)

(b) EDUSAT satellite series is specially working in the field of education. (1 mark)

(c) IRNSS satellite series is used to exactly locate the position of any place on the earth's surface. (1 mark)

(viii) (a) If we heat water from 0°C up to 4°C , it contracts instead of expanding. At 4°C its volume is minimum (due to contraction). If the water is heated further, it expands and hence its density decreases. This behaviour of water between its temperature from 0°C to 4°C is called anomalous behaviour of water. (1 mark)

(b) In cold regions, during winter, the temperature of the atmosphere falls well below 0°C . As the temperature decreases, the water at the surfaces of lakes and ponds starts contracting. Hence, its density increases and it sinks to the bottom. ($\frac{1}{2}$ mark)

This process continues till the temperature of all the water in a lake falls to 4°C . As the water at the surface cools further, i.e. its temperature falls below 4°C , it starts expanding instead of contracting. Therefore, its density decreases and it remains at the surface. ($\frac{1}{2}$ mark)

The temperature of the water at the surface continues to fall to 0°C . Finally, the water at the surface is converted into ice, but the water below the layer of ice is at 4°C . ($\frac{1}{2}$ mark)

Ice is a bad conductor of heat. Hence, the layer of the ice at the surface does not allow transfer of heat from the water to the atmosphere. As the water below the layer of ice remains at 4°C , fish and other aquatic animals and plants can survive in it.

($\frac{1}{2}$ mark)

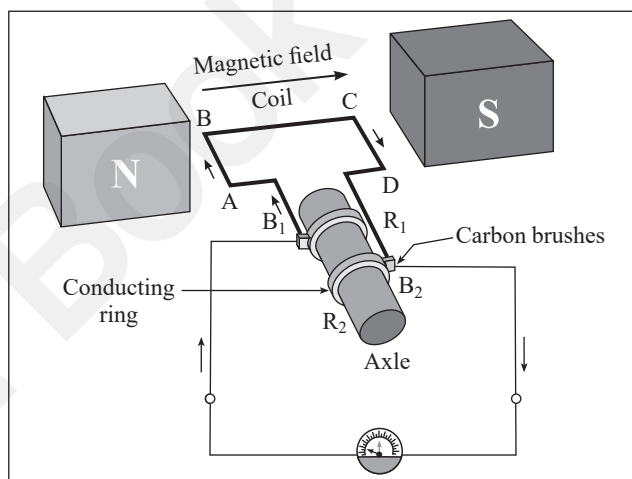
Q. 4. Answer *any one* of the following questions :

5

Note :

- This question will contain 2 subquestions carrying 5 marks each. Attempt any 1 out of 2.
- 1 question from chemistry and 1 from physics will be asked.
- Different types of questions :
 - (a) Draw a figure and give explanation.
 - (b) Correct the given incorrect diagram and draw it.
 - (c) Classify with detailed explanation.
 - (d) Read the given paragraph and answer the questions based on it.
 - (e) Complete the given incomplete table/chart and give explanation.
 - (f) Answer the questions in detail.
 - (g) Make a concept diagram based on any component or technology.
- Any other type of question may be asked.
- Here we have provided answers to all the subquestions.

- (i) Observe the following diagram and answer the questions given below :



- (a) Identify the above diagram.
(b) Write the principle on which the above appliance works.
(c) Write the working of the above appliance.
(d) Write the use of the above appliance.
- (ii) Read the following paragraph and answer the questions given below :

Carbon compounds contain many elements. The element hydrogen is present to a smaller or larger extent in majority of carbon compounds. The compounds which contain carbon and hydrogen as the only two elements are called hydrocarbons. Hydrocarbons are the simplest and the fundamental organic compounds. The smallest hydrocarbon is methane (CH_4), formed by combination of one carbon atom and four hydrogen atoms. We have already seen the structure of methane. Ethane is one more hydrocarbon. Its molecular formula is C_2H_6 .

The carbon compounds having a double bond or triple bond between two carbon atoms are called unsaturated compounds. Ethene and ethyne are unsaturated hydrocarbons. The unsaturated hydrocarbons containing a carbon-carbon double bond are called 'Alkenes'. The unsaturated hydrocarbons whose structures contain a carbon-carbon triple bond are called 'Alkynes'. Generally the unsaturated compounds are more reactive than the saturated compounds.

- (a) What is meant by hydrocarbon?
(b) Name the smallest hydrocarbon.
(c) Define unsaturated compound.
(d) Differentiate between Alkene and Alkyne.

ANSWERS

- (i) (a) Working of an electric generator. (1 mark)
(b) The working of an electric generator is based on the principle of electromagnetic induction. When the coil of an electric generator rotates in a magnetic field, a current is induced in the coil. This induced current then flows in the circuit connected to the coil. (1 mark)
(c) When the axle is rotated with a machine from outside, the coil ABCD starts rotating. Suppose the coil rotates in clockwise

direction, as observed from the side AD, then as the branch AB moves upward, the branch CD moves downward. By Fleming's right hand rule, the induced current flows in the direction $A \rightarrow B \rightarrow C \rightarrow D$ ($\frac{1}{2}$ mark) and in the external circuit, it flows from B_2 to B_1 through the galvanometer.

($\frac{1}{2}$ mark)

The induced current is proportional to the number of turns of the copper wire in the coil. After half a rotation, AB and CD interchange their places. Hence, the induced current flows in the direction $D \rightarrow C \rightarrow B \rightarrow A$.

(1 mark)

As AB is always in contact with B_1 and CD is in contact with B_2 , the current in the external circuit flows from B_1 to B_2 through the galvanometer.

($\frac{1}{2}$ mark)

Thus, the direction of the current in the external circuit is opposite to that in the previous half rotation. The process goes on repeating and alternating current is generated.

(d) This machine is used to produce electricity. (1 mark)

(ii) (a) The compound which contains carbon and hydrogen as the only two elements are called hydrocarbons. (1 mark)

(b) Methane (CH_4) is the smallest hydrocarbon. (1 mark)

(c) The carbon compound having a double or triple bond between two carbon atoms is called unsaturated compound. (1 mark)

(d)

Alkene	Alkyne
(i) The unsaturated hydrocarbons containing a carbon-carbon double bond are called alkenes.	(i) The unsaturated hydrocarbons containing a carbon-carbon triple bond are called alkynes.
(ii) Alkenes have a general formula of C_nH_{2n} .	(ii) Alkynes have a general formula of $\text{C}_n\text{H}_{2n-2}$.

(Two points : 1 mark each)

Section 2**MOST LIKELY QUESTION SETS
(ACTIVITY SETS)****Question
Set
1****MULTIPLE CHOICE QUESTIONS**

[Textbook-Based Only]

[Q. 1 (A)]

Remember :

- ☛ In question 1 (A), there will be **5 multiple choice questions** carrying 1 mark each.
- ☛ While answering these questions, you are expected to write only the alphabet corresponding to the correct choice, e.g. (1) (C); 2 (D) ...
- ☛ The alphabet that you write in the first attempt will be considered for marking. Hence, you have to be very careful when you write the answer.

Q. Select the correct option and write only the alphabet corresponding to the correct option : (1 mark each)

Chapter 1 : Gravitation

- (1) The SI unit of universal constant of gravitation G is
(A) $\frac{N \cdot m^2}{kg}$ (B) $\frac{N \cdot m^2}{kg^2}$ (C) $\frac{N \cdot m}{kg}$ (D) $\frac{N \cdot kg^2}{m^2}$
- (2) As we go above the earth's surface, the value of g
(A) increases (B) becomes zero
(C) doesn't change (D) decreases **(Sept. '21)**
- (3) If an object weighs 120 N on the moon, what will be its weight on the earth?
(A) 240 N (B) 360 N (C) 720 N (D) 600 N
- (4) The formula for escape velocity is
(A) $\sqrt{\frac{2M}{R}}$ (B) $\sqrt{\frac{2GM}{R}}$ (C) $\sqrt{\frac{GM}{R^2}}$ (D) $\sqrt{\frac{GMm}{R^2}}$ **(July '22)**
- (5) The escape velocity of a body from the earth is about
(A) 11.2 km/s (B) 11.2 m/s (C) 2.37 km/s (D) 112 km/s

Ans. (1) (B); (2) (D); (3) (C); (4) (B); (5) (A).

Chapter 2 : Periodic Classification of Elements

- (1) Lithium (Li), and potassium (K) is a Dobereiner triad.
(A) magnesium (Mg) (B) aluminium (Al)
(C) sodium (Na) (D) calcium (Ca) (Nov. '20)
- (2) Which of the following triads does not follow Dobereiner's law of triads?
(A) Li, Na, K (B) Ca, Sr, Ba (C) Be, Mg, Ca (D) Cu, Ag, Au
- (3) During Newlands' time, elements were known.
(A) 56 (B) 65 (C) 63 (D) 36
- (4) According to Mendeleev's periodic law, properties of elements are a periodic function of their
(A) atomic numbers (B) atomic masses
(C) densities (D) boiling points (March '20)
- (5) Eka-boron was subsequently named as
(A) gallium (B) germanium
(C) scandium (D) molybdenum
- (6) In which block of the modern periodic table are the nonmetals found?
(A) s-block (B) p-block (C) d-block (D) f-block (July '23 - '24)
- (7) Halogens belong to group in the modern periodic table.
(A) 15 (B) 16 (C) 17 (D) 18
- (8) Element belongs to the eighteenth group.
(A) Na (B) Cl (C) Fe (D) Ne (Sept. '21)
- (9) The number of electrons in the outermost shell of alkali metals is
(A) 1 (B) 2 (C) 3 (D) 7
- (10) Molecular formula of the chloride of an element X is XCl. This compound is a solid having high melting point. Which of the following elements be present in the same group as X?
(A) Na (B) Mg (C) Al (D) Si
- Ans.** (1) (C); (2) (D); (3) (A); (4) (B); (5) (C); (6) (B); (7) (C); (8) (D); (9) (A); (10) (A).

Chapter 3 : Chemical Reactions and Equations

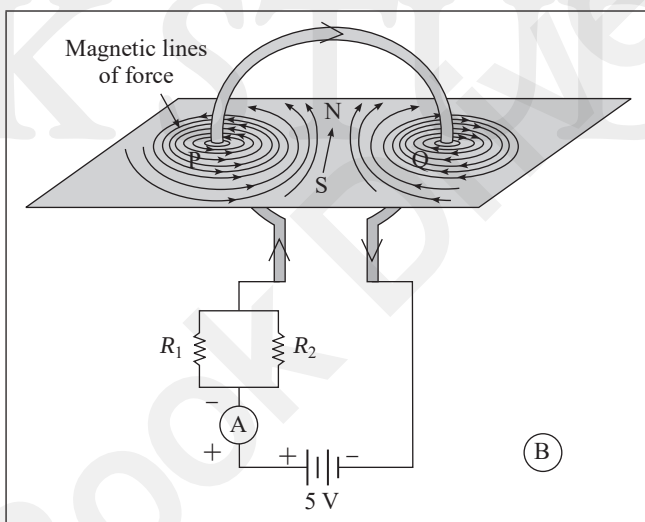
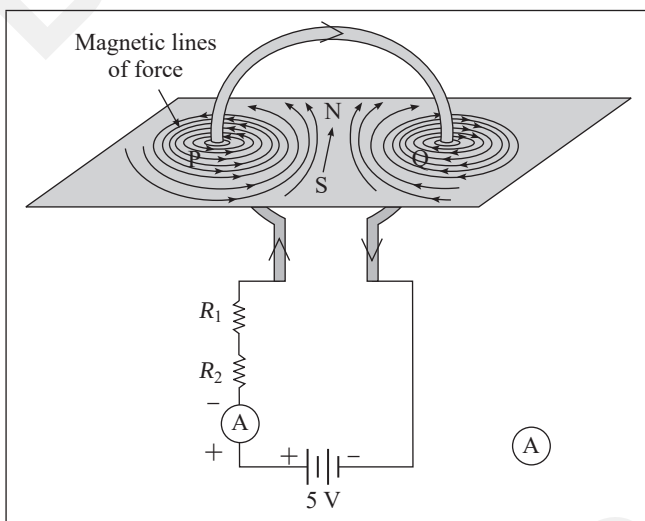
- (1) The reactions in which ions in the reactants are exchanged to form a precipitate is called as reaction.
(A) combination (B) decomposition
(C) displacement (D) double displacement (March '25)

- (2) Combustion of coal in air is a reaction.
 (A) combination (B) displacement
 (C) decomposition (D) double displacement (July '19)
- (3) Reddish brown deposit formed on iron nails kept in a solution of copper sulphate is
 (A) Cu_2O (B) Cu (C) CuO (D) CuS
- (4) In a double displacement reaction,
 (A) ions remain at rest (B) ions get liberated
 (C) ions are exchanged (D) ions are not produced
- (5) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{Heat}$ is an example of reaction.
 (A) exothermic (B) electrolysis
 (C) decomposition (D) endothermic (March '25)
- (6) The conversion of ferrous sulphate into ferric sulphate is reaction.
 (A) oxidation (B) displacement
 (C) electrolysis (D) reduction (Nov. '20)
- (7) To prevent rusting, a layer of metal is applied on iron sheets.
 (July '22)
 (A) potassium (B) sodium (C) magnesium (D) zinc
- (8) The reaction of iron nail with copper sulphate solution is reaction.
 (A) a double displacement (B) a displacement
 (C) a combination (D) a decomposition (March '19)
- Ans.** (1) (D); (2) (A); (3) (B); (4) (C); (5) (A); (6) (A); (7) (D); (8) (B).

Chapter 4 : Effects of Electric Current

- (1) is used to make a solenoid type coil in an electric bulb.
 (A) Nichrome (B) Copper
 (C) Tungsten (D) Aluminium (March '25)
- (2) The direction of the magnetic field around a straight conductor carrying current is given by
 (A) the right hand thumb rule (B) Fleming's left hand rule
 (C) Fleming's right hand rule (D) none of these

(3) Observe the following diagram and choose the correct alternative :



- (A) The intensity of magnetic field in A is larger than that in B.
 (B) The intensity of magnetic field in B is less than that in A.
 (C) The intensity of magnetic field in A and B is same.
 (D) The intensity of magnetic field in A is less than that in B

(March '19)

(4) When an electric current is passed through a solenoid, it shows magnetic lines of force similar to a

- (A) bar magnet (B) horseshoe magnet
(C) disc magnet (D) spherical magnet (July '19)

Ans. (1) (C); (2) (A); (3) (D); (4) (A).

Chapter 5 : Heat

(1) The density of water is maximum at

- (A) 0°C (B) -4°C (C) 4°C (D) 4K

(2) The phenomenon in which ice converts into liquid due to applied pressure and then reconverts into ice once the pressure is removed is called

- (A) boiling (B) regelation
(C) freezing (D) evaporation (Sept. '21)

(3) Water expands on reducing its temperature below $^{\circ}\text{C}$.

- (A) 0 (B) 4 (C) 8 (D) 12 (March '19)

(4) The SI unit of heat is

- (A) the calorie (B) the joule
(C) $\text{kcal/kg}\cdot^{\circ}\text{C}$ (D) $\text{Cal/g}\cdot^{\circ}\text{C}$ (March '24)

(5) The vapour content in the air is measured using a physical quantity called

- (A) absolute humidity (B) relative humidity
(C) dew point (D) humidity (March '20)

(6) The SI unit of specific heat capacity is

- (A) $\frac{\text{J}}{\text{kg}\cdot^{\circ}\text{C}}$ (B) $\frac{\text{J}}{\text{K}}$ (C) $\frac{\text{J}}{\text{kg}\cdot\text{K}}$ (D) $\frac{\text{J}}{\text{kg}}$

Ans. (1) (C); (2) (B); (3) (B); (4) (B); (5) (A); (6) (C).

Chapter 6 : Refraction of Light

(1) If a ray of light passes from a denser medium to a rarer medium in a straight line, the angle of incidence must be

- (A) 0° (B) 30° (C) 60° (D) 90° (March '23)

(2) A ray of light makes an angle of 50° with the surface S1 of the glass slab. Its angle of incidence will be

- (A) 50° (B) 40° (C) 140° (D) 10° (March '19)

(3) The process of separation of light into its component colours while it is passing through a medium is called

- (A) reflection (B) refraction
(C) dispersion (D) internal reflection (Nov. '20)

(4) colour is deviated the least in the spectrum of white light obtained with a glass prism.

- (A) Red (B) Yellow (C) Violet (D) Blue (March '22)

(5) For a particular value of i , the value of r becomes equal to 90° . This value of i is called

- (A) the critical angle (B) the angle of deviation
(C) the angle of refraction (D) the angle of emergence (July '22)

Ans. (1) (A); (2) (B); (3) (C); (4) (A); (5) (A).

Chapter 7 : Lenses

(1) The least distance of distinct vision is

- (A) 10 cm (B) 2.5 m (C) 25 cm (D) 2.5 cm

(2) A lens does not produce any deviation of a ray of light passing through

- (A) its centre of curvature
(B) its optical centre
(C) its principal focus
(D) an axial point at a distance $2f$ from its centre

(3) The power of a concave lens of focal length 40 cm is

- (A) +2.5 D (B) +0.25 D (C) -0.25 D (D) -2.5 D

(4) The power of a convex lens of focal length 20 cm is

- (A) +5.0 D (B) 0.20 D (C) -5.0 D (D) 0.5 D

(March '23)

(5) The power of a convex lens of focal length 25 cm is

- (A) 4.0 D (B) 0.25 D (C) -4.0 D (D) -0.4 D

(March '22)

(6) The controls the amount of light entering the eye.

- (A) iris (B) pupil (C) cornea (D) retina (July '22)

(7) For the normal human eye, the near point is at cm.

- (A) 10 (B) 20 (C) 25 (D) 30 (March '20)

Ans. (1) (C); (2) (B); (3) (D); (4) (A); (5) (A); (6) (B); (7) (C).

Chapter 8 : Metallurgy

- (1) In process a layer of molten tin is deposited on metals.
(A) Anodization (B) Tinning
(C) Galvanizing (D) Alloying **(March '24)**
- (2) Iron is
(A) more reactive than zinc
(B) more reactive than aluminium
(C) less reactive than copper
(D) less reactive than aluminium
- (3) is a good conductor of electricity.
(A) Bromine (B) Iodine
(C) Graphite (D) Sulphur **(March '23)**
- (4) In the Wilfley table method, the particles of gangue are separated by separation method.
(A) the magnetic (B) the froth floatation
(C) the hydraulic (D) the gravitational **(March '20)**
- (5) Carbonate ores are strongly heated in a limited supply of air to transform them into oxides, this process is called
(A) leaching (B) calcination
(C) roasting (D) tinning **(July '22)**
- (6) Gold plated ornaments is the example of
(A) electroplating (B) alloying
(C) anodizing (D) galvanizing **(March '22)**
- (7) is used as an anode during the electrolytic reduction of bauxite.
(A) Sulphur (B) Graphite (C) Platinum (D) Aluminium
- (8) Bronze is an alloy of
(A) copper and tin (B) copper and zinc
(C) copper and iron (D) iron and nickel

Ans. (1) (B); (2) (D); (3) (C); (4) (D); (5) (B); (6) (A); (7) (B);
(8) (A).

Chapter 9 : Carbon Compounds

- (1) The property of direct bonding between atoms of the same element to form a chain is called
(A) catenation (B) isomerism
(C) dehydration (D) polymerization

- (2) The carbon compound is used in daily life is
 (A) *edible oil* (B) *salt*
 (C) *carbon dioxide* (D) *baking soda* (March '19)
- (3) The melting point of pure ethanoic acid is
 (A) 17°C (B) 19°C (C) 15°C (D) 27°C (Nov. '20)
- (4) In homologous series, consecutive members differ from each other by group.
 (A) $-\text{CH}$ (B) $-\text{CH}_2$ (C) $-\text{CH}_3$ (D) $-\text{CH}_4$
- (5) The organic compounds having double or triple bonds in them are termed as
 (A) *saturated hydrocarbons* (B) *stable compound*
 (C) *volatile compounds* (D) *unsaturated hydrocarbons*
- (6) The molecular weights of two adjacent members in a homologous series of an alkane differ by units.
 (A) 16 (B) 20 (C) 14 (D) 12
- (7) is the general formula of alkene.
 (A) C_nH_{2n} (B) $\text{C}_n\text{H}_{2n+2}$ (C) $\text{C}_n\text{H}_{2n-2}$ (D) C_nH_{n-2}
- (8) is used to prepare carbon black.
 (A) *Methane* (B) *Ethene* (C) *Propane* (D) *Butane*
- (9) The reaction of methane with chlorine in the presence of sunlight is called
 (A) *pyrolysis* (B) *an elimination reaction*
 (C) *a substitution reaction* (D) *an addition reaction*
- (10) $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$ is
 (A) *the carboxylic acid group* (B) *the aldehyde group*
 (C) *the ketonic group* (D) *the alcohol group*
- (11) Ethanoic acid has a odour.
 (A) *rotten eggs* (B) *pungent*
 (C) *mild* (D) *vinegar-like*

Ans. (1) (A); (2) (A); (3) (A); (4) (B); (5) (D); (6) (C); (7) (A);
 (8) (A); (9) (C); (10) (A); (11) (D).

Chapter 10 : Space Missions

- (1) The astronomical object closest to us is in our galaxy.
(A) Mars (B) Venus (C) Jupiter (D) Moon (March '20)
- (2) The functioning of the satellite launch vehicle is based on
(A) Newton's first law of motion
(B) Newton's second law of motion
(C) Newton's third law of motion
(D) Newton's universal law of gravitation (March '22)
- (3) is a satellite launch vehicle.
(A) PSLV (B) GSAT (C) IRNSS (D) INSAT (Sept. '21)
- (4) The height of a medium earth orbit above the surface of the earth is
(A) 1500 km (B) 250 km (C) 45000 km (D) 25000 km (March '23)
- (5) The minimum velocity of the spacecraft to escape from the earth's gravitational force must be (Nov. '20)
(A) 112 km/s (B) 11.2 km/s (C) 1.12 km/s (D) 0.112 km/s

Ans. (1) (D); (2) (C); (3) (A); (4) (D); (5) (B).

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Question Set 2

FIND THE ODD ONE OUT

[Q. 1 (B)]

Remember :

- ☛ 'Find the odd one out' type questions are given in this set.
- ☛ Usually four terms are given. Take into consideration the relationship among them and find the one which does not fit into this group. You may be expected to write the justification for your answer.

Q. 1. Find the odd one out (and give reasons) : **(1 mark each)**

Note : The explanation to correct answers are given for better understanding. It should be written if asked in the examination.

Chapter 2 : Periodic Classification of Elements

- (1) Newlands, Moseley, Dobereiner, Mendeleev.
- (2) Fluorine, Sulphur, Bromine, Iodine.
- (3) Sodium, Aluminium, Chlorine, Carbon.

Ans.

- (1) **Moseley.** (Moseley brought out the importance of atomic number, while the others tried to classify the elements on the basis of atomic mass.)
- (2) **Sulphur.** (Sulphur is a nonmetal. Others are halogens.)
- (3) **Carbon.** (Carbon belongs to the second row, while the others belong to the third row.)

Chapter 4 : Effects of Electric Current

- (1) Voltmeter, Ammeter, Thermometer, Galvanometer. **(March '20)**
- (2) Loudspeaker, Microphone, Electric motor, Magnet. **(March '23 – '25)**

Ans.

- (1) **Thermometer.** (It measures temperature, the remaining three measure electrical quantities.)
- (2) **Magnet.** (It exerts a force on a magnetic material, the remaining three convert one form of energy into another.)

Chapter 5 : Heat

- (1) Temperature, Conduction, Convection, Radiation.

- (2) The joule, The erg, The calorie, The newton.
- (3) cal/g, cal/g·°C, kcal/kg·°C, erg/g·°C.

Ans.

- (1) **Temperature.** (It is a physical quantity. Others are modes of transfer of heat.)
- (2) **The newton.** (It is a unit of force. Others are units of energy as well as work.)
- (3) **cal/g.** (It is a unit of specific latent heat. Others are units of specific heat capacity.)

Chapter 7 : Lenses

- (1) Camera, Telescope, Peephole in door, Microscope. **(July '23)**
- (2) Myopia, Presbyopia, Hypermetropia, Spectrometer.
- (3) TV, Motion picture, Complete circle formed by a revolving burning incense stick, Colour blindness.
- (4) Planets, Stars, Satellites, Rainbow.

Ans.

- (1) **Peephole in door.** (In this case a concave lens is used. In other cases, a convex lens is used.)
- (2) **Spectrometer.** (It is an instrument; others are defects of vision.)
- (3) **Colour blindness.** (Others are examples of persistence of vision.)
- (4) **Rainbow.** (Others are celestial bodies.)

Chapter 8 : Metallurgy

- (1) Zinc, Iron, Phosphorus, Sodium **(Nov. '20)**
- (2) Sodium, Potassium, Silver, Sulphur
- (3) Boron, Chlorine, Bromine, Fluorine
- (4) Copper, Iron, Mercury, Brass
- (5) Brass, Bronze, Phosphorus, Stainless steel
- (6) Tinning, Anodization, Alloying, Froth floatation **(March '19)**

Ans.

- (1) **Phosphorus.** (Phosphorus is a nonmetal. Others are metals.)
- (2) **Sulphur.** (Sulphur is a nonmetal. Others are metals.)
- (3) **Boron.** (Boron is a metalloid. Others are nonmetals.)
- (4) **Brass.** (Brass is an alloy. Others are metals.)
- (5) **Phosphorus.** (Phosphorus is a nonmetal. Others are alloys.)

- (6) **Froth floatation.** (This process is used for concentration of ores. Others are processes of coating a thin layer of metal on the surface of other metals.)

Chapter 9 : Carbon Compounds

- (1) Methane, butane, benzene, sodium chloride
(2) Propane, methane, ethene, butane (July '22)
(3) CH_4 , C_2H_6 , C_3H_8 , CaCO_3
(4) C_2H_2 , C_3H_8 , C_2H_6 , CH_4
(5) C_2H_4 , C_4H_{10} , C_3H_8 , CH_4
(6) Polyethylene, Polysaccharide, Polystyrene, Polypropylene

Ans.

- (1) **Sodium chloride.** (Sodium chloride is an ionic compound. Others are organic compounds.)
(2) **Ethene.** (Ethene is an unsaturated hydrocarbon. Others are saturated hydrocarbons.)
(3) **CaCO_3 .** (CaCO_3 is an inorganic compound. Others are organic compounds.)
(4) **C_2H_2 .** (C_2H_2 is an unsaturated hydrocarbon. Others are saturated hydrocarbons.)
(5) **C_2H_4 .** (C_2H_4 is an unsaturated hydrocarbon. Others are saturated hydrocarbons.)
(6) **Polysaccharide.** (It is a carbohydrate. Others are man-made polymers.)

Chapter 10 : Space Missions

INSAT, GSAT, IRS, PSLV (March '22)

Ans. PSLV. (It is a satellite launch vehicle. Others are satellites.)

Question Set 3

CORRELATION

[Q. 1 (B)]

Remember :

- ☛ Taking into consideration the relation between the first two terms or groups of words, pair the third word or group of words accordingly with proper answer.

Q. Complete the correlation :

(1 mark each)

Chapter 2 : Periodic Classification of Elements

- (1) Solid : Iodine : : : Bromine.
- (2) Dobereiner : Triad : : Newlands' law :
- (3) Mendeleev's periodic table : Atomic mass : : Modern periodic table :
- (4) Group-1 : Alkali metals : : : Halogens. (March '22)
- (5) Horizontal row : Periods : : : Groups.
- (6) Chlorine : 2, 8, 7 : : Fluorine :

Ans. (1) Liquid (2) Octaves (3) Atomic number (4) Group 17
(5) Vertical columns (6) 2, 7.

Chapter 3 : Chemical Reactions and Equations

- (1) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$: Combination reaction : :
 $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$:
- (2) CuI_2 : Brown : : AgCl : (March '23)
- (3) $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$: Oxidation : :
 $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$:
- (4) $\text{C}_{12}\text{H}_{22}\text{O}_{11} \xrightarrow{\text{heat}} 12\text{C} + 11\text{H}_2\text{O}$: Decomposition
reaction : : $\text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{H}_2\text{CO}_3$:
- (5) Molecular formula of beryllium oxide : BeO : : Molecular formula of beryllium chloride : (July '19)
- (6) $\text{CuCl}_2 + 2\text{KI} \rightarrow \text{CuI}_2 + 2\text{KCl}$: Double displacement reaction : :
 $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$:

Ans. (1) Decomposition reaction (2) White (3) Reduction
(4) Combination reaction (5) BeCl_2 (6) Displacement reaction.

Chapter 4 : Effects of Electric Current

Resistance : The ohm : : Potential difference : (July '23)

Ans. The volt.

Chapter 7 : Lenses

(1) Torch : Concave lens : : Camera : (Nov. '20)

(2) Object at infinity : Image at focus F_2 : : Object between F_1 and $2F_1$: (Sept. '21)

(3) Nearsightedness : Elongated eyeball : : Farsightedness :

(4) Convex lens : Converging : : Concave lens :

(5) Object at $2F_1$ of a convex lens : Image at $2F_2$: : Object at F_1 :

Ans. (1) Convex lens (2) Image beyond $2F_2$ (3) Flattened eyeball
(4) Diverging (5) Image on the opposite side at infinity.

Chapter 8 : Metallurgy

(1) The sulphide ores are strongly heated in air : Roasting : : The carbonate ores are strongly heated in a limited supply of air :

(2) Tinning : Tin : : Galvanizing :

(3) Pressure cooker : Anodizing : : Silver plated spoons :

Ans. (1) Calcination (2) Zinc (3) Electroplating.

Chapter 9 : Carbon Compounds

(1) Alkene : $\text{C}=\text{C}$: : Alkyne : (March '20)

(2) Methane : Saturated hydrocarbon : : Propene :

Ans. (1) $\text{C}\equiv\text{C}$ (2) Unsaturated hydrocarbon.

Chapter 10 : Space Missions

(1) Hubble telescope : 569 km high from the earth's surface : : Revolving orbit of Hubble telescope : (March '19)

(2) INSAT : Weather satellite : : IRS :

(3) IRNSS : Direction showing satellite : : INSAT :

Ans. (1) Low Earth Orbit (2) Earth observation satellite (3) Weather satellite.

**Question
Set
4**

MATCH THE COLUMNS

[Q. 1 (B)]

Remember :

- ☛ Since there would be more terms in the second column, a wise selection of the terms is needed.

Q. Match the following :

(1 mark)

Chapter 1 : Gravitation

Column A	Column B
Escape velocity	(a) $\sqrt{\frac{GM}{R}}$ (b) $\sqrt{\frac{2GM}{R}}$ (c) $\sqrt{\frac{GM}{2R}}$

Ans. Escape velocity – $\sqrt{\frac{2GM}{R}}$.

Chapter 2 : Periodic Classification of Elements

(1) Column I	Column II
Dobereiner	(a) Atomic number (b) Triads (c) Atomic mass

Ans. Dobereiner – Triads.

(2) Column I	Column II
Modern periodic table	(a) Group 17 (b) Period 2 (c) Atomic number

Ans. Modern periodic table – Atomic number.

(3)	Column I	Column II
	Noble gas	(a) 18 elements (b) Eight elements (c) Helium

Ans. Noble gas – Helium.

(4)	Column I	Column II
	The <i>p</i> -block elements	(a) Lanthanides and actinides (b) Groups 1, 2 (c) Groups 13 to 18 and zero group

Ans. The *p*-block elements – Groups 13 to 18 and zero group.

Chapter 3 : Chemical Reactions and Equations

(1)	Column I	Column II
	Double displacement reaction	(a) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$ (b) $\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s}) \xrightarrow{\Delta} 12\text{C}(\text{s}) + 11\text{H}_2\text{O}(\text{g})$ (c) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$

Ans. Double displacement reaction – $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$.

(2)	Column I	Column II
	Reduction	(a) Type of chemical reaction (b) Combination with hydrogen (c) Losing hydrogen

Ans. Reduction – Combination with hydrogen.

(3)	Column I	Column II
	Oils and fats are allowed to stand in air for a long time	(a) Slow reaction (b) Rancid (c) Exothermic reaction

Ans. Oils and fats are allowed to stand in air for a long time – Rancid.

Chapter 4 : Effects of Electric Current

Match the correct pair :

Column A	Column B (July '22)
Electric current	(a) The ohm (b) The ampere (c) The volt

Ans. Electric current – The ampere.

Chapter 5 : Heat

Column A	Column B (Sept. '21)
Anomalous behaviour of water	(a) 0°C to -10°C (b) 0°C to 4°C (c) 0°C to 10°C

Ans. Anomalous behaviour of water – 0°C to 4°C .

Chapter 6 : Refraction of Light

(1) Group A	Group B (March '23)
Substance	Refractive index
Air	(a) 1.33 (b) 1.46 (c) 1.0003

Ans. Air – 1.0003.

(2) Column A	Column B (March '22 –'24)
Substance	Refractive index
Refractive index of water	(a) 1.31 (b) 1.36 (c) 1.33

Ans. Refractive index of water – 1.33.

Chapter 7 : Lenses

(1) Column A	Column B (March '25)
Simple microscope	(a) Used to observe minute objects (b) Used to see distant objects (c) Used for watch repair

Ans. Simple microscope – Used for watch repair.

(2) Column A	Column B (Sept. '21)
Power of a lens	(a) $\frac{1}{f}$ (b) $\frac{h_2}{h_1}$ (c) f

Ans. Power of a lens – $\frac{1}{f}$.

Chapter 8 : Metallurgy

(1) Column I	Column II
Gold	(a) Combustible (b) Soluble in water (c) High ductility

Ans. Gold – High ductility.

(2) Column I	Column II
Copper and tin	(a) Stainless steel (b) Brass (c) Bronze

Ans. Copper and tin – Bronze.

(3) Column I	Column II
Bauxite	(a) Mercury (b) Aluminium (c) Tin

Ans. Bauxite – Aluminium.

(4) Column I	Column II
Galvanising	(a) Pressure cooker (b) Coating of tin on copper (c) Coating of Zn on iron

Ans. Galvanising – Coating of Zn on iron.

Chapter 9 : Carbon Compounds

(1) Column I	Column II
Ethyne	(a) C_2H_6 (b) C_2H_2 (c) C_3H_6

Ans. (1) Ethyne – C_2H_2 .

(2) Column I	Column II
Aromatic hydrocarbon	(a) Propyne (b) Benzene (c) Saturated hydrocarbon

Ans. Aromatic hydrocarbon – Benzene.

(3) Column I	Column II
– COOH	(a) Aldehyde (b) Alcohol (c) Carboxylic acid

Ans. (– COOH) – Carboxylic acid.

(4) Column I	Column II (July '19)
Ethanol	(a) Hydrogen peroxide (b) Tincture iodine (c) Biogas

Ans. Ethanol – Tincture iodine.

(5) Column I	Column II
Cellulose	(a) P.V.C. pipes (b) Blankets (c) Wood

Ans. Cellulose – Wood.

**Question
Set
5**

TRUE OR FALSE

[(Q. 1 (B))]

Remember :

- ☛ The given statement may be true or false. Identify and write only the answer.
- ☛ Correcting the statement and rewriting it is not expected.

Q. State whether the following statements are True or False :

(1 mark each)

Chapter 1 : Gravitation

- (1) If the separation between two particles is decreased by a factor of 2, the gravitational force between the particles becomes double the initial force.
 - (2) At the centre of the earth, the value of the acceleration due to gravity becomes zero.
 - (3) The weight of a body is minimum at the poles.
 - (4) The value of g increases with altitude.
 - (5) The escape velocity of a body does not depend on the mass of the body.
- Ans.** (1) False (2) True (3) False (4) False (5) True.

Chapter 2 : Periodic Classification of Elements

- (1) Newlands stated the law of triads.
- (2) Newlands was the first to classify elements having similar chemical properties into groups of three.
- (3) Dobereiner stated the law of octet.
- (4) Dobereiner named the group of elements having similar properties as triad.
- (5) Eka-aluminium was later named as germanium.
- (6) The d -block elements are called transition elements.
- (7) Mendeleev arranged elements in the increasing order of their atomic masses.
- (8) All the elements beyond atomic number 92 are man-made.

Ans. (1) False (2) False (3) False (4) True (5) False (6) True (7) True (8) True.

Chapter 3 : Chemical Reactions and Equations

- (1) In a chemical equation the reactants are written on the right hand side and the products on the left hand side. **(Sept. '21)**
 - (2) The simple form of representation of a chemical reaction in words is known as word reaction.
 - (3) When calcium carbonate is heated, it decomposes into calcium oxide and oxygen gas.
 - (4) The reaction between salt and water is an example of exothermic reaction.
 - (5) The speed of a chemical reaction depends on the catalyst used in the chemical reaction.
 - (6) The rate of reaction increases on increasing the temperature.
 - (7) Rancidity is oxidation process. **(July '23)**
 - (8) Glucose combines with oxygen in our body and provides energy. The reaction is an endothermic reaction.
 - (9) A redox reaction takes place during cellular respiration. **(March '25)**
 - (10) The chemical formula of rust is $\text{Fe}_3\text{O}_4 \cdot x\text{H}_2\text{O}$.
- Ans.** (1) False (2) True (3) False (4) False (5) True (6) True
(7) True (8) False (9) True (10) False.

Chapter 4 : Effects of Electric Current

- (1) An electric motor converts mechanical energy into electrical energy. **(March '22)**
 - (2) The frequency of AC is 50 Hz. **(March '20)**
 - (3) The SI unit of electric power is the watt.
- Ans.** (1) False (2) True (3) True.

Chapter 5 : Heat

- (1) Specific latent heat of fusion is expressed in cal/g.
- (2) At dew point relative humidity is 100%.
- (3) $1 \text{ kcal} = 4.18 \text{ joules}$.

Ans. (1) True (2) True (3) False.

Chapter 6 : Refraction of Light

- (1) The incident ray and the refracted ray of light are on the opposite sides of the normal at the point of incidence.
- (2) In vacuum, the speed of light depends upon the frequency of light.

- (3) Wavelength of red light is close to 700 nm. (March '23)
(4) In a material medium, the speed of light does not depend on the frequency of light.
(5) The velocity of light is different in different media.
Ans. (1) True (2) False (3) True (4) False (5) True.

Chapter 7 : Lenses

- (1) When the incident ray is parallel to the principal axis, the refracted ray does not pass through the principal focus. (July '22)
(2) A concave lens is a converging lens. (March '19)
(3) A concave lens always forms a virtual image.
(4) Due to the light sensitive cells in the eye, we get information about the brightness or dimness of the object and the colour of the object.
(5) A simple microscope is used for watch repairs. (July '19)
Ans. (1) False (2) False (3) True (4) True (5) True.

Chapter 8 : Metallurgy

- (1) Ionic compounds in the solid state conduct electricity.
(2) Ionic compounds are soluble in water.
(3) Electrolysis is used to obtain pure metals from impure metals.
(4) Aquaregia is a mixture of conc. HCl and conc. HNO_3 in the ratio of 1 : 3.
(5) Mercury, silver and gold are very reactive metals.
(6) In electroplating, a metal is coated with another metal using electrolysis.
(7) Corrosion of metals can be stopped by detaching the air from metals.
Ans. (1) False (2) True (3) True (4) False (5) False (6) True (7) True.

Chapter 9 : Carbon Compounds

- (1) Unsaturated hydrocarbons are less reactive than saturated hydrocarbons.
(2) The chemical bonds in carbon compounds do not produce ions.
(3) Naphthalene burns with a yellow flame.
(4) Saturated fats are healthy.
(5) When vegetable oil and tincture iodine react, the colour of iodine does not change.
(6) Vinegar is a 12 – 15% aqueous solution of acetic acid.
(7) The functional group of ethanoic acid is a carboxylic group.
Ans. (1) False (2) True (3) True (4) False (5) False (6) False (7) True.

Chapter 10 : Space Missions

(True or False with Explanation : 2 marks)

- (1) If a spacecraft has to be sent away from the influence of the earth's gravitational field, its velocity must be less than the escape velocity.
- (2) A satellite needs a specific velocity to revolve in a specific orbit.
- (3) If the height of the orbit of a satellite increases, its velocity must also increase.

Ans.

- (1) False.

Explanation : The escape velocity of a body is the minimum velocity with which it should be projected from the earth's surface, so that it can escape the influence of the earth's gravitational field. This clearly shows that the given statement is false.

- (2) True.

Explanation :

Centripetal force on the satellite $\frac{mv_c^2}{R+h}$ = gravitational force exerted by the earth on the satellite $\frac{GMm}{(R+h)^2}$,

where m : mass of the satellite, v_c : critical velocity of the satellite, h : height of the satellite from the surface of the earth, M : mass of the earth, R : radius of the earth and G : gravitational constant.

$$\therefore v_c^2 = \frac{GM}{R+h} \quad \therefore v_c = \sqrt{\frac{GM}{R+h}}$$

Thus, if the value of h changes, the value of v_c also changes. It means a satellite needs to be given a specific velocity (in the tangential direction) to keep it revolving in a specific orbit.

- (3) False.

Explanation : As per the formula $v_c = \sqrt{\frac{GM}{R+h}}$, if the value of h increases, the value of v_c decreases. Hence, if the height of the satellite from the surface of the earth increases, its velocity decreases.

**Question
Set
6**
**NAME THE FOLLOWING/WRITE
THE CHEMICAL FORMULA/
ELECTRONIC CONFIGURATION**

[Q. 1 (B)]

Remember :

- ☛ This set contains questions of the type 'Write the name/molecular formula/electronic configuration'.
- ☛ Explanation is not expected.

Chapter 2 : Periodic Classification of Elements**Q. 1. Name the following :****(1 mark each)**

- (1) Two elements having a single electron in their outermost shell.
- (2) Three elements with filled outermost shell.
- (3) Three elements having 7 electrons in their outermost shell.
- (4) Horizontal rows in modern periodic table.
- (5) An alkali metal in the period 2.
- (6) Halogen in the period 3.
- (7) Four blocks in modern periodic table. (Nov. '20)
- (8) The elements in periods 2 and 3 having stable electronic configuration.
- (9) The element with electronic configuration (2, 7).

Ans. (1) (i) Hydrogen (ii) Sodium (2) (i) Helium (ii) Neon (iii) Argon
 (3) (i) Fluorine (ii) Chlorine (iii) Bromine (4) Periods (5) Lithium
 (6) Chlorine (7) s-block, p-block, d-block and f-block (8) (i) Neon
 (ii) Argon (9) Fluorine.

Q. 2. Write the name and symbol of the element from the description :**(1 mark each)**

- (1) The atom having the smallest size. (March '24)
- (2) The atom having the smallest atomic mass. (March '25)
- (3) The most electronegative atom. (March '25)
- (4) The noble gas with the smallest atomic nucleus. (March '25)
- (5) The most reactive nonmetal.

Ans. (1) Helium (He) (2) Hydrogen (H) (3) Fluorine (F) (4) Helium (He)
 (5) Fluorine (F).

Chapter 3 : Chemical Reactions and Equations

Q. Name the following :

(1 mark each)

- (1) Name the molecular formula of calcium carbonate. **(March '24)**
- (2) The product formed in the thermal decomposition of sugar.
- (3) The gas evolved when sodium metal reacts with ethanol.
- (4) The precipitate formed when barium sulphide reacts with zinc sulphate.
- (5) The oxidising agent used to oxidise ferrous sulphate.
- (6) The catalyst used to accelerate the rate of decomposition of hydrogen peroxide.

Ans. (1) CaCO_3 (2) Carbon (3) Hydrogen (4) BaSO_4 (Barium sulphate)
(5) Potassium permanganate (KMnO_4) (6) Manganese dioxide (MnO_2).

Chapter 4 : Effects of Electric Current

Q. Name the following :

(1 mark each)

- (1) An alloy used to prepare a coil of high resistance for use in electric appliances such as an electric heater.
- (2) Constituents of the alloy used to make a fuse wire.

Ans. (1) Nichrome (2) Lead and tin.

Chapter 7 : Lenses

Q. Name the following :

(1 mark each)

- (1) The lens which forms a real image or a virtual image depending on the position of the object.
- (2) The lens which produces magnification always less than 1.

Ans. (1) A convex lens (2) A concave lens.

Chapter 8 : Metallurgy

Q. Name the following :

(1 mark each)

- (1) The non metal having electrical conductivity.
- (2) The reagent that dissolves noble metals.
- (3) The oxide that forms salt and water by reacting with both acid and base.
- (4) The process of heating the sulphide ore to a high temperature in the excess of air.

- (5) The process of heating the carbonate ore to a high temperature in limited air.
- (6) The device used for grinding an ore.
- (7) Compounds which are added to lower the fusion temperature.
- (8) Alloy of sodium with mercury.
- (9) Molecular formula of common ore of aluminium.
- (10) The two metals which can be cut with knife. **(March '25)**
- (11) A sound is produced when certain metals are struck. Name this property. **(March '25)**
- (12) The non-metallic substance which is a good conductor of electricity. **(March '25)**

Ans. (1) Graphite (2) Aqua regia (3) Aluminium oxide (Al_2O_3) (4) Roasting (5) Calcination (6) Grinding mill (7) Cryolite (AlF_3 , 3NaF) and fluorspar (8) Sodium amalgam (9) $\text{Al}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ (10) Lithium, Sodium, Potassium (Any two) (11) Sonority (12) Graphite.

Chapter 9 : Carbon Compounds

Q. Name the following :

(1 mark each)

- (1) The functional group in ether and halogen.
- (2) The number of double bonds in benzene.
- (3) The higher homologue of hexane.
- (4) A nomenclature system based on the structure of the compounds and it was accepted all over the world.
- (5) IUPAC name of sodium acetate.
- (6) The main component of natural gas.
- (7) Two oxidising compounds.
- (8) Two carbon compounds used in day-to-day life. **(March '20)**

Ans. (1) Functional groups : Ether : $-\text{O}-$ Halogen : $-\text{X}$ ($-\text{Cl}$, $-\text{Br}$, $-\text{I}$) (2) Three (3) Heptane C_7H_{16} (4) International Union of Pure and Applied Chemistry (IUPAC) (5) Sodium ethanoate (6) Methane (7) Potassium permanganate, Potassium dichromate (8) Methane, sugar.

Chapter 10 : Space Missions

Name the first artificial satellite sent by Russia in space.

(March '20)

Ans. Sputnik.

**Question
Set
7**

SCIENTIFIC REASONS

[Q. 2 (A) 2 marks each]

Remember :

- ☛ You have to attempt any two out of three questions.
- ☛ There will be at least one question on chemistry and at least one question on physics.
- ☛ The distribution of marks is : two points, 1 mark each.

Q. Give scientific reasons :

(2 marks each)

Chapter 1 : Gravitation

The weight of an object changes from place to place though its mass is constant. (March '20)

Ans. (1) The mass (m) of an object is the amount of matter present in it. It is a measure of the inertia of the object. Hence, it is constant.

(2) The weight of an object is the force with which the earth attracts the body towards the centre of the earth. Weight, $W = mg$. As the value of the acceleration due to gravity (g) changes from place to place, the weight of an object changes from place to place.

Chapter 2 : Periodic Classification of Elements

(1) Zero group elements (inert gases) are called noble gases.

Ans. (1) In the atoms of the inert gas elements (zero group elements), all the electronic shells, including the outermost shell, are completely filled.

(2) The electronic configuration is stable and these elements do not lose or accept electrons. These elements do not take part in chemical reactions. These elements are gases. Hence, they are called noble gases.

(2) Atomic radius goes on increasing down a group.

(Sept. '21; March '22)

Ans. (1) The size of an atom is indicated by its radius.

(2) While going down a group a new shell is added. Therefore, the distance between the outermost electron and the nucleus goes on increasing. These electrons experience lesser pull from the nucleus. Thus, atomic radius goes on increasing down a group.

(3) Elements belonging to the same group have the same valency.

(March '20 – '23; Sept. '21)

Ans. (1) The valency of an element is determined by the number of valence electron in the outermost shell of an atom of an element.

(2) All the elements in a group have the same number of valence electrons. Therefore, elements in the same group should have the same valency. For example, the elements of group I contain only one valence electron; the valency of elements of group I is one. Similarly, for group II, the valency is two.

(4) While going from left to right within a period, the size of atom decreases. **(March '25)**

Ans. (1) In a period while going from left to right, atomic radius goes on decreasing and the atomic number increases one by one, that means positive charge on the nucleus increases by one unit at a time.

(2) However, the additional electron is added to the same outermost shell. Due to the increased nuclear charge the electrons are pulled towards the nucleus to a greater extent, as a result the size of atom decreases i.e. atomic radius decreases.

(5) Metallic character goes on decreasing while going from left to right in a period. **(July '23)**

Ans. (1) Metals have a tendency to lose the valence electrons to form cations. This tendency of an element is called the metallic character of the element.

(2) While going from left to right within a period the outermost shell remains the same and electrons are added to the same shell. However, the positive charge on the nucleus goes on increasing while the atomic radius goes on decreasing and thus the effective nuclear charge goes on increasing. As a result of this the tendency of atom to lose electrons decreases, i.e. electropositivity decreases. Thus, metallic character goes on decreasing within a period from left to right.

(6) The third period contains only eight elements even though the electron capacity of the third shell is 18.

Ans. (1) In the modern periodic table, there are seven horizontal rows called periods. In a period, elements are arranged in an increasing order of their atomic numbers. The third row contains eight elements and the electron capacity of the third shell is 18.

(2) In the third period, while moving from left to right, atomic number increases. Hence, the number of electrons in the shell increases. The number

of elements present in the third period is decided on the basis of electronic configuration and octet rule.

Atomic number	11	12	13	14	15	16	17	18
Element	Na	Mg	Al	Si	P	S	Cl	Ar

Argon (Ar) is the last element of the third period and has a capacity of maximum 18 electrons. Its octet of electrons is completed and as argon belongs to the zero group, the third shell contains 18 electrons.

Chapter 3 : Chemical Reactions and Equations

(1) Physical states of reactants and products are mentioned while writing a chemical equation.

Ans. (1) While writing a chemical equation, gaseous, liquid and solid states are symbolised as (g), (l) and (s) respectively.

(2) This is done to make it more informative and to emphasise that those reactions occur in that manner only under those conditions. Hence, physical states of reactants and products are mentioned while writing a chemical equation.

(2) It takes time for pieces of Shahabad tile to disappear in HCl, but its powder disappears rapidly.

Ans. (1) The rate of a reaction depends upon the size of the particles of the reactant taking part in the reaction. The smaller the size of the reactants particles, the more is their total surface area and the greater is the rate of reaction.

(2) In the reaction of dil. HCl with pieces of Shahabad tile, CO_2 effervescence is formed and the tile disappears slowly. On the other hand, CO_2 effervescence forms at greater rate with Shahabad tile powder and it disappears rapidly.

(3) Concentrated hydrochloric acid reacts more vigorously with calcium carbonate than dilute hydrochloric acid.

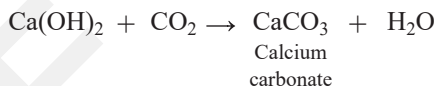
Ans. (1) The rate of a reaction increases with the concentration of the reactant.

(2) As concentrated hydrochloric acid contains more number of HCl molecules than those in an equal volume of dilute HCl, concentrated HCl reacts more vigorously with calcium carbonate than dilute hydrochloric acid.

(4) When the gas formed on heating lime stone is passed through freshly prepared lime water, the lime water turns milky. (March '24)

Ans. (1) When lime stone is heated, calcium oxide (CaO) and carbon dioxide (CO_2) are formed.

(2) This carbon dioxide gas is passed through freshly prepared lime water, insoluble calcium carbonate and water are formed. In this reaction, lime water turns milky.



(5) When silver articles exposed to air turns blackish after sometime.

(March '24)

Ans. (1) On exposure to air, silver reacts with hydrogen sulphide in the air.

(2) As a result a layer of silver sulphide (Ag_2S) is formed on the surface of silver articles which is blackish. Hence the silver articles turn blackish on exposure to air.

(6) Zinc powder reacts much faster with dil. H_2SO_4 than does granulated zinc of the same mass.

Ans. (1) In a reaction, the rate of the reaction depends upon the particle size of the solid reactant as the reaction takes place on the surface only. The smaller the particles, the more is their total surface area and hence the higher is the rate of the reaction.

(2) Hence, zinc powder reacts much faster with dil. H_2SO_4 than does granulated zinc.

(7) Grills of doors and windows are always painted before they are used.

Ans. (1) Grills of doors and windows are made from iron. Iron has a tendency to undergo corrosion.

(2) Paint does not allow air or moisture to come in contact with iron surface.

Therefore, to prevent rusting of iron, grills of doors and windows are always painted before they are used.

(8) It is recommended to use an airtight container for storing oil for a long time.

(March '22)

Ans. (1) If edible oil is allowed to stand for a long time, it undergoes air oxidation; it becomes rancid and its smell and taste changes.

(2) Rancidity in the foodstuff cooked in oil or ghee is prevented by using antioxidants. The process of oxidation of foodstuff can also be slowed down by storing it in an airtight container.

Chapter 4 : Effects of Electric Current

(1) For electric power transmission, copper or aluminium wire is used.

(March '23 – '25)

Ans. (1) Copper and aluminium are good conductors of electricity.

(2) Copper and aluminium have very low resistivity. Hence, when an electric current flows through a wire of copper or aluminium, heat produced is comparatively low. Therefore, for electric power transmission, copper or aluminium wire is used.

(2) In the electric equipment producing heat e.g. iron, electric heater, boiler, toaster, etc. an alloy such as Nichrome is used, not pure metals.

(March '19; July '19)

OR

The coils in heating devices such as a toaster and electric iron are made of an alloy such as Nichrome, rather than a pure metal.

Ans. (1) The working of heating devices such as a toaster and an electric iron is based on the heating effect of electric current, i.e. conversion of electric energy into heat by passage of electric current through a metallic conductor.

(2) An alloy, such as Nichrome, has high resistivity and it can be heated to a high temperature without oxidation, in contrast to pure metals. Therefore, the coils in heating devices such as a toaster and an electric iron are made of an alloy, such as Nichrome, rather than a pure metal.

(3) A magnetic needle shows decreasing deviation of its angle as the distance from a current-carrying conductor is increased. **(July '22)**

Ans. (1) At a given place, deviation of a magnetic needle depends on the intensity of the magnetic field at that place. The greater the intensity, the greater is the angle of deviation.

(2) As we go away from a current-carrying conductor, the intensity of the magnetic field produced by the current goes on decreasing. Hence, the magnetic needle shows decreasing deviation of its angle.

(4) The material used for fuse has low melting point.

OR

A fuse should be made of a material of low melting point.

Ans. (1) A fuse is used to protect a circuit and the appliances connected in the circuit by stopping the flow of an excessive electric current. For this, a fuse is connected in series in the circuit.

(2) When the current in the circuit passes through the fuse, its temperature increases. When the current exceeds the specified value, the fuse must melt to break the circuit. For this, the material used for a fuse has low melting point.

(5) It is necessary to have earthing connections in home electrical circuits. (Sept. '21)

Ans. (1) The potential difference between the live wire and the neutral wire is 220 V. The 'earth' wire is connected to the ground.

(2) Due to a fault in the equipment or if the plastic coating on the live wire and neutral wire gives away, the two wires come in contact with each other. In that case a large amount of current flows through the wires producing heat. To prevent this excess flow of current and to prevent the shock, the earthing wire is used. Thus, it is necessary to have earthing connections in home electrical circuits.

Chapter 5 : Heat

(1) If crushed ice is pressed and then the pressure is released, a lump of ice is formed.

Ans. (1) When crushed ice is pressed, its melting point is lowered and some ice melts to form water. (2) When pressure is released, the melting point becomes normal and the water freezes to form ice forming a lump.

(2) In cold countries, in winter, even when the water of lakes freezes, aquatic animals and plants can survive.

Ans. (1) In cold countries, in winter, a layer of ice is formed on the surface of lakes when the atmospheric temperature falls below 0°C . However, below this layer, there is water at 4°C . (2) Ice, being a bad conductor of heat, does not allow transfer of heat from this water to the atmosphere. Hence, aquatic animals and plants can survive in this water.

Chapter 6 : Refraction of Light

(1) The sun is seen on the horizon a little before sunrise. OR

We see the sun even before it emerges above the horizon. (July '22)

OR

The sun is seen on the horizon for sometime even after sunset.

Ans. (1) The earth is surrounded by an atmosphere which is denser near the surface of the earth. When the rays of light from the sun enter the earth's atmosphere from outer space, they travel from a rarer medium to a denser medium. Hence, they bend towards the normal on refraction.

(2) Hence, even when the sun is below the horizon while rising or setting, its rays reach us due to refraction and it appears to be on the horizon. Therefore, the sun is seen on the horizon a little before sunrise as well as for some time even after sunset.

(2) Stars twinkle but we do not see the twinkling of planets.

(March '20)

OR

Stars appear to be twinkling at night.

(Nov. '20)

Ans. (1) As a star is far away from the earth, it appears as a point source of light. Air is always in motion. The density and temperature of air change continuously resulting in the change in the refractive index of air in a given region of space. Hence, there is a continuous change in the apparent position and brightness of a star. Hence, the star appears to twinkle.

(2) Compared to stars, planets are relatively closer to the earth. Hence, a planet appears as a collection of a large number of point sources. Hence, though the refractive index of air in a given region of space changes continuously, there is no change in the average apparent position and brightness of a planet. Hence, we do not see the twinkling of planets.

Chapter 7 : Lenses

(1) A person suffering from myopia (nearsightedness) uses spectacles of concave lenses.

Ans. (1) A person suffering from myopia can see nearby objects clearly as the image of a nearby object is formed on the retina, but cannot see distant objects clearly as the image of a distant object is formed in front of the retina instead of on the retina.

(2) A concave lens diverges the rays of light passing through it. When spectacles of concave lenses of appropriate power are used, the parallel rays coming from a distant object are diverged to proper extent before they are incident on the eye lens. Therefore, after the converging action of the eye lens, the image of a distant object is formed on the retina of the eye and hence the distant object can be seen clearly.

(2) A person suffering from hypermetropia (farsightedness) uses spectacles of convex lenses.

Ans. (1) A person suffering from hypermetropia can see distant objects clearly as the image of a distant object is formed on the retina, but cannot see nearby objects clearly as the image of a nearby object would be formed behind the retina instead of on the retina.

(2) A convex lens converges the rays of light passing through it. When spectacles of convex lenses of appropriate power are used, the rays of light coming from a nearby object are converged to proper extent before they are incident on the eye lens. Therefore, after the converging action of the eye lens,

the image of a nearby object is formed on the retina of the eye and hence the nearby object can be seen clearly.

(3) In old age, a bifocal lens is necessary for some persons.

Ans. (1) Some people, in old age, suffer from myopia (nearsightedness) as well as hypermetropia (farsightedness).

(2) Myopia is corrected using a concave lens of appropriate power. Hypermetropia is corrected using a convex lens of appropriate power. Therefore, they need a bifocal lens.

(4) You cannot enjoy watching a movie from a very short distance from the screen in a cinema hall.

Ans. (1) The less the distance between the screen in a cinema hall and the person watching the movie, the more is the intensity of light falling on the eye.

(2) This results in great contraction of the pupil of the eye causing a strain. Hence, you cannot enjoy watching a movie from a very short distance from the screen in a cinema hall.

(5) When a burning stick of incense is moved fast in a circle, a circle of red light is seen.

Ans. (1) The impression of the image on the retina lasts for about $\frac{1}{16}$ th of a second after the removal of the object.

(2) If a burning stick of incense is revolved at a rate of more than sixteen revolutions per second, we see a circle of red light due to persistence of vision.

Chapter 8 : Metallurgy

(1) Calcium floats on water during the reaction with water.

Ans. (1) Calcium reacts with water less vigorously hence the heat evolved is not sufficient for hydrogen to catch fire.

(2) Instead, calcium floats on water because the bubbles of hydrogen gas formed stick to the surface of the metal.

(2) Generally the ionic compounds have high melting points.

Ans. (1) The ionic compounds exist in solid state and are hard due to strong electrostatic force of attraction between oppositely charged ions.

(2) The intermolecular force of attraction is high in ionic compounds and large energy is required to overcome it. Therefore, ionic compounds have high melting points.

(3) When zinc granules are added to copper sulphate solution, the blue coloured solution turns colourless.

Ans. (1) Zinc is more reactive than copper.

(2) When zinc granules are added to copper sulphate solution, they displace copper from the copper sulphate solution to form zinc sulphate solution. As zinc sulphate is colourless, the blue coloured solution of copper sulphate disappears.

(4) When an iron nail is dipped into a copper sulphate solution, a shiny coat of copper is formed on the nail.

Ans. (1) Iron is more reactive than copper.

(2) When an iron nail is dipped into a copper sulphate solution, iron displaces copper from the copper sulphate solution. The copper so liberated deposits on the iron nail. As a result, a shiny coat of copper is formed on the nail.

(5) Lemon or tamarind is used for cleaning copper vessels turned greenish. **(March '23)**

Ans. (1) Copper undergoes oxidation in air to form black copper oxide. Copper oxide reacts slowly with carbon dioxide in air and gains a green coat. This green substance is copper carbonate.

(2) Lemon and tamarind contain acid. The acid dissolves the green coating of basic copper carbonate present on the surface of a tarnished copper utensil and makes it shiny again.

(6) Anodes need to be replaced from time to time during the electrolysis of alumina. **(July '19)**

Ans. (1) During electrolysis of alumina, the oxygen liberated at the carbon anode reacts with graphite rods (carbon anode) and forms carbon dioxide.

(2) As the anodes get oxidised during electrolysis of alumina, they are continuously eroded. Hence, it is necessary to replace anodes from time to time.

(7) Cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added to the electrolytic mixture containing pure alumina.

Ans. (1) Alumina has very high melting point ($>2000^\circ\text{C}$). Cryolite (Na_3AlF_6) and fluorspar (CaF_2) lower the fusion temperature of the mixture containing alumina from 2000°C to 1000°C , thereby saving electrical energy.

(2) They increase the conductivity and the mobility of the fused mixture. Hence, cryolite and fluorspar are added to the electrolytic mixture containing pure alumina.

Chapter 9 : Carbon Compounds

(1) Carbon atoms are capable of forming an unlimited number of compounds.

Ans. (1) Carbon has the property of catenation. Two or more carbon atoms can share some of their valence electrons to form (single, double and triple) bonds.

(2) The straight chains or branched chains or rings may have different shapes and sizes. This results in formation of many compounds. Hence, carbon atoms are capable of forming an unlimited number of compounds.

(2) Generally most of the carbon compounds are bad conductor of electricity. *(July '22)*

Ans. (1) Most of the carbon compounds lack ionic compounds.

(2) The chemical bonds in carbon compounds do not produce ions, hence, carbon compounds are bad conductor of electricity.

(3) Ethylene is an unsaturated hydrocarbon.

Ans. (1) Ethylene ($\text{CH}_2 = \text{CH}_2$) contains a double bond between carbon atoms.

(2) Thus, the valencies of the two carbon atoms are not fully satisfied by single covalent bonds. Hence, ethylene is an unsaturated hydrocarbon.

(4) Naphthalene burns with a yellow flame.

Ans. (1) Naphthalene is an unsaturated compound. In unsaturated hydrocarbon, the proportion of carbon is larger than that of saturated hydrocarbon. As a result, some unburnt carbon particles are also formed during combustion of unsaturated compounds.

(2) In the flame, these unburnt hot carbon particles emit yellow light and therefore the flame appears yellow. Hence, naphthalene burns with a yellow flame.

(5) The colour of iodine disappears in the reaction between vegetable oil and iodine.

Ans. (1) Vegetable oil (unsaturated compound) contains a multiple bond as its functional group. It undergoes addition reaction to form a saturated compound as the product.

(2) The addition reaction of vegetable oil with iodine takes place instantaneously at room temperature. The colour of iodine disappears in this reaction. This iodine test indicates the presence of a multiple bond in vegetable oil.

(6) In some countries, ethanol is used as an additive to increase the efficiency of petrol. *(March '25)*

Ans. (1) On combustion in sufficient air ethanol gives carbon dioxide and water as the only products.

(2) In this way, ethanol is a clean fuel. Therefore, in some countries, ethanol is used as an additive fuel to increase the efficiency of petrol.

Geostationary satellites are not useful for studies of polar regions.

OR

Why are geostationary satellites not useful for studies of polar region?

(July '19)

Ans. (1) A geostationary satellite revolves in the equatorial plane of the earth. It revolves in the same sense as the rotation of the earth and its period of revolution equals the period of rotation of the earth. It is stationary as observed from the earth.

(2) It does not fly above the polar regions. Hence, geostationary satellites are not useful for studies of polar regions.

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Question Set 8

NUMERICAL PROBLEMS

[Q. 2 (B) (2 marks) and Q. 3 (3 marks)]

Remember :

- ☛ Write the formula, substitute the data, calculate and write the answer (with appropriate unit).
- ☛ To be able to solve numerical problems, write the formulae in chapters 1, 4, 5, 6, 7 and 10 and memorise them.

Q. Solve the following problems :

Chapter 1 : Gravitation

(1) What would be the value of g on the surface of the earth if its mass was twice and its radius half of what it is now? (March '19) (3 marks)

Solution : Data : $M_2 = 2M_1$, $R_1 = R_2/2$

Earth's gravitational acceleration, $g = \frac{GM}{R^2}$

$$\therefore g_1 = \frac{GM_1}{R_1^2}, g_2 = \frac{GM_2}{R_2^2}$$

$$\therefore \frac{g_2}{g_1} = \left(\frac{M_2}{M_1}\right) \left(\frac{R_1}{R_2}\right)^2 = 2(2)^2 = 8$$

$$\therefore g_2 = 8g_1$$

Thus, the value of g on the surface of the earth would be eight times the present value.

(2) The radius of planet A is half the radius of planet B. If the mass of A is M_A , what must be the mass of B so that the value of g on B is half that of its value on A? (2 marks)

Solution : Data : $R_A = R_B/2$, $g_B = \frac{1}{2} g_A$, $M_B = ?$

$$g = \frac{GM}{R^2} \quad \therefore g_A = \frac{GM_A}{R_A^2} \quad \text{and} \quad g_B = \frac{GM_B}{R_B^2}$$

$$\therefore \frac{g_B}{g_A} = \left(\frac{M_B}{M_A} \right) \left(\frac{R_A}{R_B} \right)^2 \quad \therefore \frac{1}{2} = \left(\frac{M_B}{M_A} \right) \left(\frac{1}{2} \right)^2 = \frac{1}{4} \left(\frac{M_B}{M_A} \right)$$

$$\therefore \frac{M_B}{M_A} = \frac{4}{2} = 2$$

$$\therefore M_B = 2M_A.$$

This is the mass of B in terms of the mass of A.

(3) An object thrown vertically upwards reaches a height of 500 m. What was its initial velocity? How long will the object take to come back to the earth? Assume $g = 10 \text{ m/s}^2$. (July '19) (3 marks)

Solution : Data : $h = 500 \text{ m}$, $g = 10 \text{ m/s}^2$, $v = 0 \text{ m/s}$, $u = ?$,

t (for the object going up) + t (for the object coming down) = ?

(i) As the object moves upward,

$$v^2 = u^2 + 2as$$

$$= u^2 + 2(-g)h \quad (\because a = -g)$$

Now, $v = 0 \text{ m/s}$

$$\therefore u^2 = 2gh = 2 \times 10 \text{ m/s}^2 \times 500 \text{ m}$$

$$\therefore u^2 = (100 \times 100) \text{ m/s}^2$$

$$\therefore u = 100 \text{ m/s (initial velocity of the body)}$$

(ii) Also, $v = u + at = u - gt$

For $v = 0 \text{ m/s}$, $u = gt$

$$\therefore 100 \text{ m/s} = 10 \text{ m/s}^2 \times t$$

$$\therefore t \text{ (for the object going up)} = 10 \text{ s}$$

Now, t (for the object coming down) = t (for the object going up) = 10 s

$$\therefore t \text{ (for the object going up)} + t \text{ (for the object coming down)}$$

$$= 10 \text{ s} + 10 \text{ s} = 20 \text{ s}$$

It will take 20 s for the object to come back to the earth.

(4) A tennis ball is thrown up and reaches a height of 5 m before coming down. What was its initial velocity?

(Assume $g = 10 \text{ m/s}^2$) (Sept. '21) (2 marks)

For reference, see the solved problem (3) above.

Ans. $u = 10 \text{ m/s}$ (Initial velocity of the body).

(5) A metal ball of mass 5 kg falls from a height of 490 m. How much time will it take to reach the ground? ($g = 9.8 \text{ m/s}^2$) (March '19) (2 marks)

Solution : Data : $s = 490 \text{ m}$, $a = g = 9.8 \text{ m/s}^2$,

$$u = 0 \text{ m/s}, \quad s = ut + \frac{1}{2} at^2$$

$$\therefore 490 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2 = 4.9 t^2$$

$$\therefore t^2 = \frac{490}{4.9} = 100$$

$$\therefore t = 10 \text{ s} \quad \text{This is the required time.}$$

(6) A ball falls off a table and reaches the ground in 1 s. Assuming $g = 10 \text{ m/s}^2$, calculate its speed on reaching the ground and the height of the table. (3 marks)

Solution : Data : $t = 1 \text{ s}$, $g = 10 \text{ m/s}^2$, $u = 0 \text{ m/s}$, $s = ?$, $v = ?$

$$(i) v = u + at = u + gt = 0 \text{ m/s} + 10 \text{ m/s}^2 \times 1 \text{ s} = 10 \text{ m/s}$$

$\therefore$ The speed of the ball on reaching the ground = 10 m/s.

$$(ii) s = ut + \frac{1}{2} gt^2 = \frac{1}{2} gt^2 \text{ for } u = 0 \text{ m/s}$$

$$\therefore s = \frac{1}{2} \times 10 \text{ m/s}^2 \times (1 \text{ s})^2 = 5 \text{ m}$$

$\therefore$ The height of the table = 5 m.

(7) An iron ball of mass 3 kg is released from a height of 125 m and falls freely to the ground. Assuming that the value of g is 10 m/s^2 , calculate
(a) the time taken by the ball to reach the ground (July '23) (2 marks)
(b) the velocity of the ball on reaching the ground.

(Nov. '20; March '25) (3 marks)

Solution : Data : $s = h = 125 \text{ m}$, $a = g = 10 \text{ m/s}^2$, $u = 0 \text{ m/s}$, $t = ?$, $v = ?$

(a) According to Newton's second equation of motion,

$$s = ut + \frac{1}{2} at^2$$

$$\therefore 125 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\therefore 125 = 5 t^2$$

$$\therefore t^2 = \frac{125}{5} = 25$$

$$\therefore t = 5 \text{ s}$$

The ball takes 5 s to reach the ground.

- (b) According to Newton's first equation of motion,

$$v = u + at$$

$$\therefore v = 0 + 10 \times 5 = 50 \text{ m/s}$$

The velocity of the ball on reaching the ground is 50 m/s.

(8) The mass and weight of an object on the earth are 5 kg and 49 N respectively. What will be their values on the moon? Assume that the acceleration due to gravity on the moon is $1/6$ th of that on the earth.

(July '22) (3 marks)

Solution : Data : $m = 5 \text{ kg}$, $W = 49 \text{ N}$,

$$g_M = \frac{g_E}{6}, m \text{ (on the moon)} = ?, W \text{ (on the moon)} = ?$$

- (i) The mass of the object on the moon = the mass of the object on the earth = **5 kg**

(ii) $W = mg$

$$\therefore \frac{W_M}{W_E} = \frac{m g_M}{m g_E} = \frac{g_M}{g_E} = \frac{1}{6}$$

$$\therefore W_M = \frac{W_E}{6} = \frac{49 \text{ N}}{6}$$

$$= 8.167 \text{ N (weight of the object on the moon).}$$

(9) Let the period of revolution of a planet at a distance R from a star be T . Prove that if it was at a distance of $2R$ from the star, its period of revolution will be $\sqrt{8}T$. (2 marks)

Proof : $T = \frac{2\pi}{\sqrt{GM}} r^{3/2}$, where T = period of revolution of a planet around

the Sun, M = mass of the Sun, G = gravitational constant and r = radius of the orbit assumed to be circular = distance of the planet from the Sun.

For $r = R$ and $T = T_1$,

$$\therefore T_1 = \frac{2\pi}{\sqrt{GM}} R^{3/2}$$

For $r = 2R$ and $T = T_2$,

$$\therefore T_2 = \frac{2\pi}{\sqrt{GM}} (2R)^{3/2} = \frac{2\pi}{\sqrt{GM}} R^{3/2} \times 2^{3/2} = T_1 \times 2^{3/2}$$

$$\therefore T_2 = T_1 \sqrt{8} = \sqrt{8} T.$$

(10) Calculate the escape velocity on the surface of the moon, given the mass and radius of the moon to be 7.34×10^{22} kg and 1.74×10^6 m respectively. (Given : $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$) (March '23) (3 marks)

Solution : Data : $M = 7.34 \times 10^{22}$ kg, $R = 1.74 \times 10^6$ m,
 $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$

Escape velocity on the surface of the moon,

$$\begin{aligned} v_{\text{esc}} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 7.34 \times 10^{22}}{1.74 \times 10^6}} \\ &= \sqrt{\frac{13.34 \times 7.34 \times 10^5}{1.74}} = \sqrt{\frac{1.334 \times 7.34 \times 10^6}{1.74}} \\ &= 2.372 \times 10^3 \text{ m/s} \\ &= 2.372 \text{ km/s.} \end{aligned}$$

Chapter 4 : Effects of Electric Current

(1) Two tungsten bulbs of wattage 50 W and 60 W power work on 220 V potential difference. If they are connected in parallel, how much current will flow in the main conductor? (March '19) (2 marks)

Solution : Data : $P_1 = 50$ W, $P_2 = 60$ W, $V = 220$ V, $I = ?$

$$P = VI \quad \therefore I = \frac{P}{V}$$

$$\therefore I_1 = \frac{P_1}{V} \text{ and } I_2 = \frac{P_2}{V}$$

Current in the main conductor, $I = I_1 + I_2$ (parallel connection)

$$\begin{aligned} &= \frac{P_1}{V} + \frac{P_2}{V} = \frac{P_1 + P_2}{V} = \frac{50 \text{ W} + 60 \text{ W}}{220 \text{ V}} = \frac{110}{220} \text{ A} = \frac{1}{2} \text{ A} \\ &= 0.5 \text{ A.} \end{aligned}$$

(2) Who will spend more electrical energy? 500 W TV set in 30 minutes, or 600 W heater in 20 minutes? (Sept. '21) (3 marks)

Solution : Data : $P_1 = 500$ W, $t_1 = 30$ minutes = $\frac{30}{60}$ h, $P_2 = 600$ W,

$$t_2 = 20 \text{ minutes} = \frac{20}{60} \text{ h}$$

Electrical energy used = Pt

$$\text{TV set : } P_1 t_1 = 500 \text{ W} \times \frac{30}{60} \text{ h} = \frac{500}{2} \text{ W.h} = 250 \text{ W.h}$$

$$\text{Heater : } P_2 t_2 = 600 \text{ W} \times \frac{20}{60} \text{ h} = 200 \text{ W.h}$$

Thus, the TV set will spend more electrical energy than the heater.

(3) An electric bulb is connected to a source of 250 volts. The current passing through it is 0.27 A. What is the power of the bulb? (2 marks)

Solution : Data : $V = 250 \text{ V}$, $I = 0.27 \text{ A}$, $P = ?$

$$P = VI = 250 \text{ V} \times 0.27 \text{ A} = 67.5 \text{ W}$$

The power of the bulb = 67.5 W.

(4) An electric iron of 1100 W is operated for 2 hours daily. What will be the electrical consumption expenses for that in the month of April? (The electric company charges ₹ 5 per unit of energy.) (3 marks)

Solution : Data : $P = 1100 \text{ W}$, $t = 2 \times 30 = 60 \text{ h}$, ₹ 5 per unit of energy, expenses = ?

$$N = \frac{Pt}{1000 \text{ W.h/unit}} = \frac{1100 \text{ W} \times 60 \text{ h}}{1000 \text{ W.h/unit}} \\ = 66 \text{ units}$$

$$\therefore \text{Electrical consumption expenses} = 66 \text{ units} \times ₹ 5 \text{ per unit} \\ = ₹ 330.$$

Chapter 5 : Heat

(1) A copper sphere of 100 g mass is heated to raise its temperature to 100°C and is released in water of mass 195 g and temperature 20°C in a copper calorimeter. If the mass of the calorimeter is 50 g, what will be the maximum temperature of water? (Specific heat of copper = $0.1 \text{ cal/g}^\circ\text{C}$.)

(Nov. '20) (3 marks)

Solution : Data : $m_1 = 100 \text{ g}$, $c_1 = 0.1 \text{ cal/g}^\circ\text{C}$, $T_1 = 100^\circ\text{C}$,

$m_2 = 195 \text{ g}$, $c_2 = 1 \text{ cal/g}^\circ\text{C}$, $T_2 = 20^\circ\text{C}$, $m_3 = 50 \text{ g}$, $c_3 = 0.1 \text{ cal/g}^\circ\text{C}$,

$T_3 = 20^\circ\text{C}$, $T = ?$

Heat lost by the copper sphere,

$$Q_1 = m_1 c_1 (T_1 - T) = 100 \times 0.1 \times (100 - T)$$

Heat absorbed by the water,

$$Q_2 = m_2 c_2 (T - T_2) = 195 \times 1 \times (T - 20)$$

Heat absorbed by the calorimeter,

$$Q_3 = m_3 c_3 (T - T_3) = 50 \times 0.1 \times (T - 20)$$

Now, $Q_1 = Q_2 + Q_3$

$$\therefore 1000 - 10 T = 195 T - 195 \times 20 + 5T - 100$$

$$\therefore 210 T = 1000 + 3900 + 100 = 5000$$

$$\therefore T = \frac{5000}{210} = \frac{500}{21} = 23.8^\circ\text{C}$$

This is the maximum temperature of water.

(2) Equal heat is given to two objects A and B of mass 1 g each. The temperature of A increases by 3°C and that of B by 5°C . Which object has more specific heat? And by what factor? (2 marks)

Solution : Data : $m = 1$ g, $\Delta T_1 = 3^\circ\text{C}$, $\Delta T_2 = 5^\circ\text{C}$, Q same

Here, $Q = mc_1 \Delta T_1 = mc_2 \Delta T_2$

$$\therefore \frac{c_1}{c_2} = \frac{\Delta T_2}{\Delta T_1} = \frac{5^\circ\text{C}}{3^\circ\text{C}} = \frac{5}{3}$$

Thus, $c_1 > c_2$

The specific heat of A is more than that of B and

$$\frac{\text{specific heat of A}}{\text{specific heat of B}} = \frac{5}{3}$$

Chapter 6 : Refraction of Light

(Given : $c = 3 \times 10^8$ m/s)

(1) If the speed of light in a medium is 1.5×10^8 m/s, what is the absolute refractive index of the medium? (Speed of light in vacuum = 3×10^8 m/s)

(July '19) (2 marks) (March '22) (3 marks)

Solution : Data : $v = 1.5 \times 10^8$ m/s, $c = 3 \times 10^8$ m/s, $n = ?$

$$n = \frac{c}{v} = \frac{3 \times 10^8 \text{ m/s}}{1.5 \times 10^8 \text{ m/s}} = 2.$$

This is the absolute refractive index of the medium.

(2) The refractive index of water is $4/3$ and the speed of light in air is 3×10^8 m/s. Find the speed of light in water. (2 marks)

Solution : Data : ${}_1n_2 = 4/3$, $v_1 = 3 \times 10^8$ m/s, $v_2 = ?$

$${}_1n_2 = \frac{v_1}{v_2} \quad \therefore v_2 = \frac{v_1}{{}_1n_2}$$

$$\therefore v_2 = \frac{3 \times 10^8 \text{ m/s}}{4/3} = \frac{9 \times 10^8 \text{ m/s}}{4} = 2.25 \times 10^8 \text{ m/s}$$

The speed of light in water = 2.25×10^8 m/s.

(3) If the absolute refractive indices of glass and water are $3/2$ and $4/3$ respectively, what is the refractive index of glass with respect to water? (2 marks)

Solution : Data : $n_g = \frac{3}{2}$, $n_w = \frac{4}{3}$, ${}_wn_g = ?$

$$n_g = \frac{c}{v_g}, \quad n_w = \frac{c}{v_w}, \quad {}_wn_g = \frac{v_w}{v_g}$$

$$\therefore {}_wn_g = \frac{n_g}{n_w} = \frac{3/2}{4/3} = \frac{3 \times 3}{4 \times 2} = \frac{9}{8}$$

This is the refractive index of glass with respect to water.

Chapter 7 : Lenses

(1) When a pin of height 3 cm is fixed at 10 cm from a convex lens, the height of the virtual image formed is 12 cm. Find the focal length of the lens. (3 marks)

Solution : Data : Convex lens, $h_1 = 3$ cm, $h_2 = 12$ cm (virtual image), $u = -10$ cm, $f = ?$

$$\frac{h_2}{h_1} = \frac{v}{u}$$

$$\therefore \frac{12 \text{ cm}}{3 \text{ cm}} = \frac{v}{-10 \text{ cm}} \quad \therefore v = -40 \text{ cm}$$

$$\text{Now, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\therefore \frac{1}{f} = \frac{1}{-40 \text{ cm}} - \frac{1}{-10 \text{ cm}} = \frac{1}{10 \text{ cm}} - \frac{1}{40 \text{ cm}} = \frac{4-1}{40 \text{ cm}} = \frac{3}{40 \text{ cm}}$$

$$\therefore f = \frac{40}{3} \text{ cm} = 13.33 \text{ cm (approximately)}$$

The focal length of the lens = 13.33 cm [approximately].

(2) An object of height 4 cm is placed in front of a concave lens of focal length 40 cm. If the object distance is 60 cm, find the position and height of the image. (3 marks)

Solution : Data : $f = -40 \text{ cm}$ (concave lens), $u = -60 \text{ cm}$,

$$h_1 = 4 \text{ cm}, v = ?, h_2 = ?$$

$$(i) \frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\therefore \frac{1}{v} = \frac{1}{-40 \text{ cm}} + \frac{1}{-60 \text{ cm}} = -\left(\frac{1}{40 \text{ cm}} + \frac{1}{60 \text{ cm}}\right)$$

$$= -\left(\frac{3+2}{120 \text{ cm}}\right) = -\frac{5}{120} \text{ cm}$$

$$\therefore v = -\frac{120 \text{ cm}}{5} = -24 \text{ cm}$$

The image is formed at 24 cm from the lens. It is on the same side as the object.

$$(ii) \frac{h_2}{h_1} = \frac{v}{u} \quad \therefore h_2 = \frac{v}{u} h_1$$

$$\therefore h_2 = \left(\frac{-24 \text{ cm}}{-60 \text{ cm}}\right) \times 4 \text{ cm} = 0.4 \times 4 \text{ cm} = 1.6 \text{ cm}$$

The height of the image is 1.6 cm.

(3) Three lenses having powers 2 D, 2.5 D and 1.7 D are kept touching in a row. What is the total power of the lens combination? (2 marks)

Solution : Data : $P_1 = 2 \text{ D}$, $P_2 = 2.5 \text{ D}$, $P_3 = 1.7 \text{ D}$, $P = ?$

Total power of the lens combination,

$$P = P_1 + P_2 + P_3 = 2 \text{ D} + 2.5 \text{ D} + 1.7 \text{ D} = \mathbf{6.2 \text{ D}}.$$

(4) Doctor has prescribed a lens having power +1.5 D for correction of eye defect. What is the focal length of the lens? What is the type of the lens and what must be the defect of vision? (July '19) (3 marks)

Solution : Data : $P = +1.5 \text{ D}$, $f = ?$

$$\text{Focal length of the lens, } f = \frac{1}{P} = \frac{1}{1.5 \text{ D}}$$

$$= \frac{10}{15} \text{ m} = 0.6667 \text{ m} \doteq \mathbf{0.67 \text{ m}}$$

***P* is positive. This shows that the lens is convex. The defect of vision is farsightedness (hypermetropia).**

(5) Surabhi from Std. X uses spectacles. The power of the lenses in her spectacles is 0.5 D. Answer the following questions from the given information : (March '19) (3 marks)

- (i) Identify the type of lenses used in her spectacles.**
- (ii) Identify the defect of vision Surabhi is suffering from.**
- (iii) Find the focal length of the lenses used in her spectacles.**

Ans. (i) Convex lenses are used in the spectacles used by Surabhi.

(ii) Surabhi is suffering from hypermetropia (farsightedness).

(iii) Focal length of the lenses used in her spectacles

$$= \frac{1}{\text{power of the lens}} = \frac{1}{0.5 \text{ D}} = \mathbf{2 \text{ m.}}$$

(6) The focal length of a convex lens is 25 cm. What is its power?

(July '22) (2 marks)

Solution : Data : $f = 25 \text{ cm} = 0.25 \text{ m}$

$$\text{Power of the lens, } P = \frac{1}{f} = \frac{1}{0.25 \text{ m}} = \mathbf{4 \text{ D.}}$$

Chapter 10 : Space Missions

(1) The orbit of a satellite is exactly 35780 km above the earth's surface and its tangential velocity is 3.08 km/s.

How much time will the satellite take to complete one revolution around the earth? (Radius of the earth = 6400 km) (March '24) (3 marks)

Solution : Data : $h = 35780 \text{ km} = 3.578 \times 10^7 \text{ m}$,

$$R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m},$$

$$v_c = 3.08 \text{ km/s} = 3.08 \times 10^3 \text{ m/s}$$

$$\begin{aligned}
 T &= \frac{2\pi(R+h)}{v_c} = \frac{2 \times 3.14 \times (0.64 \times 10^7 + 3.578 \times 10^7)}{3.08 \times 10^3} \\
 &= \frac{6.28 \times 4.218 \times 10^4}{3.08} = 86003 \text{ s} = \frac{86003}{3600} \text{ h} = 23.89 \text{ h} \\
 &= 23 \text{ hours } 53 \text{ minutes}
 \end{aligned}$$

This is the required time.

(2) If the mass of a planet is eight times the mass of the earth and its radius is twice the radius of the earth, what will be the ratio of the escape velocity on the earth to the escape velocity on the planet? (July '19) (2 marks)

Solution : Data : $M_p = 8M_E$, $R_p = 2R_E$, $v_{\text{escP}}/v_{\text{escE}} = ?$

$$\text{Escape velocity, } v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

$$\begin{aligned}
 v_{\text{escP}} &= \sqrt{\frac{2GM_p}{R_p}} \quad \text{and} \quad v_{\text{escE}} = \sqrt{\frac{2GM_E}{R_E}} \\
 \therefore \frac{v_{\text{escP}}}{v_{\text{escE}}} &= \sqrt{\frac{M_p}{M_E} \times \frac{R_E}{R_p}} = \sqrt{8 \times \frac{1}{2}} = \sqrt{4} = 2.
 \end{aligned}$$

(3) If the height of a satellite completing one revolution around the earth in T seconds is h_1 metres, then what would be the height of a satellite taking $2\sqrt{2} T$ seconds for one revolution? (3 marks)

Ans. Data : Time : T seconds, Height : h_1 metres, h_2 (for time $2\sqrt{2} T$ seconds) = ?

Let us assume the height of the satellite completing one revolution in $2\sqrt{2} T$ seconds as h_2 .

$$T = \frac{2\pi r}{v_c}, \text{ i.e. } T = \frac{2\pi (R+h_1)}{\sqrt{\frac{GM}{(R+h_1)}}} \quad \therefore T = 2\pi \sqrt{\frac{(R+h_1)^3}{GM}} \quad \dots (1)$$

To be calculated : Height h_2 in time $2\sqrt{2} T$ seconds

$$2\sqrt{2} T = 2\pi \sqrt{\frac{(R+h_2)^3}{GM}} \quad \dots (2)$$

From Eqs. (1) and (2),

$$\therefore \frac{T}{2\sqrt{2}T} = \frac{2\pi \sqrt{\frac{(R+h_1)^3}{GM}}}{2\pi \sqrt{\frac{(R+h_2)^3}{GM}}} \quad \therefore \frac{1}{\sqrt{8}} = \frac{\sqrt{(R+h_1)^3}}{\sqrt{(R+h_2)^3}}$$

$$\therefore \frac{1}{2} = \frac{(R+h_1)}{(R+h_2)} \quad \therefore R+h_2 = 2R+2h_1 \quad \therefore h_2 = R+2h_1.$$

(4) If the mass of a planet is 8 times that of the earth and its radius is twice the radius of the earth, what will be the escape velocity for that planet? (Escape velocity for the earth = 11.2 km/s)

Solution : Data : Mass of the planet = $8M_E$, radius of the planet, $R_P = 2R_E$,

escape velocity for the earth, $v_{\text{escE}} = 11.2 \text{ km/s}$

escape velocity for the planet, $v_{\text{escP}} = ?$

$$v_{\text{escP}} = \sqrt{\frac{2GM_P}{R_P}} = \sqrt{\frac{2G(8M_E)}{2R_E}} = \sqrt{\frac{8}{2} \times \frac{2GM_E}{R_E}} = \sqrt{\frac{8}{2}} \times \sqrt{\frac{2GM_E}{R_E}}$$

$$= \sqrt{4} \times v_{\text{escE}} \quad \therefore v_{\text{escE}} = \sqrt{\frac{2GM_E}{R_E}}$$

$$= 2 v_{\text{escE}} = 2 \times 11.2$$

$$v_{\text{escP}} = \mathbf{22.4 \text{ km/s.}}$$

Question Set 9

LAWS, RULES, UNITS, PRINCIPLES, DEFINITIONS AND EXPLANATION

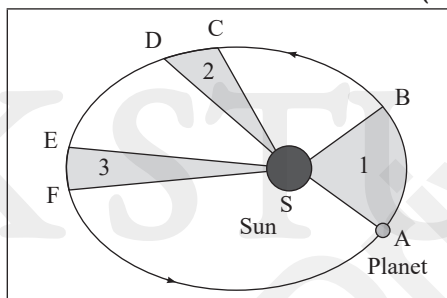
[Q. 1 (B), 2 (B) and 3]

Remember :

- ☛ This question set is based on **laws, rules, units, definitions, principles** and **explanation** with an example.
- ☛ A diagram may be expected.

Chapter 1 : Gravitation

(1) Observe the given figure showing the orbit of the planet moving around the Sun and write the three laws related to it. (March '20) (3 marks)



The orbit of a planet moving around the Sun

OR

State the laws related to the given diagram. (March '22) (3 marks) OR

State Kepler's three laws of motion.

(July '23) (3 marks)

Ans. Kepler's first law : The orbit of a planet is an ellipse with the Sun at one of the foci.

The above figure shows the elliptical orbit of a planet revolving around the Sun. S denotes the position of the Sun.

Kepler's second law : The line joining the planet and the Sun sweeps equal areas in equal intervals of time.

$$\text{Area ASB} = \text{area CSD} = \text{area ESF}.$$

Kepler's third law :

The square of the period of revolution of a planet around the Sun is directly proportional to the cube of the mean distance of the planet from the Sun.

Thus, if r is the average distance of the planet from the Sun and T is its period of revolution, then

$$T^2 \propto r^3, \text{ i.e. } \frac{T^2}{r^3} = \text{constant} = K.$$

(2) Name the laws shown in the given figure :

[See the diagram given in Q. (1).] (Sept. '21) (1 mark)

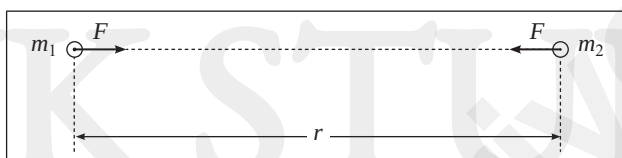
Ans. Kepler's laws.

(3) State Newton's universal law of gravitation. (March '24) (1 mark)

Express it in mathematical form. (2 marks)

Ans. Newton's universal law of gravitation : Every object in the Universe attracts every other object with a definite force. This force is directly proportional to the product of the masses of the two objects and inversely proportional to the square of the distance between them.

Mathematical form : Consider two objects of masses m_1 and m_2 . We assume that the objects are very small spheres of uniform density and the distance r between their centres is very large compared to the radii of the spheres.



Gravitational force between two objects

The magnitude (F) of the gravitational force of attraction between the objects is directly proportional to $m_1 m_2$ and inversely proportional to r^2 .

$$\therefore F \propto \frac{m_1 m_2}{r^2} \quad \therefore F = G \frac{m_1 m_2}{r^2},$$

where G is the constant of proportionality, called the universal gravitational constant.

(4) (i) Determine the SI unit of the universal constant of gravitation from the formula for the gravitational force between two particles. (ii) Define G (universal gravitational constant).

Ans. (i) According to Newton's law of gravitation, the gravitational force between two particles is

$$F = G \frac{m_1 m_2}{r^2},$$

where m_1 and m_2 are the masses of the two particles, r is the distance between them and G is the universal constant of gravitation.

$$\therefore G = \frac{Fr^2}{m_1m_2}$$

The SI unit of force is the newton (N), that of distance is the metre (m) and that of mass is the kilogram (kg).

$$\therefore \text{The SI unit of } G \text{ is } \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}.$$

$$(ii) F = G \frac{m_1m_2}{r^2} \therefore G = \frac{Fr^2}{m_1m_2}. \text{ If we take } m_1 = m_2 = \text{unit mass and}$$

$r = \text{unit distance}$, numerically, $G = F$, i.e. G (universal gravitational constant) represents the magnitude of the gravitational force of attraction between two unit masses, separated by a unit distance.

(5) What is the acceleration due to gravity?

OR

Define acceleration due to gravity.

(1 mark)

Ans. The acceleration produced in a body due to the gravitational force of the earth is called the acceleration due to gravity.

(6) What is centripetal force?

OR

Define centripetal force.

(1 mark)

Ans. In uniform circular motion of a body, the force acting on the body is directed towards the centre of the circle. This force is called centripetal force.

(7) What is gravitational potential energy?

OR

Define gravitational potential energy.

Write the formula for it.

(2 marks)

Ans. The energy stored in a body due to the gravitational force between the body and the earth is called the gravitational potential energy.

Gravitational potential energy of a body of mass $m = -\frac{GMm}{R+h}$, where

$G = \text{gravitational constant}$, $M = \text{mass of the earth}$, $R = \text{radius of the earth}$, $h = \text{height of the body from the surface of the earth}$.

(8) Will the value of g be the same everywhere on the surface of the earth? Justify your answer.

(July '22) (2 marks)

Ans. (1) No. (2) As the earth is not exactly spherical in shape, the distance of a point on the earth's surface from the earth's centre is different from place to place. (3) Due to the earth's rotation about its axis the earth bulges at the equator and flattens at the poles. Thus, the equatorial radius of the earth is greater than the polar radius. (4) The value of g is, therefore, highest at the poles and lowest at the equator.

Chapter 2 : Periodic Classification of Elements

(1) In Dobereiner's triad containing Li, Na, K, if the atomic masses of lithium and potassium are 6.9 and 39.1, then what is the atomic mass of sodium? (March '19) (2 marks)

Ans. The atomic mass of sodium is equal to the average of the atomic masses of Li and K, i.e. $\frac{6.9 + 39.1}{2} = \frac{46}{2} = 23$.

The atomic mass of sodium is 23.

(2) Illustrate Newlands' law of octaves with a suitable example. (2 marks)

Ans. (1) Newlands' law of octaves states that when the elements are arranged in the order of their increasing atomic masses, every eighth element has properties similar to those of the first.

(2) Illustration : If the first 21 elements, except inert gases, are arranged in the order of their increasing atomic masses we have octaves as given below :

H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe

It is found that Na is the eighth element from Li and both of them have similar properties. Similarly, the elements, in the following pairs show similar properties : C and Si, Na and K, Mg and Ca, F and Cl.

(3) State Mendeleev's periodic law. (1 mark)

Ans. When the elements are arranged in the order of their increasing atomic masses, Mendeleev found that the elements with similar physical and chemical properties repeat after a definite interval. On the basis of these findings Mendeleev stated the periodic law as follows :

The physical and chemical properties of elements are a periodic function of their atomic masses.

(4) State the modern periodic law. (Nov. '20) (1 mark)

Ans. The chemical and physical properties of elements are a periodic function of their atomic numbers.

(5) What is meant by the modern periodic table? (1 mark)

Ans. The classification of elements resulting from an arrangement of the elements in an increasing order of their atomic numbers is the modern periodic table.

(6) What is meant by periodic trends in the modern periodic table?

(1 mark)

Ans. When the properties of elements in a period or a group of the modern periodic table are compared, certain regularity is observed in their variations. It is called the periodic trends in the modern periodic table. The periodic trends is observed in properties of elements, namely, valency, atomic size and metallic-nonmetallic character.

(7) Define period.

(March '23) (1 mark)

Ans. The horizontal row of the elements in the periodic table of the elements is called a period.

(8) Define group.

(March '23) (1 mark)

Ans. The vertical column of elements in the periodic table of the elements is called a group.

Chapter 3 : Chemical Reactions and Equations

(1) What is meant by a chemical reaction?

(1 mark)

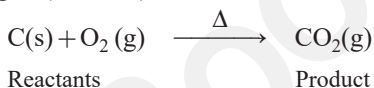
Ans. A process in which some substances undergo bond breaking and are transformed into new substances by formation of new bonds is called a chemical reaction.

(2) Explain the term reactant and product giving examples. (3 marks)

Ans. (1) The substance which undergoes bond breaking while taking part in a chemical reaction is called a reactant.

(2) The substance formed as a result of a chemical reaction by formation of new bonds is called a product.

(3) Example : In a chemical reaction, the formation of carbon dioxide gas takes place by combustion of coal in air. In this reaction, coal (carbon) and oxygen (from air) are the reactants while carbon dioxide is the product.



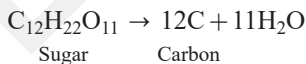
(3) What is meant by a decomposition reaction?

OR

Write the chemical reaction of decomposition of sugar on heating.

(March '25) (2 marks)

Ans. The chemical reaction in which two or more products are formed from a single reactant is called a decomposition reaction.



(4) What is meant by a thermal decomposition? (1 mark)

Ans. The reaction in which a compound is decomposed by heating it to a high temperature is called a thermal decomposition.

(5) What is meant by an electrolytic decomposition? (1 mark)

Ans. The reaction in which a compound is decomposed by passing an electric current through its solution or molten mass is called an electrolytic decomposition.

(6) What is meant by a displacement reaction? (1 mark)

Ans. The reaction in which the place of the ion of a less reactive element in a compound is taken by another more reactive element by formation of its own ions, is called a displacement reaction.

(7) What is meant by a double displacement reaction? (1 mark)

Ans. The reaction in which the ions in the reactants are exchanged to form a precipitate is called a double displacement reaction.

(8) Explain the following terms with examples :

(i) Endothermic reaction. (2 marks)

(ii) Exothermic reaction. (2 marks)

Ans. (i) Endothermic reaction : The reaction in which heat is absorbed is called an endothermic reaction.

When $\text{KNO}_3(\text{s})$ dissolves in water, there is absorption of heat during the reaction and the temperature of the solution falls.



(ii) Exothermic reaction : The reaction in which heat is given out is called an exothermic reaction.

When $\text{NaOH}(\text{s})$ dissolves in water, there is evolution of heat leading to a rise in temperature.



(9) What do you mean by a slow speed reaction? OR

Define : Slow speed reaction. (1 mark)

Ans. The reaction which requires long time for completion i.e. occurs slowly is called a slow speed reaction.

(10) Define : Fast speed reaction. (1 mark)

Ans. The reaction which is completed in very short time, i.e. it occurs rapidly is called a fast speed reaction.

(11) Define : Oxidation reaction. (1 mark)

Ans. The chemical reaction in which a reactant combines with oxygen or loses hydrogen to form the product is called an oxidation reaction.

(12) What do you mean by oxidant? Explain with a suitable example.

(2 marks)

Ans. The chemical substance which brings about an oxidation reaction by making oxygen available is called oxidant or oxidising agent.

(1) In the combustion of carbon, oxygen is an oxidant.

(2) In the oxidation of ethyl alcohol, potassium dichromate is used as an oxidant.

(13) Define : Reduction reaction.

(1 mark)

Ans. The chemical reaction in which a reactant gains hydrogen and loses oxygen to form the product is called a reduction reaction.

(14) What do you mean by reductant? Explain with a suitable example.

(2 marks)

Ans. The chemical substance which brings about reduction by making hydrogen available is called a reductant.

In the preparation of methane from carbon, hydrogen is a reductant.

(15) What is corrosion?

(1 mark)

Ans. The slow process of decay or oxidation of metals due to various components of atmosphere is known as corrosion.

Iron rusts and a reddish layer is collected on it. This is corrosion of iron.

(16) Define : Rancidity.

(1 mark)

Ans. When oil or fat or left over cooking oil for making foodstuff undergoes oxidation if stored for a long time, it is found to have foul odour. This process is called rancidity.

Chapter 4 : Effects of Electric Current

(1) Define electric power.

(1 mark)

Ans. Electric power is the electric work done per unit time.

OR

Electric power is the time rate at which electric energy is used.

(2) State the formula for electric power. Hence, obtain its SI unit.

(3 marks)

Ans. Electric power (P) = $\frac{\text{electric work (W) done or electric energy used}}{\text{time (t)}}$

The SI unit of work is the joule and that of time is the second. Hence, the SI unit of power is the joule per second. It is given the special name : the watt (W), named after James Watt. One watt equals one joule per second.

$$W = VIt = I^2Rt = \frac{V^2}{R}t \quad \therefore P = W/t = VI = I^2R = V^2/R.$$

Here, V is the potential difference applied across an electrical appliance, R is the resistance of the appliance and I is the current through the appliance.

(3) What is one kilowatt·hour?

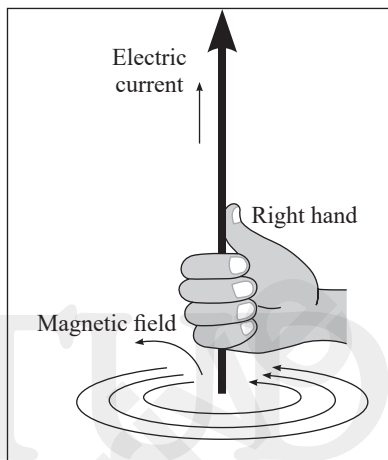
(1 mark)

Ans. One kilowatt·hour is the electric energy used in one hour by an electrical appliance of power one kilowatt. It is equal to 3.6×10^6 J.

(4) State the right hand thumb rule. Draw a neat labelled diagram to illustrate it.

(3 marks)

Ans. Imagine that you have held a current-carrying straight conductor in your right hand in such a way that your thumb points in the direction of the current. Then turn your fingers around the conductor. The direction of the fingers is the direction of the magnetic lines of force produced by the current.



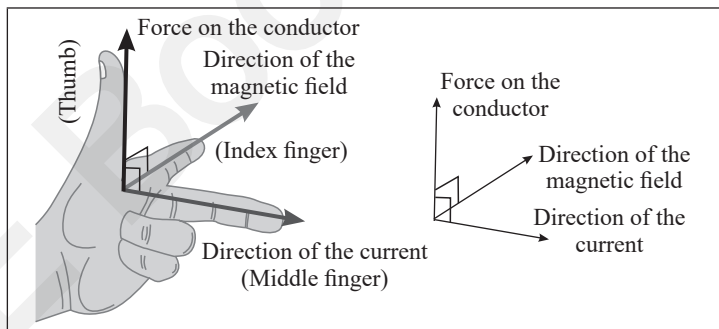
Right hand thumb rule

(5) Write Fleming's left hand rule.

Draw a neat labelled diagram to illustrate it.

(3 marks)

Ans. Fleming's left hand rule : The left hand thumb, index finger and the middle finger are stretched so as to be perpendicular to each other. If the index finger is in the direction of the magnetic field and the middle finger points in the direction of the current, then the direction of the thumb is the direction of the force on the conductor.



Fleming's left hand rule

(6) What is electric motor?

(1 mark)

Ans. A device which converts electric energy into mechanical energy is called electric motor.

(7) What is electromagnetic induction?

(1 mark)

Ans. The process by which a changing magnetic field in a conductor induces a current in another conductor is called electromagnetic induction. A current can be induced in a conductor either by moving it in a magnetic field or by changing the magnetic field around the conductor.

(8) State Faraday's law of induction.

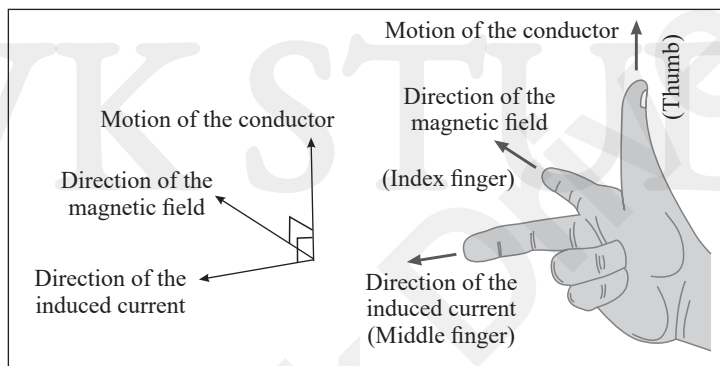
(1 mark)

Ans. Whenever the number of magnetic lines of force passing through a coil changes, a current is induced in the coil.

(9) State Fleming's right hand rule. Draw a neat labelled diagram to illustrate it.

(3 marks)

Ans. Stretch the thumb, the index finger and the middle finger of the right hand in such a way that they are perpendicular to each other.



Fleming's right hand rule

In this position, the thumb indicates the direction of the motion of the conductor, the index finger the direction of the magnetic field and the middle finger shows the direction of the induced current.

Chapter 5 : Heat

(1) Define latent heat of fusion.

(1 mark) OR

What is latent heat of fusion?

(1 mark)

Ans. When a solid is converted into liquid at constant temperature (melting point of the substance) the amount of heat absorbed by it is called the latent heat of fusion.

(2) Define specific latent heat of fusion.

(1 mark) OR

What is specific latent heat of fusion?

(1 mark)

Ans. The amount of heat energy absorbed at constant temperature by a unit mass of solid to convert into liquid phase is called the specific latent heat of fusion.

(3) Define boiling point of a liquid. (1 mark)

Ans. The constant temperature at which a liquid transforms into gaseous state is called the boiling point of the liquid.

(4) What is specific latent heat of vaporization? (1 mark)

Ans. The amount of heat energy absorbed at constant temperature by a unit mass of liquid to convert into gaseous phase is called the specific latent heat of vaporization.

(5) Define regelation. (1 mark)

Ans. The phenomenon in which the ice converts to liquid due to applied pressure and then re-converts to ice once the pressure is removed is called regelation.

(6) Define dew point temperature. (1 mark)

Ans. If the temperature of unsaturated air is decreased, a temperature is reached at which the air becomes saturated with water vapour. This temperature is called the dew point temperature.

(7) Define absolute humidity. (1 mark)

State its unit. (½ mark)

Ans. The mass of water vapour present in a unit volume of air is called absolute humidity. Generally it is expressed in kg/m^3 .

(8) Define relative humidity. (1 mark) OR

What is relative humidity? (1 mark)

Write the formula for percentage relative humidity. (½ mark)

Ans. The ratio of the actual mass of water vapour content in the air for a given volume and temperature to that required to make the same volume of air saturated with water vapour at the same temperature is called the relative humidity.

Percentage relative humidity = $\left[\frac{\text{the actual mass of water vapour content in the air for a given volume and temperature}}{\text{the mass of water vapour required to make the same volume of air saturated with water vapour at the same temperature}} \right] \times 100$.

(9) Define the kilocalorie. (1 mark)

Ans. The amount of heat necessary to raise the temperature of 1 kg of water by 1°C from 14.5°C to 15.5°C is called one kilocalorie.

(10) Define the calorie.

(1 mark)

Ans. The amount of heat necessary to raise the temperature of 1 g of water by 1°C from 14.5°C to 15.5°C is called one calorie.

(11) Define specific heat capacity.

(1 mark)

Ans. The amount of heat energy required to raise the temperature of a unit mass of an object by 1°C is called the specific heat capacity of that object.

(12) Determine the unit of specific heat capacity.

(2 marks)

Ans. Specific heat capacity

$$= \frac{\text{heat absorbed by an object}}{\text{mass of the object} \times \text{rise in temperature of the object}}$$

$$\therefore \text{Unit of specific heat capacity} = \frac{\text{unit of heat}}{\text{unit of mass} \times \text{unit of temperature}}$$

In SI, heat is expressed in joule (J), mass in kg and temperature in kelvin (K).

$\therefore$ SI unit of specific heat capacity = $\frac{\text{J}}{\text{kg} \cdot \text{K}}$. Other units of specific heat capacity : J/kg.°C, erg/g.°C, cal/g.°C, kcal/kg.°C.

(13) Name and state the principle used to measure the specific heat capacity of a substance.

(March '25) (2 marks)

Ans. The principle of heat exchange is used to measure the specific heat capacity of a substance. This principle is as follows : If a system of two objects is isolated from the environment by keeping it inside a heat resistant box, then no energy can leave the box or enter the box. In this situation, heat energy lost by the hot object = heat energy gained by the cold object.

Chapter 6 : Refraction of Light

(1) What is meant by refraction of light?

(1 mark)

Ans. The change in the direction of propagation of light when it passes obliquely from one transparent medium to another is called refraction of light.

(2) Define angle of incidence and angle of refraction. (1 mark each)

Ans. (1) The angle made by the incident ray of light with the normal to the surface at the point of incidence is called the angle of incidence.

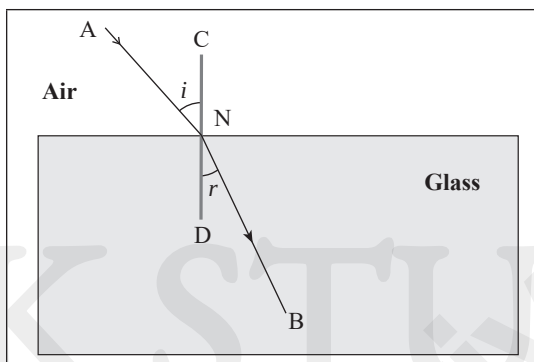
(2) The angle made by the refracted ray of light with the normal to the surface at the point of incidence is called the angle of refraction.

(3) State the laws of refraction of light. (Sept.'21; March '22) (2 marks)

Ans. Laws of refraction of light : (1) The incident ray and the refracted ray are on the opposite sides of the normal to the surface at the point of incidence and all the three i.e. the incident ray, the refracted ray and the normal are in the same plane.

(2) For a given pair of media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant (Snell's law). This constant is called the refractive index of the second medium with respect to the first medium.

(4) Observe the given figure and answer the following questions :



(a) Name the process represented by the figure.

(b) State the two laws related to the process. (March '20) (3 marks)

Ans. (a) Refraction of light.

(b) See the answer to Q. (3).

(5) Define the refractive index of the second medium with respect to the first medium. (1 mark)

Ans. The refractive index of the second medium with respect to the first medium is defined as the ratio of the sine of the angle of incidence to the sine of the angle of refraction when the ray of light is obliquely incident at the boundary separating the two media and travels from the first medium to the second medium.

OR

The refractive index of the second medium with respect to the first medium is defined as the ratio of the magnitude of the velocity of light in the first medium to the magnitude of the velocity of light in the second medium.

(6) Define absolute refractive index. (1 mark)

Ans. The absolute refractive index of a medium is defined as the ratio of the magnitude of the velocity of light in vacuum to the magnitude of the velocity of light in the medium.

(7) What do you mean by dispersion of light? (1 mark)

What is a spectrum of light? (1 mark)

Ans. The process of separation of light into its component colours while passing through a medium is called dispersion of light.

The band of coloured components of a light beam is called spectrum.

(8) Define total internal reflection of light. (1 mark)

Ans. When light travels from a denser medium to a rarer medium, if the angle of incidence is greater than the critical angle, there is no refraction of light and all the light is reflected in the denser medium. This is called total internal reflection of light.

(9) Define critical angle. (1 mark)

Ans. When light travels from a denser medium to a rarer medium, the angle of incidence for which the angle of refraction becomes 90° , is called the critical angle.

Chapter 7 : Lenses

(1) With reference to spherical lenses, state the meaning of the following terms or define the following terms : (i) centre of curvature (ii) radii of curvature (iii) principal axis (iv) optical centre (O) (v) principal focus (vi) focal length. (1 mark each)

Ans.

(i) Centre of curvature (C) : The centres of the spheres whose parts form the surfaces of a lens are called the centres of curvature of the lens. A lens has two centres of curvature C_1 and C_2 for its two spherical surfaces.

(ii) Radii of curvature (R_1 , R_2) : The radii of the spheres whose parts form surfaces of a lens are called the radii of curvature of the lens.

(iii) Principal axis : The imaginary straight line passing through the two centres of curvature of a lens is called the principal axis of the lens.

(iv) Optical centre : The point inside a lens on the principal axis, through which light rays pass without changing their path is called the optical centre of the lens.

(v) Principal focus (F) : When light rays parallel to the principal axis are incident on a convex lens, they converge at a point on the principal axis. This point is called the principal focus (F) of the convex lens. Light rays travelling parallel to the principal axis of a concave lens diverge after refraction in such a way that they appear to be coming out of a point on the principal axis. This point is called the principal focus of the concave lens. A lens has two principal foci F_1 and F_2 .

(vi) **Focal length (f)** : The distance between the optical centre and the principal focus of a lens is called the focal length (f) of the lens.

(2) State the Cartesian sign convention for refraction of light (image formation) by a lens. (3 marks)

Ans. Cartesian sign convention for refraction of light (image formation) by a lens :

In this case, the optical centre (O) of the lens is taken as the origin and the principal axis of the lens is taken as X-axis of the coordinate system.

(1) The object is always placed at the left of the lens.

All distances parallel to the principal axis are measured from the optical centre of the lens.

(2) All distances measured to the right of the origin are taken as positive while distances measured to the left of the origin are taken as negative.

(3) Distances measured perpendicular to and above the principal axis are taken as positive.

(4) Distances measured perpendicular to and below the principal axis are taken as negative.

(5) The focal length of a convex lens is positive and that of a concave lens is negative.

(3) What is a lens formula? State it. (1 mark)

Ans. The relationship between the object distance (u), image distance (v) and focal length (f) of a lens is called the lens formula and is given as

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}.$$

(4) What is meant by the magnification produced by a lens? State the formulae for it. (2 marks)

Ans. The magnification (M) produced by a lens is the ratio of the height of the image (h_2) to the height of the object (h_1).

$$M = \frac{h_2}{h_1} \quad \dots (1)$$

$$\text{Also, } M = \frac{v \text{ (image distance)}}{u \text{ (object distance)}} \quad \dots (2)$$

(5) Define the SI unit of power of a lens. (1 mark)

Ans. The SI unit of power of a lens is the dioptre.

One dioptre is the power of a lens whose focal length is one metre.

$$1 \text{ dioptre (D)} = \frac{1}{1 \text{ metre (m)}}$$

(6) What is the power of a lens?

OR

Define power of a lens.

(July '22) (1 mark)

Ans. The capacity of a lens to converge or diverge incident rays is called its power. The power (P) of a lens is the inverse of the focal length (f) of the lens,

$$P = \frac{1}{f}.$$

(7) What is meant by power of accommodation of the eye? (1 mark)

Ans. The ability of the eye lens to adjust its focal length is called the power of accommodation of the eye.

(8) What is meant by the minimum distance of distinct vision?

(1 mark)

Ans. The minimum distance from the normal eye, at which an object is clearly visible without stress on the eye is called the minimum distance of distinct vision.

(9) What is the minimum distance of distinct vision for normal human eye? (July '22) (1 mark)

Ans. The minimum distance of distinct vision for normal human eye is 25 cm.

(10) State any two rules used for drawing ray diagrams for the formation of an image by a convex lens. (July '19) (2 marks)

Ans. Rules used for drawing ray diagrams for the formation of an image by a convex lens :

(1) When the incident ray is parallel to the principal axis, the refracted ray passes through the principal focus.

(2) When the incident ray passes through the principal focus, the refracted ray is parallel to the principal axis.

(3) When the incident ray passes through the optical centre of the lens, it passes without changing its direction.

(Any two rules)

Chapter 8 : Metallurgy

(1) What is meant by an ionic bond?

(1 mark)

Ans. The cation and anion being oppositely charged, there is an electrostatic force of attraction between them, this force of attraction between cation and anion is called the ionic bond.

(2) Explain the following terms :

(1 mark each)

Ans. (i) Minerals : The naturally occurring compounds of metals along with other impurities are known as minerals.

Examples : Rocks are composed of mixtures of minerals. Talc and granite are minerals.

(ii) Gangue (March '22) : Ores contain metal compounds with some of the impurities like soil, sand, rocky material, etc. These impurities are called gangue.

(iii) Ores (March '22) : The minerals from which metals are extracted profitably and conveniently are called ores.

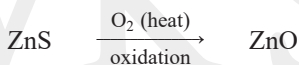
Examples : Bauxite ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$), Cinnabar (HgS).

(iv) Metallurgy (March '22) : The process of extraction of metals in their pure form from their ores and then their further purification by different methods of purification is called metallurgy.

(v) Concentration of ores : The process of separating gangue from the other ores is called concentration of ores.

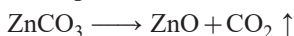
(vi) Roasting : The process of heating an ore to a high temperature in excess of air and converting it into its oxide is called roasting.

Examples : ZnS (zinc blend), PbS (Galena)



(vii) Calcination : The process of heating an ore in a limited supply of air and converting it into its oxide is called calcination.

Example : Zinc carbonate (ZnCO_3)



(viii) Refining : The metal obtained by chemical reduction contains impurities. The process of electrolysis used to obtain pure metals from impure metals is known as refining.

(3) Explain the term corrosion with a suitable example. (2 marks) OR

What is corrosion?

(1 mark)

Ans. The process in which a metal is destroyed gradually by the action of air, moisture or a chemical (like an acid) on its surface is called corrosion. OR

Corrosion is degradation of a material due to reaction with its environment.

Chapter 9 : Carbon Compounds

(1) What is meant by catenation power?

(1 mark)

Ans. Carbon has a unique ability to form strong covalent bonds with other carbon atoms; this results in formation of big molecules. This property of carbon is called catenation power.

(2) What is meant by a covalent bond?

(1 mark) OR

Explain the term covalent bond with an example. (2 marks)

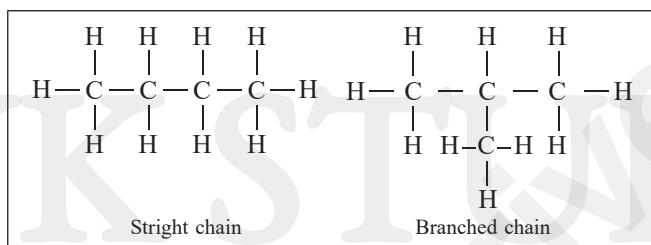
Ans. The chemical bond formed by sharing of two valence electrons between the two atoms is called covalent bond.

Example : Two hydrogen atoms share their electrons with each other to form H_2 molecule. One covalent bond is formed between two hydrogen atoms by sharing of two electrons.

(3) Explain the term structural isomerism with an example. (2 marks)

Ans. The phenomenon in which compounds having different structural formulae have the same molecular formula is called structural isomerism.

Butane is represented by two different compounds as their structural formulae are different. These two different structural formulae have the same molecular formula i.e. C_4H_{10} .



(4) Explain the term hetero atom in a carbon compound with an example. (2 marks)

Ans. Carbon compounds are formed by formation of bonds of carbon with other elements such as halogens, oxygen, nitrogen, sulphur. The atoms of these elements substitute one or more hydrogen atoms in the hydrocarbon chain and thereby the tetravalency of carbon is satisfied. The atom of the element which is substituted for hydrogen is referred to as a hetero atom. Sometimes hetero atoms are not alone but exist in the form of certain groups of atoms.

Examples :

Structural formula	Hetero atom
$-X$ ($-Cl$, Br , $-I$)	Halogen
$-O-H$	Oxygen
$-O-$	Oxygen
$-NH_2$	Nitrogen
$\begin{array}{c} O \\ \\ -C- \end{array}$	Oxygen

(5) Explain the following terms :

(2 marks each)

(i) Oxidants : Substances that can give oxygen to other substances are called oxidants.

Example : Potassium permanganate, potassium dichromate are oxidants.

(ii) Reduction : The chemical reaction in which a reactant gains hydrogen and loses oxygen to form the product is called the reduction reaction.

Example : Hydrogenation transforms vegetable oil into vanaspati ghee.

(6) Explain the following terms :

(2 marks each)

(i) Alkane : In hydrocarbon, the four valencies of carbon atom are satisfied only by the single bonds. Such a compound is called alkane.

Example : Methane (CH_4), Ethane (C_2H_6).

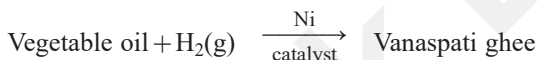
(ii) Alkene : The unsaturated hydrocarbon containing a carbon-carbon double bond is called alkene.

Example : Ethene ($\text{CH}_2 = \text{CH}_2$)

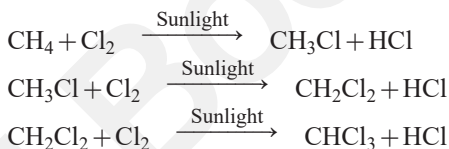
(iii) Alkyne : The unsaturated hydrocarbon containing a carbon-carbon triple bond is called alkyne.

Example : Ethyne C_2H_2 ($\text{CH} \equiv \text{CH}$).

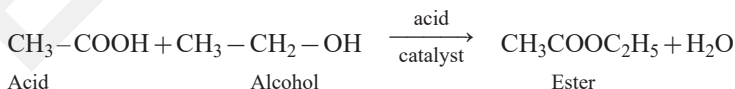
(iv) Addition reaction : When a carbon compound combines with another compound to form a product that contains all the atoms in both the reactants; it is called an addition reaction.



(v) Substitution reaction : The reaction in which the place of one type of atom/group in a reactant is taken by another atom/group of atoms, is called a substitution reaction.



(vi) Esterification : A carboxylic acid reacts with an alcohol in presence of an acid catalyst, an ester is formed. The reaction is known as esterification.



(vii) **Saponification** : When an ester reacts with the alkali, i.e. sodium hydroxide, the corresponding alcohol and sodium salt of carboxylic acid are obtained. This reaction is called saponification. It is used in the preparation of soap.

Ester + Sodium hydroxide $\rightarrow$ Sodium salt of carboxylic acid + Alcohol.

Chapter 10 : Space Missions

What is meant by space debris?

(Sept. '21) (1 mark)

Ans. In space, nonessential objects such as the parts of launchers and satellites, revolving around the earth are called the space debris.

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**Question
Set
10**

**SHORT NOTES OR
EXPLAIN THE TERM**

[Q. 2 (B) (2 marks) OR Q. 3 (3 marks)]

Remember :

- ☛ In this question set, **short notes** are included.
- ☛ The short note should be brief and point-wise.
- ☛ A diagram may be expected.

Chapter 1 : Gravitation

Q. Explain the term the earth's gravitational force.

OR

Write a short note on the earth's gravitational force. (2 marks)

Ans. The earth attracts every object towards it because of the gravitational force. As the earth's centre of mass is at its centre, the gravitational force exerted by the earth on any object is always directed towards the earth's centre. Hence, an object released from a point above the earth's surface falls vertically downward towards the earth. If an object is thrown vertically upward, its velocity goes on decreasing due to the earth's gravitational force on the object. At one stage, the velocity of the body becomes zero and later the body falls back to the earth.

Chapter 2 : Periodic Classification of Elements

Q. Write a short note on the position of isotopes in the Mendeleev's periodic table and the modern periodic table. (3 marks)

Ans. Isotopes were discovered long time after Mendeleev put forth the periodic table. A challenge was posed in placing isotopes in Mendeleev's periodic table, as isotopes have the same chemical properties but different atomic masses. Isotopes do not find separate places in this table.

Moseley found out that atomic number is a fundamental property of an element rather than its atomic mass. The atomic number of any element is increased by one unit (number) from the atomic number of the preceding element. In the modern periodic table, the elements are arranged in the order of their increasing atomic numbers. This solved the problem of discrepancy in the pairs of isotopes of elements observed in Mendeleev's periodic table. The isotopes of $_{17}\text{Cl}^{35}$ and $_{17}\text{Cl}^{37}$ were placed in the same group as both have the same atomic number.

Chapter 3 : Chemical Reactions and Equations

Q. What is the reaction called when oxidation and reduction takes place simultaneously? Explain with one example. OR

Explain redox reaction with a suitable example. OR

Write a note on redox reaction. (2 marks) OR

What is redox reaction? Explain with the help of a balanced chemical equation. (July '22) (3 marks)

Ans. The reaction which involves simultaneous oxidation and reduction is called an oxidation-reduction or redox reaction.

In a redox reaction, one reactant gets oxidized while the other gets reduced during the reaction.

Redox reaction = Reduction + Oxidation

In redox reaction, the reductant is oxidized by the oxidant and the oxidant is reduced by the reductant.

Example : $\text{CuO(s)} + \text{H}_2\text{(g)} \longrightarrow \text{Cu(s)} + \text{H}_2\text{O}$

In this reaction, oxygen is removed from copper oxide; therefore it is a reduction of CuO, while hydrogen accepts oxygen to form water that means oxidation of hydrogen takes place. Thus, oxidation and reduction occur simultaneously.

Other examples of redox reactions :



Hydrogen
Sulphide

Sulphur



Manganese
dioxide

Manganese
chloride

Chapter 7 : Lenses

Q. Write a note on persistence of vision. (Sept. '21) (2 marks)

Ans. Persistence of vision : We see an object when its image is formed on the retina. The image disappears when the object is removed from our sight. But this is not instantaneous and the image remains imprinted on the retina for about $\frac{1}{16}$ th of a second after the removal of the object. The sensation on the retina

persists for a while. This effect is known as the persistence of vision. It is due to persistence of vision that we continue to see the object in its position for about $\frac{1}{16}$ th of a second after it is removed.

Example : When a burning stick of incense is moved fast in a circle, a circle of red light is seen.

Chapter 8 : Metallurgy

(1) Write short notes on the following :

(2 marks each)

(i) Galvanizing.

Ans. The process of coating a thin layer of zinc on iron or steel is called galvanization.

In galvanization, an iron object is dipped into molten zinc. A thin layer of zinc is formed all over the iron object.

Examples : Shiny iron nails, pin, iron pipes.

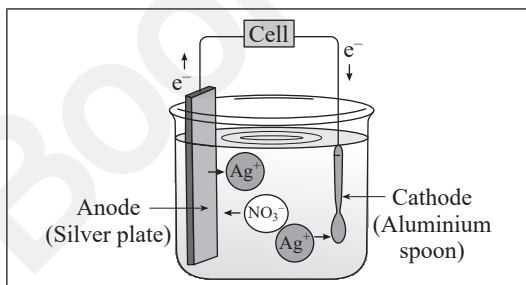
In this method corrosion of zinc occurs first because zinc is more electropositive than iron. After a few years zinc layer goes away and the iron layer gets exposed and starts rusting.

(ii) Tinning.

Ans. The process of coating a thin layer of tin (molten tin) on copper or brass is called tinning. Cooking vessels made of copper and brass get a greenish coating due to corrosion. The greenish substance is copper carbonate and it is poisonous. If butter milk or curry is placed in such a vessel it gets spoiled. Therefore, these vessels are coated with tin to prevent corrosion.

(iii) Electroplating.

Ans.

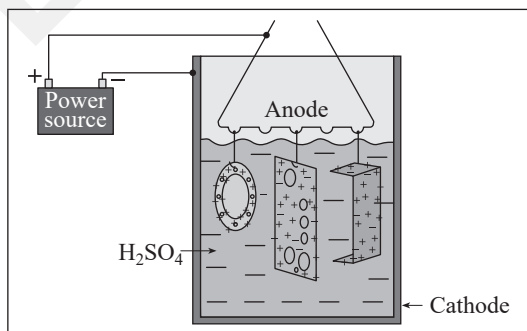


The process in which a less reactive metal is coated on a more reactive metal by electrolysis is called electroplating.

Examples : Silver-plated spoon, gold-plated jewellery.

(iv) Anodizing.

Ans.



The anodizing technique is an application of electrolysis. In this method, metals like copper and aluminium are coated with a strong film of their oxides by means of electrolysis. This thin film protects the metals from corrosion. The protection can be further increased by making the oxide layer thicker during the anodization.

Examples : Kitchen articles such as anodized pressure cooker and anodized pan.

(v) Alloying.

(March '20)

Ans. A homogeneous mixture of two or more metals or a metal and a nonmetal in a definite proportion is called an alloy. The physical properties of an alloy are different from those of its constituents. Alloys are corrosion-resistant. Alloy decreases the intensity of corrosion of metals.

Examples : Brass is made from copper and zinc. 90% copper and 10% tin are used to make an alloy called bronze. Stainless steel is made from 74% iron, 8% carbon and 18% chromium.

Chapter 9 : Carbon Compounds

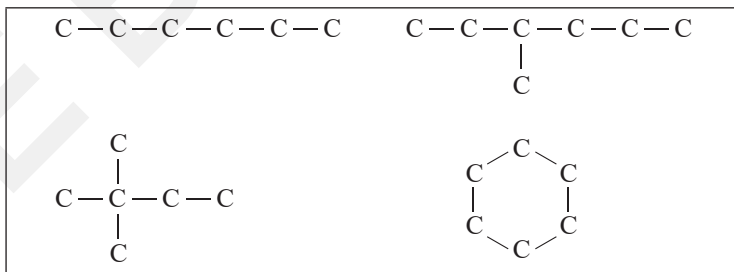
(1) Write a note on catenation power.

(3 marks)

Ans. (1) Carbon has a unique ability to form strong covalent bonds with other carbon atoms; this results in formation of big molecules. This property of carbon is called its catenation power.

(2) Carbon shows catenation. Two or more carbon atoms can share their valence electrons and bond with each other. Thus, carbon chains can be straight or branched or closed chain ring structure forming large molecules. The covalent bond between two carbon atoms is strong and therefore stable. Carbon is bestowed with catenation power due to the strong and stable covalent bonds.

(3) Hence, carbon atoms can form an unlimited number of compounds.



Chain structures of organic compounds

(2) Explain the term functional group with an example. (3 marks)

Ans. (1) A compound acquires specific chemical properties due to the hetero atoms or the groups of atoms that contain hetero atoms, irrespective of the length and nature of the carbon chain in that compound. Therefore these hetero atoms or the groups of atoms containing hetero atoms are called functional groups.

All organic compounds are derivatives of hydrocarbons. The derivatives are formed by replacing one or more H-atom/atoms of hydrocarbon by some other hetero atom or groups of atoms containing hetero atoms. After replacement, a new compound is formed which has properties different from the parent hydrocarbon.

Examples : For methane, if one hydrogen atom is replaced by an —OH group, methyl alcohol (CH_3OH) is formed. The —OH group is known as the alcoholic functional group.

Functional groups in some organic compounds :

1. Alcohol : —OH (hydroxy group)

2. Aldehyde : $\begin{array}{c} \text{H} \\ | \\ \text{—C=O} \end{array}$

3. Ketone : $\begin{array}{c} \text{O} \\ || \\ \text{—C—} \end{array}$

4. Carboxylic acid : —COOH

(3) What is meant by a homologous series?

(Nov. '20; March '22) (1 mark)

State the two characteristics of a homologous series. (2 marks)

Ans. Though the length of the carbon chains in some carbon compounds is different, their chemical properties are very much similar due to the presence of the same functional group in them. The series of compounds formed by joining the same functional group in place of a particular hydrogen atom on the chains

having sequentially increasing length is called a homologous series. Two adjacent members of the series differ by one $-\text{CH}_2-$ (methylene) unit and their masses differ by 14 units.

The homologous series of straight chain alkanes can be represented by the general formula $\text{C}_n\text{H}_{2n+2}$. The members of this series are as follows :

Methane – CH_4 Ethane – C_2H_6	These differ by one – CH_2 unit
Ethane – C_2H_6 Propane – C_3H_8	These differ by one – CH_2 unit
Butane – C_4H_{10} Pentane – C_5H_{12}	These differ by one – CH_2 unit

Characteristics of homologous series :

(1) In homologous series while going in an increasing order of the length of carbon chain (a) one methylene unit ($-\text{CH}_2-$) is added (b) the molecular mass increases by 14 u (c) number of carbon atoms increases by one.

(2) Chemical properties of members of a homologous series show similarity due to the presence of the same functional group in them.

(3) Each member of the homologous series can be represented by the same general molecular formula.

(4) While going in an increasing order of the length there is gradation in the physical properties such as boiling and melting points.

Chapter 10 : Space Missions

What is the contribution of India in space technology?

(July '23) (3 marks)

Ans. (1) Indian Space Research Organisation (ISRO) has designed and built two important launchers : Polar Satellite Launch Vehicle (PSLV) and Geosynchronous Satellite Launch Vehicle (GSLV).

(2) Many satellites in INSAT and GSAT series are active in telecommunication, television broadcasting, meteorological services, disaster management and in monitoring and management of natural resources. EDUSAT is used specifically for education.

(3) Satellites in IRNSS series are used for navigation. IRS series is used for disaster management.

**Question
Set
11**

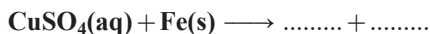
**BALANCED CHEMICAL
EQUATIONS**
[Q. 2 (B)]

Remember :

- ☛ Write the balanced chemical reaction with appropriate names of the reactants and products. Mention the reaction conditions above the arrow ($\rightarrow$).

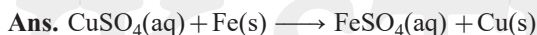
Q. Explain the following reactions giving their balanced chemical equations : (2 marks each)

(1) Complete the given chemical reaction.



Name the type of the reaction.

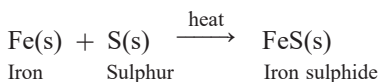
(March '20)



Type of the reaction : Displacement reaction.

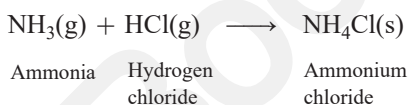
(2) Iron reacts with sulphur.

Ans. Iron reacts with sulphur to form iron sulphide.



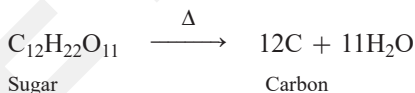
(3) Ammonia gas reacts with hydrogen chloride gas.

Ans. When ammonia gas reacts with hydrogen chloride gas, it forms the salt ammonium chloride in gaseous state, but immediately it gets transformed into the solid state.



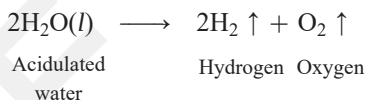
(4) Sugar is heated.

Ans. When sugar is heated, it decomposes to form carbon (black substance).



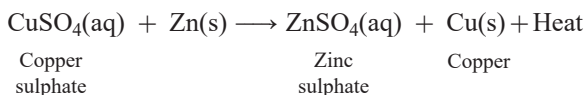
(5) Electric current is passed through acidulated water.

Ans. When an electric current is passed through acidulated water, it decomposes into hydrogen and oxygen gas.



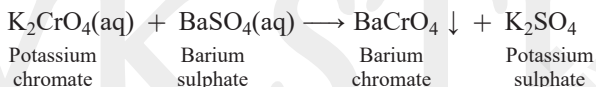
(6) Zinc powder is added to copper sulphate solution.

Ans. When zinc powder is added to copper sulphate solution, more reactive zinc displaces less reactive copper from copper sulphate solution. Colourless zinc sulphate is formed with evolution of heat.



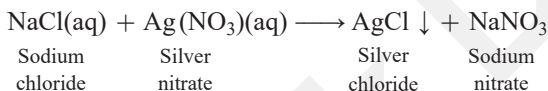
(7) Potassium chromate solution is added to barium sulphate solution.

Ans. When potassium chromate solution is added to barium sulphate solution, yellow precipitate of barium chromate is formed.



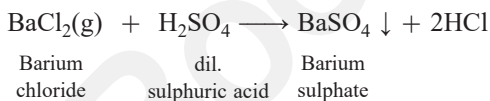
(8) Sodium chloride solution is mixed with silver nitrate solution.

Ans. When sodium chloride solution is mixed with silver nitrate solution, white precipitate of silver chloride is formed.



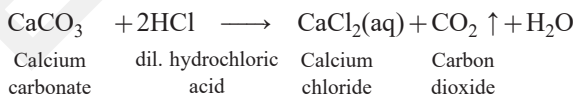
(9) Dilute sulphuric acid is added to barium chloride solution.

Ans. When dilute sulphuric acid is added to barium chloride solution, white precipitate of barium sulphate is formed.



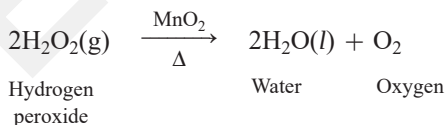
(10) Calcium carbonate (lime stone) is treated with dil. hydrochloric acid.

Ans. When calcium carbonate (lime stone) is treated with dil. hydrochloric acid, carbon dioxide gas is formed.



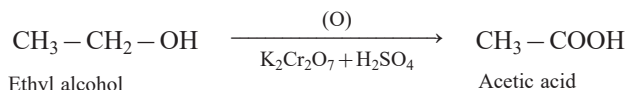
(11) Hydrogen peroxide is decomposed in the presence of manganese dioxide (MnO_2).

Ans. When hydrogen peroxide is decomposed in the presence of manganese dioxide (MnO_2), water and oxygen are formed.



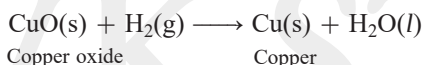
(12) Ethyl alcohol is treated with acidified potassium dichromate.

Ans. When ethyl alcohol is treated with acidified potassium dichromate, acetic acid is formed. This is oxidation reaction.



(13) Hydrogen gas is passed over black copper oxide.

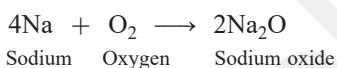
Ans. When hydrogen gas is passed over black copper oxide, a reddish coloured layer of copper is formed.



Chapter 8 : Metallurgy

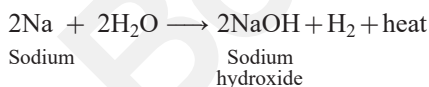
(1) Out of sodium and sulphur, which is a metal? Explain its reaction with oxygen. (March '19)

Ans. Sodium is a metal and sulphur is a nonmetal. Sodium reacts with oxygen in air at room temperature to form sodium oxide.



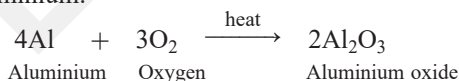
(2) Sodium reacts with water.

Ans. When sodium reacts with water, it evolves hydrogen which immediately catches fire producing a lot of heat.



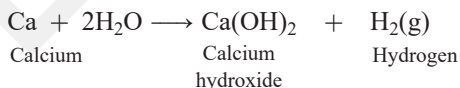
(3) Aluminium is exposed to air.

Ans. When aluminium is exposed to air, it develops a thin oxide layer of aluminium.



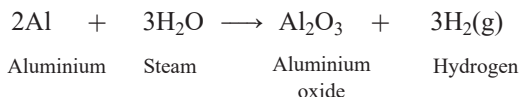
(4) Calcium reacts with water.

Ans. Calcium reacts with water less vigorously to form hydrogen gas and calcium hydroxide. In this reaction, the heat evolved is not sufficient for hydrogen to catch fire.



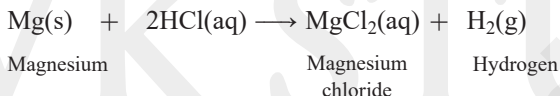
(5) Steam is passed over aluminium.

Ans. When steam is passed over aluminium, hydrogen gas and aluminium oxide are formed.



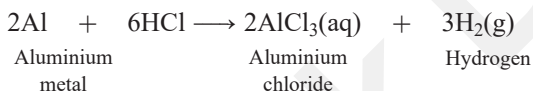
(6) Magnesium reacts with dil. hydrochloric acid.

Ans. When magnesium reacts with dil. hydrochloric acid, magnesium chloride is formed and hydrogen gas is evolved.



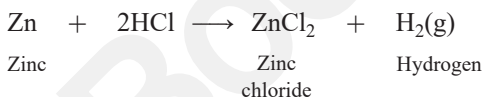
(7) Aluminium is treated with dil. hydrochloric acid.

Ans. When aluminium is treated with dil. hydrochloric acid, aluminium chloride and hydrogen gas are formed.



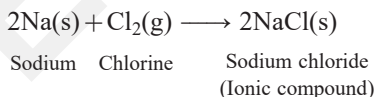
(8) Zinc reacts with dil. hydrochloric acid.

Ans. When zinc reacts with dil. hydrochloric acid, zinc chloride and hydrogen gas are formed.



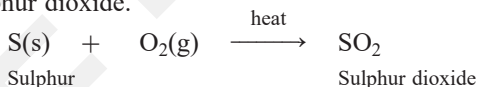
(9) Sodium metal is reacted with chlorine gas.

Ans. When sodium metal is reacted with chlorine, sodium chloride, an ionic compound, is formed.



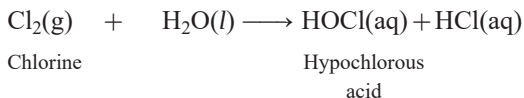
(10) Sulphur burns in air.

Ans. When sulphur burns in air, it combines with oxygen to form acidic sulphur dioxide.



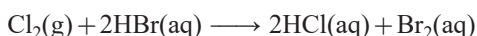
(11) Chlorine dissolves in water.

Ans. When chlorine dissolves in water, hypochlorous acid is formed.



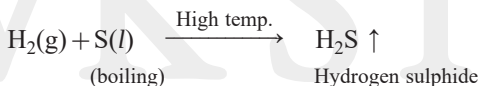
(12) Chlorine is treated with hydrobromic acid.

Ans. When chlorine is treated with hydrobromic acid, chlorine displaces bromine from hydrobromic acid.



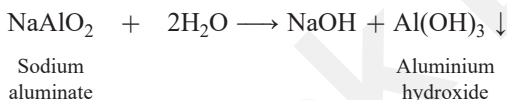
(13) Hydrogen gas is passed over boiling sulphur.

Ans. When hydrogen gas is passed over boiling sulphur, sulphur combines with hydrogen to form hydrogen sulphide which has rotten egg smell.



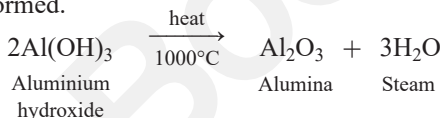
(14) Sodium aluminate is treated with water.

Ans. When sodium aluminate is treated with water, it is hydrolysed to give a precipitate of aluminium hydroxide.



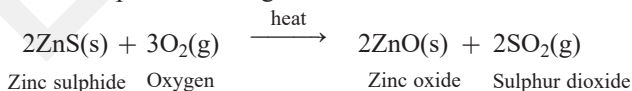
(15) Dry aluminium hydroxide is ignited at 1000°C.

Ans. When dry aluminium hydroxide is ignited at 1000°C, alumina (Al_2O_3) is formed.



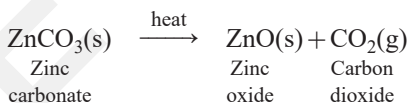
(16) Zinc sulphide is heated strongly in excess of air.

Ans. When zinc sulphide is heated strongly in excess of air, it forms zinc oxide and sulphur dioxide gas.



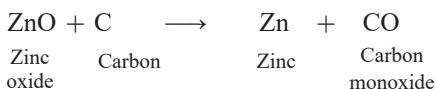
(17) Zinc carbonate is heated strongly in a limited supply of air.

Ans. When zinc carbonate is heated strongly in a limited supply of air, it gives zinc oxide and carbon dioxide.



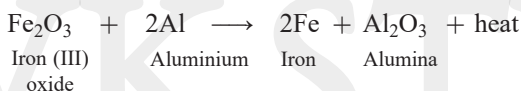
(18) Zinc oxide is treated with carbon.

Ans. When zinc oxide is treated with carbon, it is reduced to zinc. In this reaction, carbon acts as a reducing agent.



(19) Iron (III) oxide (ferric oxide) is heated with aluminium.

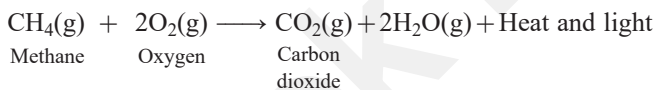
Ans. When iron (III) oxide (ferric oxide) is heated with aluminium, it gives aluminium oxide and iron. This reaction is highly exothermic and it is known as thermit reaction.



Chapter 9 : Carbon Compounds

(1) Methane is burnt in air.

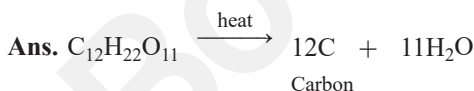
Ans. When methane is burnt in air, carbon dioxide and water are formed. The reaction is exothermic with release of a large amount of heat and light.



(2) Complete the following reaction :

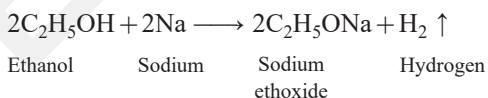


(March '23)



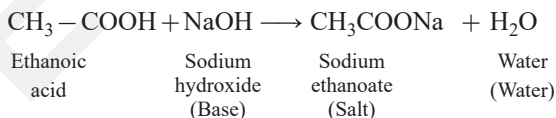
(3) A piece of sodium is dropped in ethanol.

Ans. When a piece of sodium is dropped in ethanol at room temperature, sodium ethoxide is formed and hydrogen gas is liberated.



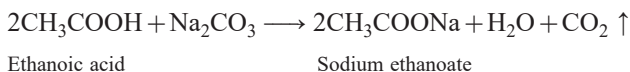
(4) Ethanoic acid is treated with sodium hydroxide.

Ans. When ethanoic acid is treated with sodium hydroxide, neutralization takes place to form sodium acetate (sodium ethanoate) and water.



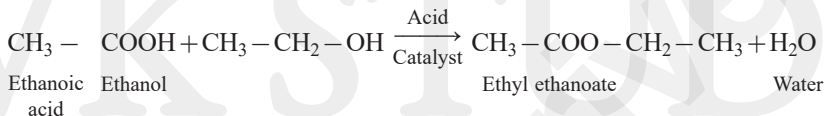
(5) Ethanoic acid is treated with sodium carbonate.

Ans. When ethanoic acid is treated with sodium carbonate, sodium ethanoate, water and carbon dioxide is formed.



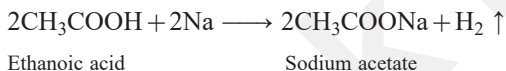
(6) Ethanoic acid is treated with ethanol.

Ans. When ethanoic acid is treated with ethanol in the presence of an acid catalyst, an ester, i.e. ethyl ethanoate is formed.



(7) A piece of sodium is dropped in ethanoic acid.

Ans. When a piece of sodium is dropped in ethanoic acid, sodium acetate is formed and hydrogen gas is liberated.



Question Set 12

DISTINGUISH BETWEEN

[Q. 2 (B) and 3]

Remember :

- ☛ In this question set the points of **distinction between** two terms are to be written.
- ☛ Write minimum 2 points of distinction.

Chapter 1 : Gravitation

Q. Distinguish between : (2 marks, Two points : 1 mark each)

Mass and Weight.

(March '23 – '25)

Ans.

Mass	Weight
1. The mass of a body is the amount of matter present in it.	1. The weight of a body is the force with which the earth attracts it.
2. It has magnitude, but not direction.	2. It has both magnitude and direction.
3. It does not change from place to place.	3. It changes from place to place.
4. It can never be zero.	4. It is zero at the centre of the earth.

(Any two points)

Chapter 2 : Periodic Classification of Elements

Mendeleev's periodic table and Modern periodic table.

Ans.

Mendeleev's periodic table	Modern periodic table
1. In this table, the elements are arranged in the order of their increasing atomic weights.	1. In this table, the elements are arranged in the order of their increasing atomic numbers.
2. In this table, the position of an element is based on its properties and atomic weight.	2. In this table, the position of an element is based on its electronic configuration.
3. There are 8 groups in this table.	3. There are 18 groups in this table.

4. In this table, some elements having similar properties are found in different groups, while those having different properties are sometimes found in the same group.	4. In this table, the elements belonging to the same group show similar chemical properties.
5. Isotopes do not find separate places in this table.	5. Isotopes of an element can be placed at the same place as their atomic number is the same.

Chapter 4 : Effects of Electric Current

Electric motor and Electric generator.

Ans.

Electric motor	Electric generator
1. A battery is used in an electric motor to pass a current through the coil.	1. A battery is not used in an electric generator.
2. In this case, a current-carrying coil is set into rotation due to the magnetic field.	2. In this case, a potential difference and hence a current is produced when the coil is set into rotation in the magnetic field by an external agent.
3. Split rings are used in an electric motor.	3. Rings used in an AC generator are not split.
4. In this case, electric energy is converted into mechanical energy.	4. In this case, mechanical energy is converted into electric energy.

(Any two points)

Chapter 7 : Lenses

(1) Concave lens and Convex lens.

Ans.

Concave lens	Convex lens
1. A concave lens has its surfaces curved inwards.	1. A convex lens has its surfaces puffed up outwards.
2. It is thicker at the edges than in the middle.	2. It is thicker in the middle than at the edges.

3. It can form only a virtual image.	3. It can form a real image as well as a virtual image.
4. It can form only a diminished image.	4. It can form a magnified, diminished or the same sized image (relative to the object) depending on the position of the object.

(Any two points)

(2) Farsightedness (Hypermetropia) and Nearsightedness (Myopia).

(Sept. '21) (3 marks)

Ans.

Farsightedness (Hypermetropia)	Nearsightedness (Myopia)
1. In hypermetropia, a human eye can see distant objects distinctly but is unable to see nearby objects clearly.	1. In myopia, a human eye can see near objects distinctly, but is unable to see distant objects clearly.
2. In this case, the curvature of the cornea and the eye lens decreases (possible reason).	2. In this case, the curvature of the cornea and the eye lens increases (possible reason).
3. In this case, the distance between the eye lens and the retina decreases (possible reason).	3. In this case, the distance between the eye lens and the retina increases (possible reason).
4. In this case, the image of a nearby object would be formed behind the retina.	4. In this case, the image of a distant object is formed in front of the retina.

(Any three points)

(3) Simple microscope and Compound microscope.

Ans.

Simple microscope	Compound microscope
1. In a simple microscope, only one convex lens is used.	1. In a compound microscope, two convex lenses, objective and eyepiece, are used.

2. In this case, the object is placed within the focal length of the convex lens.	2. In this case, the object is placed beyond the focal length of the objective lens.
3. Its magnifying power is much less than that of a compound microscope.	3. Its magnifying power is much greater than that of a simple microscope.
4. It is used to observe minute parts of a watch, to read words in small print, etc.	4. It is used to observe blood corpuscles, plant and animal cells, etc.

(Any two points)

Chapter 9 : Carbon Compounds

Saturated hydrocarbons and Unsaturated hydrocarbons.

Ans.

Saturated hydrocarbons	Unsaturated hydrocarbons
1. In saturated hydrocarbons, the carbon atoms are linked to each other only by single covalent bonds.	1. In unsaturated hydrocarbons, the valencies of carbon atoms are not fully satisfied by single covalent bonds.
2. They contain only a single bond.	2. They contain carbon to carbon double or triple bonds.
3. They are chemically less reactive.	3. They are chemically more reactive.
4. Substitution reaction is a characteristic property of these hydrocarbons.	4. Addition reaction is a characteristic property of these hydrocarbons.
5. Their general formula is C_nH_{2n+2} .	5. Their general formula is C_nH_{2n} or C_nH_{2n-2} .

Chapter 4 : Effects of Electric Current

(1) Explain the application of heating effect of electric current in an electric bulb.

Ans. In an electric bulb, there is a filament of metal such as tungsten having high melting point (3410°C). When an electric current is passed through the filament, it becomes hot and emits light. The bulbs are usually filled with chemically inactive gases such as nitrogen and argon to prevent oxidation of the filament and hence prolong their life.

(2) Explain the application of heating effect of electric current in an electric iron.

Ans. In an electric iron, a coil of high resistance is held between mica sheets and placed inside a heavy metal block provided with a handle made of an insulator such as plastic. When an electric current is passed through the coil, it becomes hot. Mica is a good conductor of heat. Hence, heat produced in the coil is transferred to the metal block which can then be used for ironing clothes.

Mica is a bad conductor of electricity. Hence, there is no electrical contact between the coil and the metal block. Therefore, the person using the iron does not get an electric shock even if he or she happens to touch it by chance.

(3) Explain the application of heating effect of electric current in a fuse.

Ans. A fuse protects electrical circuits and appliances by stopping the flow of electric current when it exceeds a specified value. For this, it is connected in series with the appliance (or circuit) to be protected. A fuse is a piece of wire made of an alloy of low melting point (e.g. an alloy of lead and tin). If a current larger than the specified value flows through the fuse, its temperature increases enough to melt it. Hence, the circuit breaks and the appliance is protected from damage.

(4) State the uses/applications of an electric motor.

Ans. Uses/applications of an electric motor : (1) In domestic appliances such as a mixer, a blender, a refrigerator and washing machine. (2) In an electric fan, a hair dryer, a record player, a tape recorder and a blower. (3) In an electric car, a rolling mill, an electric crane, an electric lift, a pump, a computer and an electric train.

Chapter 5 : Heat

Q. What is the use of the calorimeter? (March '24) (1 mark)

Ans. The calorimeter is used for determination of specific heat capacity.

Chapter 6 : Refraction of Light

Q. Name the devices in which total internal reflection of light is used. (1 mark)

Ans. (1) Total internal reflecting prisms are used in a camera, binoculars, periscope.

(2) Total internal reflection of light is used in optical fibres.

Chapter 7 : Lenses

(1) State uses of a concave lens.

Ans. (1) Concave lenses are used for proper working of medical equipment, scanner, CD player – the instruments that employ laser rays.

(2) One or more concave lenses are used in a small safety device, fitted in the peep hole in a door, due to which we can see a large area outside the door.

(3) Concave lenses are used in spectacles to correct nearsightedness (myopia).

(4) A concave lens is used to spread light emitted by the small bulb in a torch over a wide area.

(5) A concave lens is used in front of the eyepiece or inside the eyepiece fitted in a camera, telescope and microscope – the instruments employing convex lenses.

(2) State uses of a convex lens.

Ans. Convex lenses are used in a simple microscope, compound microscope, refracting telescope, camera, projector, spectroscope, spectacles for correcting farsightedness (hypermetropia) and binoculars.

(3) State two uses of a compound microscope. (2 marks)

Ans. Uses of a compound microscope :

(1) It is used to observe blood corpuscles, plant and animals cells, microorganisms like bacteria, etc.

(2) It is used in a pathological laboratory to observe blood, urine, etc.

(3) It is a part of a travelling microscope used for measurement of very small distance.

Chapter 8 : Metallurgy

Q. State the general properties of ionic compounds.

(July '22; March '23) (5 marks)

Ans. (1) Ionic compounds are solids and hard due to strong electrostatic

force of attraction between oppositely charged ions. (2) They are generally brittle. When pressure is applied they break into pieces. (3) They have high melting and boiling points, because intermolecular force of attraction is high in ionic compounds. (4) They are soluble in water and insoluble in solvents such as kerosine and petrol. (5) Ionic compounds cannot conduct electricity when in solid state as they are electrically neutral. They conduct electricity in the molten state and also in an aqueous solution.

Chapter 9 : Carbon Compounds

(1) State the physical properties of ethyl alcohol/ethanol. ($\frac{1}{2}$ mark each)

Ans. (1) Ethanol is a colourless liquid and it is soluble in water in all proportions and has pleasant odour.

(2) The boiling point of ethanol is 78°C and the freezing point is -114°C .

(3) It is combustible and burns with a blue flame.

(4) An aqueous solution of ethanol is neutral to litmus paper.

(2) What is meant by vinegar and gasohol? What are their uses?

(July '19) (2 marks)

Ans. (1) Vinegar is a 5 – 8% aqueous solution of acetic acid. It is used as preservative in pickles. It is used to cook meat. It is used as a salad dressing.

(2) To increase the efficiency of petrol, it is mixed with 10% anhydrous ethanol. Such a fuel is called gasohol. It is used as a fuel in cars and other vehicles.

(3) State the uses of ethanol.

(July '22) (1 mark each)

Ans. (1) Ethanol is used as an important commercial solvent in industries.

(2) It is used in medicines such as tincture iodine, cough mixture and also in many tonics.

(3) Ethanol is used as a clean fuel.

(4) State the properties of ethanoic acid.

($\frac{1}{2}$ mark each)

Ans. (1) Ethanoic acid is a colourless liquid with boiling point 118°C and melting point 17°C . It has a pungent odour.

(2) Its aqueous solution is acidic and turns blue litmus red.

(3) A 5% – 8% aqueous solution of acetic acid is used as vinegar.

(4) It is a weak acid.

**Question
Set
14**

QUESTIONS BASED ON EXAMPLES

[Q. 2 (B)]

Remember :

- ☛ In this question set, questions based on examples are included.
- ☛ Each subquestion will carry 2 marks. Write two/four examples.

Chapter 6 : Refraction of Light

Q. Give two examples of refraction observed in the atmosphere.

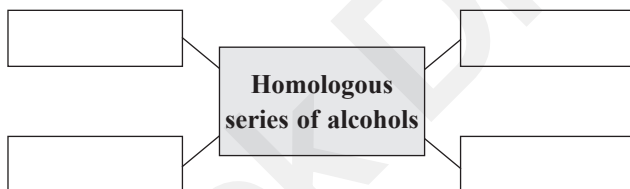
(1 mark)

Ans. Atmospheric refraction of light results in many interesting optical phenomena such as (1) twinkling of a star (2) advanced Sunrise and delayed Sunset.

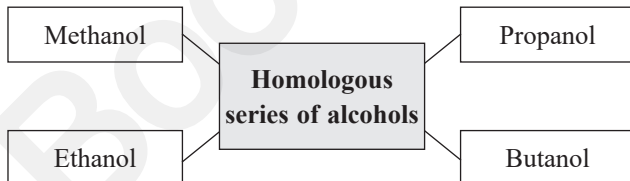
Chapter 9 : Carbon Compounds

Q. Write names of first four homologous series of alcohols.

(March '19) (2 marks)



Ans.



Question Set 15

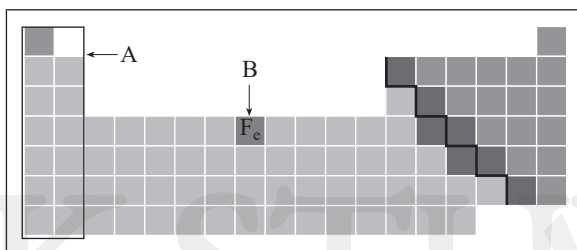
QUESTIONS BASED ON DIAGRAMS

[For Q. 3 and Q. 4]

Note : In the Board examination, you may be asked to draw a diagram and label its parts OR only part of the diagram or incorrect diagram may be given and you will be expected to complete it OR correct it and redraw it as the case may be OR a complete diagram may be given and you will be expected to label its parts.

Chapter 2 : Periodic Classification of Elements

Q. Observe the following figure and answer the questions based on it :



(A) Identify the block shown by box A and write the electronic configuration of any one element of this block.

(B) Identify the block of elements denoted by letter B and write its period number. **(2 marks)**

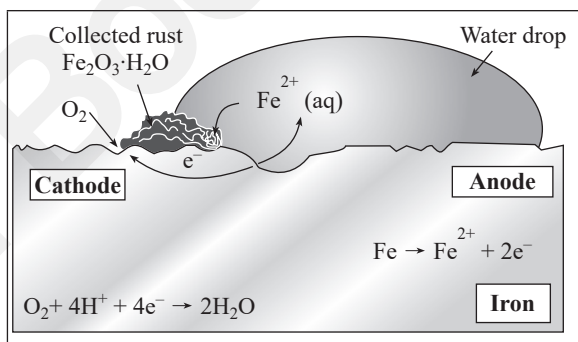
Ans. (A) The block shown by box A is the s-block. Electronic configuration of Mg : 2, 8, 2.

(B) The block of elements denoted by letter B is the d-block and its period number is 4.

Chapter 3 : Chemical Reactions and Equations

Q. Observe the following picture and answer the following questions :

(March '20) (5 marks)



- (a) What is rust?
 (b) Write the chemical formula of rust. (Sept. '21)
 (c) Write the reaction of oxidation of iron at anode. (Sept. '21)
 (d) Write the reaction of oxidation of iron at cathode.
 (e) What is corrosion?

Ans. (a) When iron is exposed to moist air, it forms reddish layer of hydrated ferric oxide; this is called rust.

(b) Chemical formula of rust : $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$

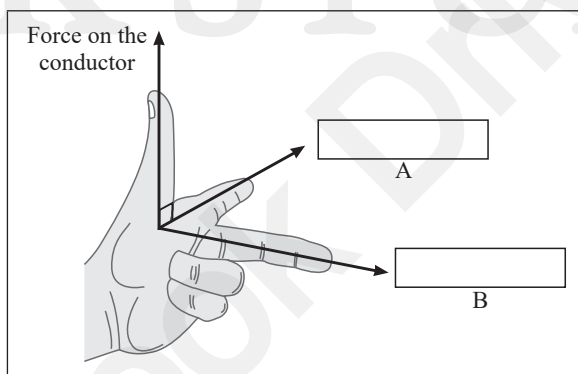
(c) Reaction at anode : $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$

(d) Reaction at cathode : $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O(l)}$

(e) The slow process of decay or damage or oxidation of a metal due to various components of atmosphere is known as corrosion.

Chapter 4 : Effects of Electric Current

Q. 1. Observe the given figure of Fleming's left hand rule and write the labels of 'A' and 'B' : (March '22) (2 marks)

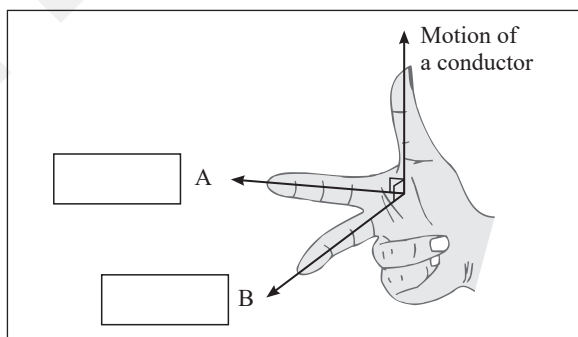


Ans. A : Direction of the magnetic field

B : Direction of the current

[Note : Do not draw the diagram in the answer book.]

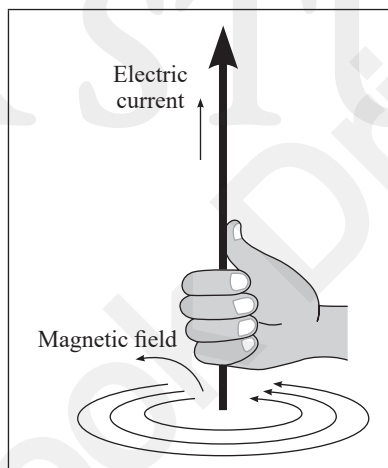
Q. 2. Observe the given figure of Fleming's right hand rule and write the labels of A and B correctly. **(March '20) (2 marks)**



Ans. A : Direction of the magnetic field
B : Direction of the induced current.

[Note : Do not draw the diagram in the answer book.]

Q. 3. Identify the law from the given figure : **(July '22)**



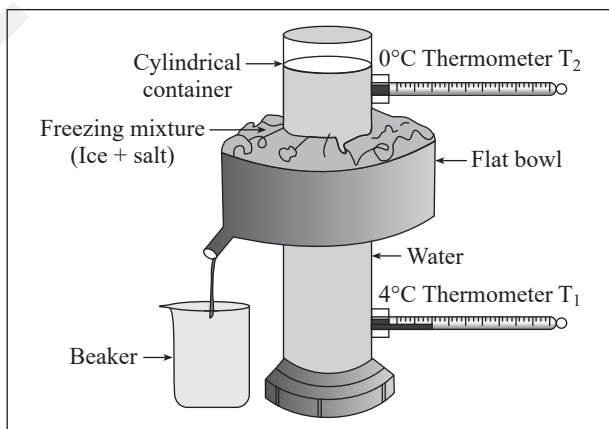
Ans. Right hand thumb rule.

Chapter 5 : Heat

Q. 1. Draw a neat labelled diagram of Hope's apparatus.

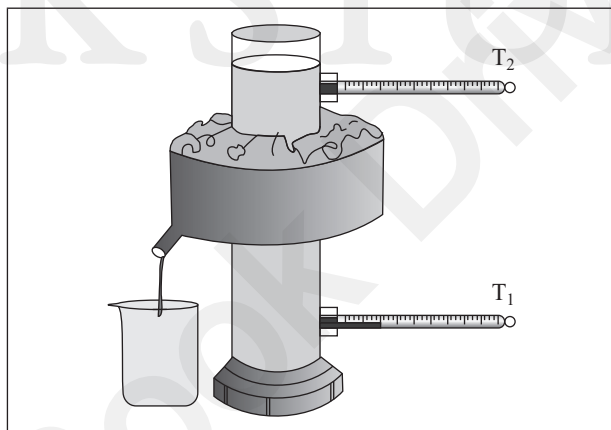
(March '22) (2 marks)

Ans.



Q. 2. Observe the given diagram and answer the following questions :

(July '23) (5 marks)



- What is the name of the given apparatus?
- Which phenomenon is studied with the help of this apparatus?
- What are the final temperatures in thermometers T_1 and T_2 ?
- At what temperature is the density of water maximum?
- Give *one* example of the above phenomenon in nature.

Ans. (a) Hope's apparatus

(b) Anomalous behaviour of water

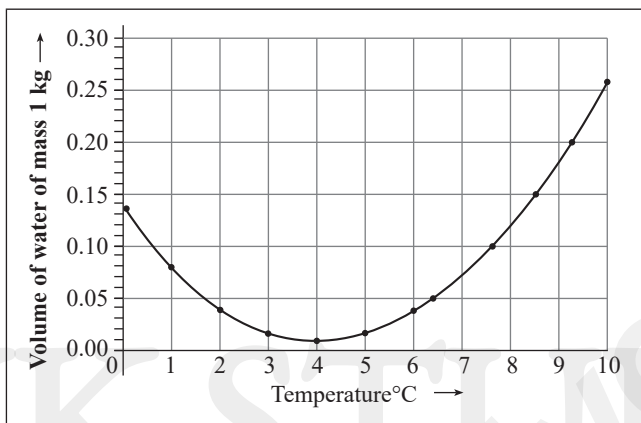
(c) $T_1 : 4^{\circ}\text{C}$, $T_2 : 0^{\circ}\text{C}$

(d) 4°C

(e) Even when the temperature becomes less than 0°C aquatic plants and animals can survive.

Q. 3. Observe the given graph and answer the following questions :

(March '20) (2 marks)



(a) Name the process represented in the figure.

(b) At what temperature does this process take place?

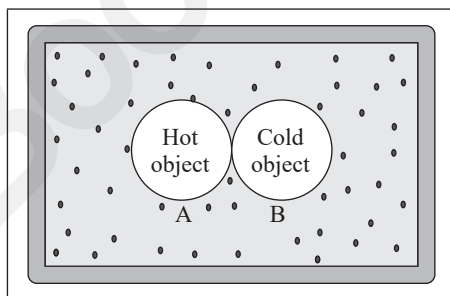
Ans. (a) Anomalous behaviour of water.

(b) Temperature range : 0°C to 4°C .

[Notes : (1) The unit of volume is not given in the figure. But it is not needed in answering the question (3) **Q. 3. (b)** should be as follows : State the temperature range in which this process takes place.]

Q. 4. Answer the questions with reference to the given figure :

(Sept. '21) (3 marks)



(a) From where to where is heat transferred?

(b) Which principle do we learn about from this process?

(c) Which property of the substance is measured using this principle?

Ans. (a) Heat is transferred from the hot object (A) to the cold object (B).

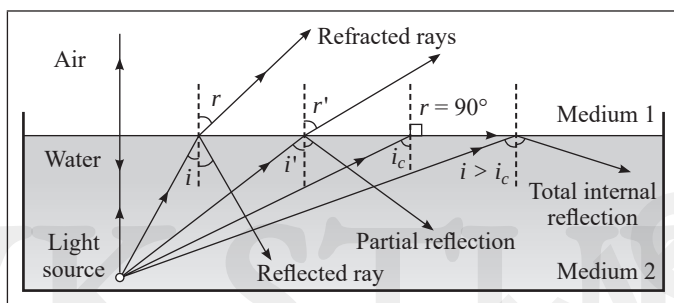
(b) Principle of heat exchange.

(c) Specific heat of a substance.

Chapter 6 : Refraction of Light

Q. With a neat labelled diagram, explain the terms total internal reflection and critical angle. (5 marks)

Ans. The following figure shows propagation of light from water (denser medium) to air (rarer medium).



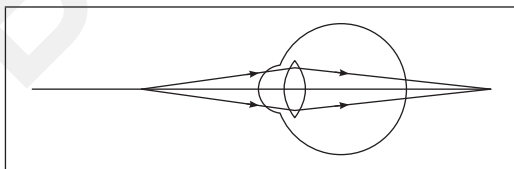
The ray of light incident at the boundary separating the two media bends away from the normal on refraction. Here, the angle of refraction r is greater than the angle of incidence i . Now ${}_w n_a = \frac{\sin i}{\sin r} < 1$. Where, ${}_w n_a$ is the refractive index of air with respect to water. As ${}_w n_a$ is constant, r increases as i increases. For $r = 90^\circ$, the ray travels along the boundary. If i is increased further, as r cannot be greater than 90° , light does not enter air. There is no refraction of light and all the light enters water on reflection. This is called total internal reflection.

For $r = 90^\circ$, ${}_w n_a = \frac{\sin i}{\sin 90^\circ} = \sin i$. This angle i is called the critical angle.

Chapter 7 : Lenses

Q. 1. Identify the defect of vision shown in the figure :

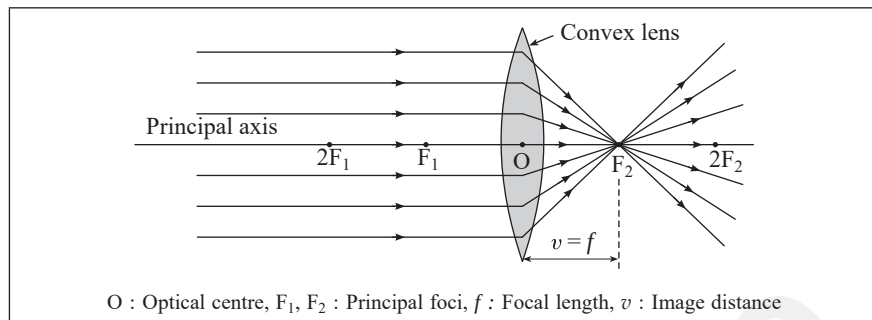
(July '22) (1 mark)



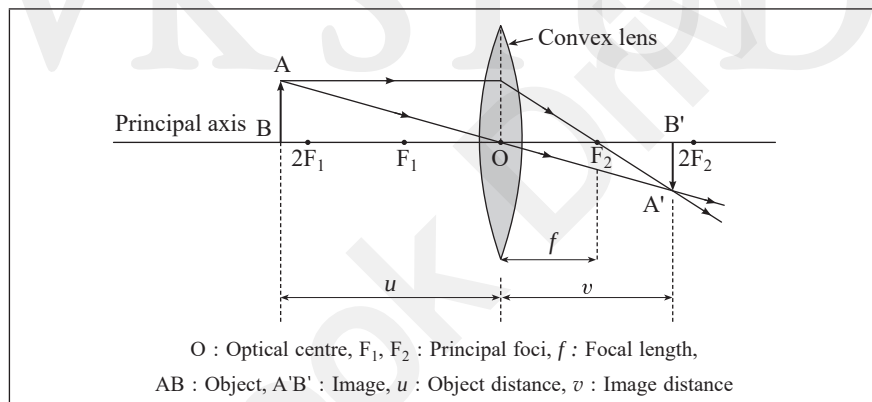
Ans. Hypermetropia or farsightedness.

Q. 2. Draw neat and well labelled ray diagrams for image formation by a convex lens when an object is (1) at infinity (2) beyond $2F_1$ (3) at $2F_1$ (July '23) (4) between F_1 and $2F_1$ (5) at focus F_1 (6) between focus F_1 and optical centre O . (2 marks each)

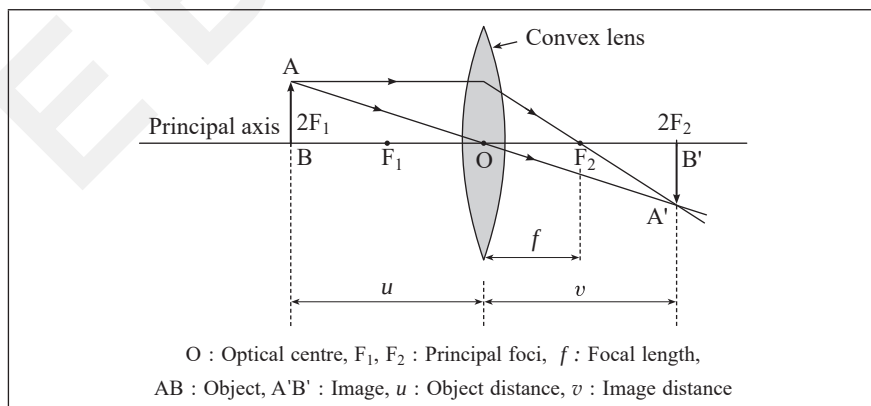
Ans. (1) Object at infinity, image at F_2 :



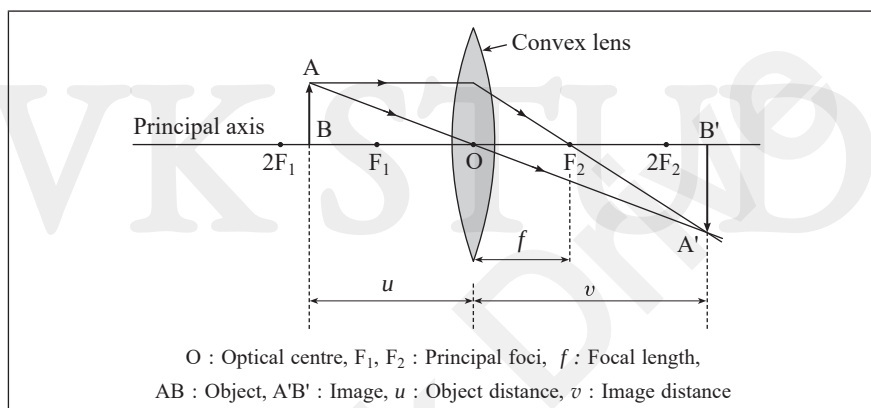
(2) Object beyond $2F_1$, image between F_2 and $2F_2$:



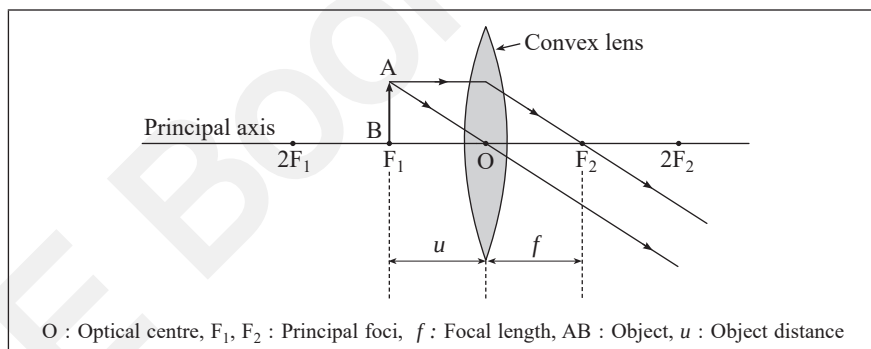
(3) Object at $2F_1$, image at $2F_2$:



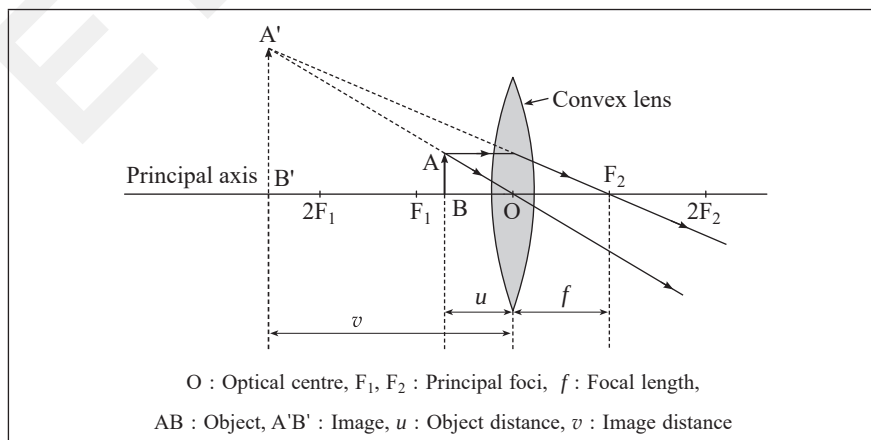
(4) Object between F_1 and $2F_1$, image beyond $2F_2$:



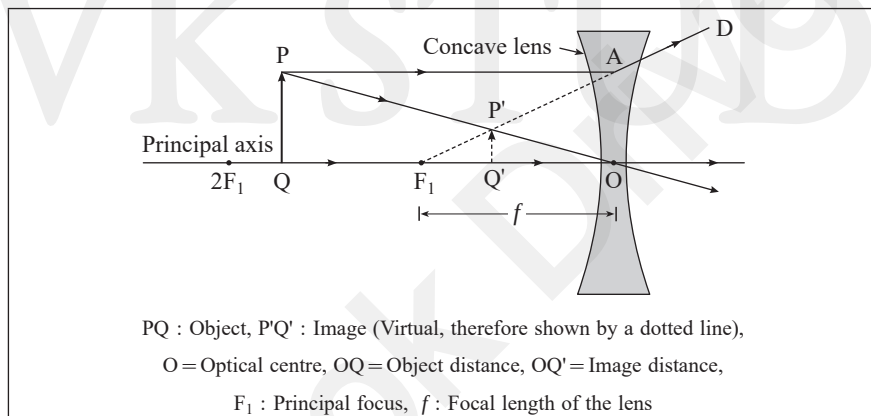
(5) Object at focus F_1 , image at infinity :



- (6) Object between focus F_1 and optical centre O , image on the same side as object with $v > u$:



- Q. 3. Draw a ray diagram to show image formation by a concave lens.**

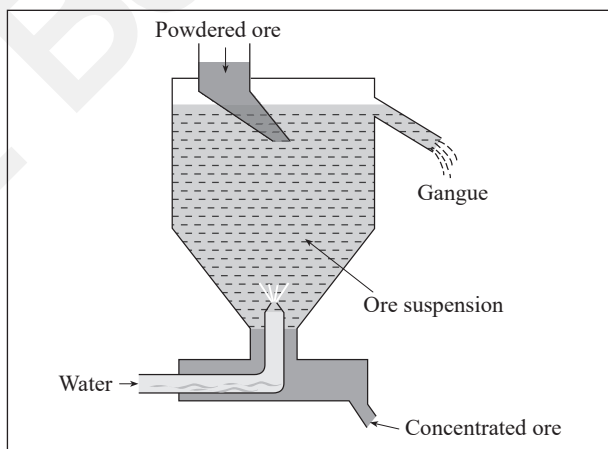


Chapter 8 : Metallurgy

- (1) With a neat labelled diagram describe the hydraulic separation method. (3 marks)

Ans. (1) The hydraulic separation method is based on the working of a mill. This is a tapering vessel similar to that used in a grinding mill. It opens in a tank-like container that is tapering on the lower side.

(2) The tank has an outlet for water on the upper side and a water inlet on the lower side.



Finely ground ore is added to the tank. A fast stream of water is released in the tank from the lower side.

(3) The lighter gangue particles flow out along with the water stream from the outlet on the upper side of the tank and are collected separately; simultaneously the heavy particles of the ore are collected at the bottom from the lower side of the tank. This method is based on the law of gravitation, wherein particles of the same size are separated by their weight with the help of water.

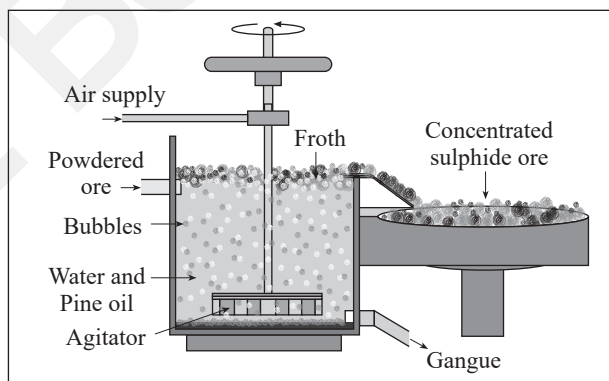
(2) With a neat labelled diagram describe the froth floatation method.

(3 marks)

Ans. (1) The froth floatation method is based on the two opposite properties, hydrophilic and hydrophobic, of the particles. The metal sulphide particles get wet mainly with oil due to their hydrophobic property. The gangue particles get wet with water due to the hydrophilic property.

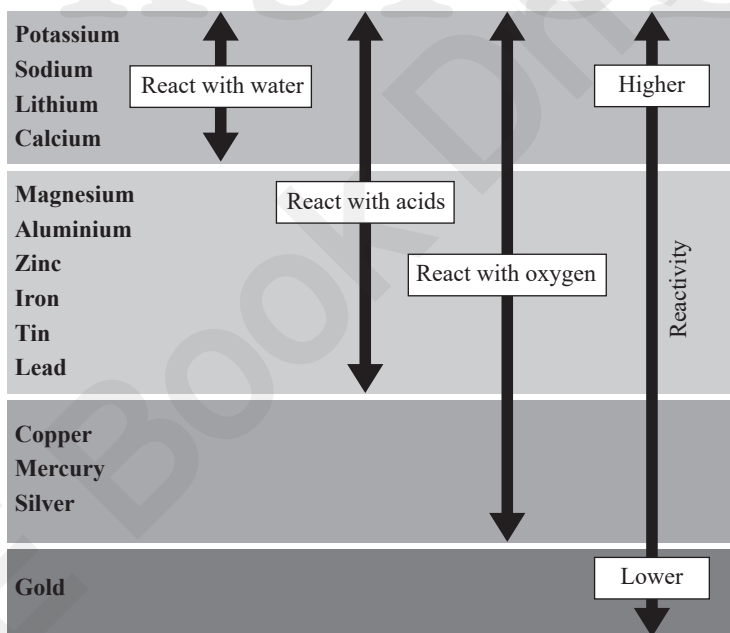
(2) In this method the finely ground mineral is put into a big tank containing a lot of water. The finely powdered ore and vegetable oil such as pine oil, eucalyptus oil are mixed with water for formation of froth.

(3) The pressurised air is blown through the mixture. There is an agitator rotating around its axis at the centre of the floatation tank. The agitator is used as per the requirement.



(4) Bubbles are formed due to the blown air. A foam is formed from oil, water and air bubbles together, due to the agitation. This foam rises to the surface of the water and floats. Hence this method is called the froth floatation method. Sulphide minerals float with the foam on water as they get and can be removed. The gangue particles are wetted by water and settle down at the bottom. This method is used for concentration of zinc blend (ZnS) and copper pyrite (CuFeS_2).

(3) Observe the given figure of reactivity series of metals and answer the following questions :



Reactivity series of metals

- (a) Name any two metals which react with water.
 (b) Name any two moderately reactive metals.
 (c) Name the most highly reactive metal and the most less reactive metal.

(March '20) (3 marks)

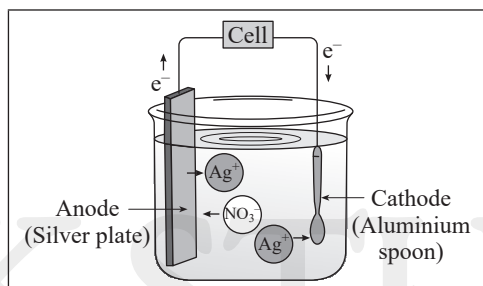
Ans. (a) Metals which react with water : Potassium, Sodium.

(b) Moderately reactive metals : Magnesium, Aluminium.

(c) The most highly reactive metal : Potassium.

The most less reactive metal : Gold.

(4) Answer the following questions with the help of the given diagram :



(a) Name the process shown in the diagram.

(b) How does this process take place?

(c) Give two examples in which this process is used.

(March '25)

Ans. (a) The process shown in the diagram is electroplating.

(b) In this process, a less reactive metal is coated on a more reactive metal by electrolysis.

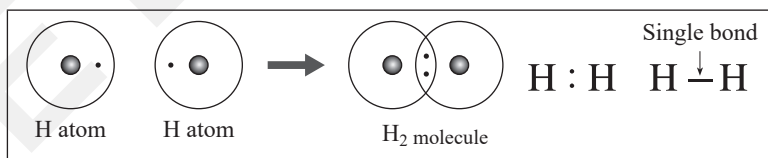
(c) Silver plated spoons, gold plated ornaments are the examples in which this process is used.

Chapter 9 : Carbon Compounds

(1) Draw the electron dot structure and line structure of hydrogen molecule.

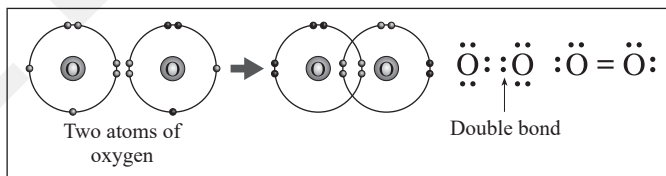
(2 marks)

Ans.



(2) Draw the electron dot structure and line structure of oxygen molecule. (2 marks)

Ans.

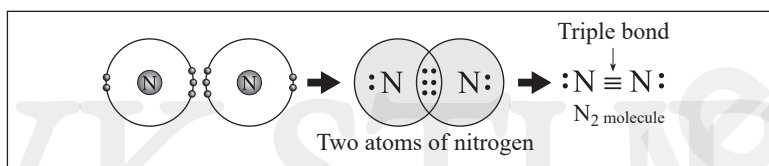


(3) (a) The atomic number of nitrogen is 7. How many electrons are present in the valence shell of nitrogen?

(b) Molecular formula of nitrogen is N_2 . Draw the electron dot structure and line structure of a nitrogen molecule. (July '22) (3 marks)

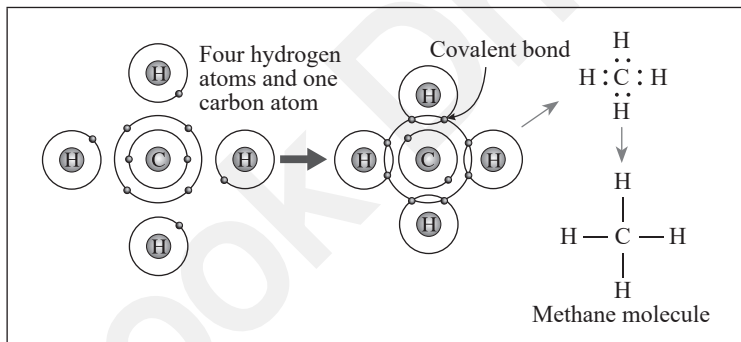
Ans. (a) Five electrons are present in the valence shell of nitrogen.

(b)

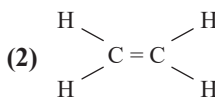
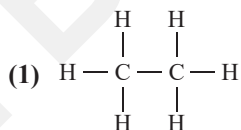


(4) Draw the electron dot structure and line structure of methane molecule. (2 marks)

Ans.



(5) (a) Identify the saturated and unsaturated hydrocarbons from the given structural formula : (March '22) (2 marks)

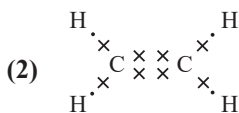
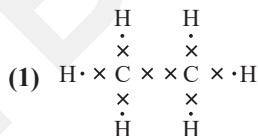


Ans. (1) Saturated hydrocarbons (2) Unsaturated hydrocarbons

(b) Draw the electron dot structure for (1) and (2).

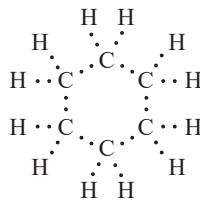
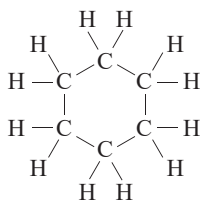
(2 marks)

Ans.



(c) Draw the electron dot structure of cyclohexane.

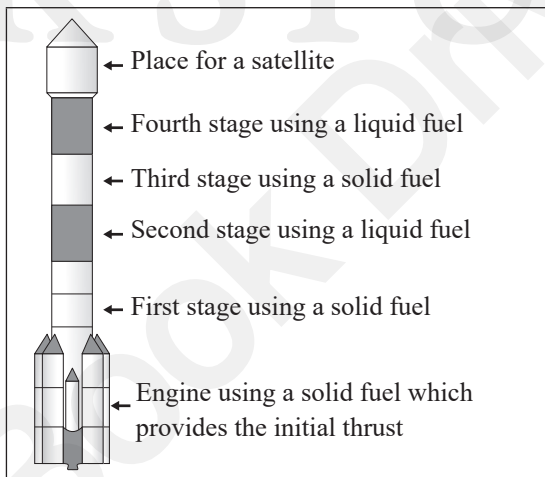
Ans. Cyclohexane : Molecular formula : C_6H_{12}



Chapter 10 : Space Missions

Q. Draw a neat labelled diagram : Structure of PSLV made by ISRO.

(2 marks)



Question Set 16

QUESTIONS BASED ON THE TABLE AND FLOW CHART

[Q. 2 (B), Q. 3 and Q. 4]

Remember :

- ☛ In this question set, incomplete tables or incomplete flow charts are given.
- OR
- ☛ You will be required to rearrange the entries in the table, depending on correlation.
- ☛ Any four or six blank spaces should be correctly filled to procure 2 or 3 marks.

Q. Study the relation between the entries in the following table and rewrite the chart. (3 marks)

Chapter 1 : Gravitation

I	II	III
Mass	m/s^2	Zero at the centre of the earth
Weight	kg	Measure of inertia
Acceleration due to gravity	$\text{N}\cdot\text{m}^2/\text{kg}^2$	Same in the entire universe
Gravitational constant	N	Depends on height

Ans.

I	II	III
Mass	kg	Measure of inertia
Weight	N	Depends on height
Acceleration due to gravity	m/s^2	Zero at the centre of the earth
Gravitational constant	$\text{N}\cdot\text{m}^2/\text{kg}^2$	Same in the entire universe

Chapter 2 : Periodic Classification of Elements

(1) Rearrange the columns 2 and 3 so as to match with the column 1.

Column 1	Column 2	Column 3
i. Triad	a. Lightest and negatively charged particle in all the atoms	1. Mendeleev
ii. Octave	b. Concentrated mass and positive charge	2. Thomson
iii. Atomic number	c. Average of the first and the third atomic mass	3. Newlands
iv. Period	d. Properties of the eighth element similar to the first	4. Rutherford
v. Nucleus	e. Positive charge on the nucleus	5. Dobereiner
vi. Electron	f. Sequential change in molecular formulae	6. Moseley

Ans.

Column 1	Column 2	Column 3
i. Triad	Average of the first and the third atomic mass	Dobereiner
ii. Octave	Properties of the eighth element similar to the first	Newlands
iii. Atomic number	Positive charge on the nucleus	Moseley
iv. Period	Sequential change in molecular formulae	Mendeleev
v. Nucleus	Concentrated mass and positive charge	Rutherford
vi. Electron	Lightest and negatively charged particle in all the atoms	Thomson

(2) Complete the flow chart :

(2 marks each)

(1) Na → ... → ... → Si → ... → ... → Cl → Ar

Ans. Na → Mg → Al → Si → P → S → Cl → Ar

(2) ... → ... → Ar → ... → ... → Rn → Og

Ans. He → Ne → Ar → Kr → Xe → Rn → Og

(3)	Shell	...	...	...	...
		↓	↓	↓	↓
	Electron capacity	2	8	18	→ 32

Ans.	Shell	<u>K</u>	<u>L</u>	<u>M</u>	<u>N</u>
		↓	↓	↓	↓
	Electron capacity	2	8	18	→ 32

Chapter 3 : Chemical Reactions and Equations

(1) Match the columns in the following table :

Reactants	Products	Type of chemical reaction
$\text{BaCl}_2(\text{aq}) + \text{ZnSO}_4(\text{aq})$	$\text{H}_2\text{CO}_3(\text{aq})$	Displacement
$2\text{AgCl}(\text{s})$	$\text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$	Combination
$\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s})$	$\text{BaSO}_4\downarrow + \text{ZnCl}_2(\text{aq})$	Decomposition
$\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	$2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$	Double displacement

Ans.

Reactants	Products	Type of chemical reaction
$\text{BaCl}_2(\text{aq}) + \text{ZnSO}_4(\text{aq})$	$\text{BaSO}_4\downarrow + \text{ZnCl}_2(\text{aq})$	Double displacement
$2\text{AgCl}(\text{s})$	$2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$	Decomposition
$\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s})$	$\text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$	Displacement
$\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	$\text{H}_2\text{CO}_3(\text{aq})$	Combination

(2)

Reactants	Products	Type of chemical reaction
$\text{Fe} + \text{S}$	$\text{NaCl} + \text{H}_2\text{O}$	Oxidation
$\text{CuSO}_4 + \text{Zn}$	2CuO	Neutralization
$2\text{Cu} + \text{O}_2$	$\text{ZnSO}_4 + \text{Cu}$	Displacement
$\text{HCl} + \text{NaOH}$	FeS	Combination

Ans.

Reactants	Products	Type of chemical reaction
$\text{Fe} + \text{S}$	FeS	Combination
$\text{CuSO}_4 + \text{Zn}$	$\text{ZnSO}_4 + \text{Cu}$	Displacement
$2\text{Cu} + \text{O}_2$	2CuO	Oxidation
$\text{HCl} + \text{NaOH}$	$\text{NaCl} + \text{H}_2\text{O}$	Neutralization

Chapter 7 : Lenses

(1) Complete the following table for convex lens :

(March '22) (3 marks)

Sr. No.	Position of the object	Position of the image	Nature of the image
1.	Beyond $2F_1$		
2.		At infinity	
3.			Real, inverted and enlarged

Ans.

Sr. No.	Position of the object	Position of the image	Nature of the image
1.	Beyond $2F_1$	<u>Between F_2 and $2F_2$</u>	<u>Real and inverted</u>
2.	<u>On focus F_1</u>	At infinity	<u>Real and inverted</u>
3.	<u>Between F_1 and $2F_1$</u>	<u>Beyond $2F_2$</u>	Real, inverted and enlarged

(2) Match the columns in the following table :

(3 marks)

Column 1	Column 2	Column 3
Farsightedness	Nearby object can be seen clearly	Bifocal lens
Presbyopia	Faraway object can be seen clearly	Concave lens
Nearsightedness	Problem of old age	Convex lens

Ans.

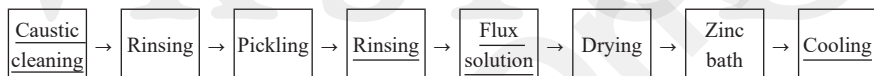
Column 1	Column 2	Column 3
Farsightedness	Faraway object can be seen clearly	Convex lens
Presbyopia	Problem of old age	Bifocal lens
Nearsightedness	Nearby object can be seen clearly	Concave lens

Chapter 8 : Metallurgy

• Complete the steps involved in the galvanizing process :



Ans.



Chapter 9 : Carbon Compounds

(1) Complete the following table :

(March '19) (3 marks)

Common name	Structural formula	IUPAC name
Ethylene	$\text{CH}_2 = \text{CH}_2$	
.....	CH_3COOH	Ethanoic acid
Methyl alcohol		Methanol

Ans.

Common name	Structural formula	IUPAC name
Ethylene	$\text{CH}_2 = \text{CH}_2$	<u>Ethene</u>
<u>Acetic acid</u>	CH_3COOH	Ethanoic acid
Methyl alcohol	<u>CH_3OH</u>	Methanol

(2) Complete the following table :

(March '23) (5 marks)

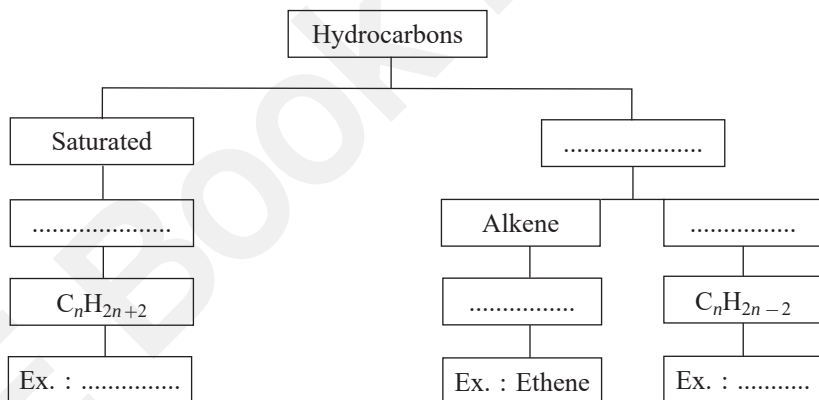
Sr. No.	Common Name	Structural Formula	IUPAC Name
1	Ethylene	$\text{CH}_2 = \text{CH}_2$	
2	Acetylene		Ethyne
3	Acetic acid	$\text{CH}_3 - \text{COOH}$	
4	Methyl alcohol		Methanol
5		$\text{CH}_3 - \text{CO} - \text{CH}_3$	Propane-2-one

Ans.

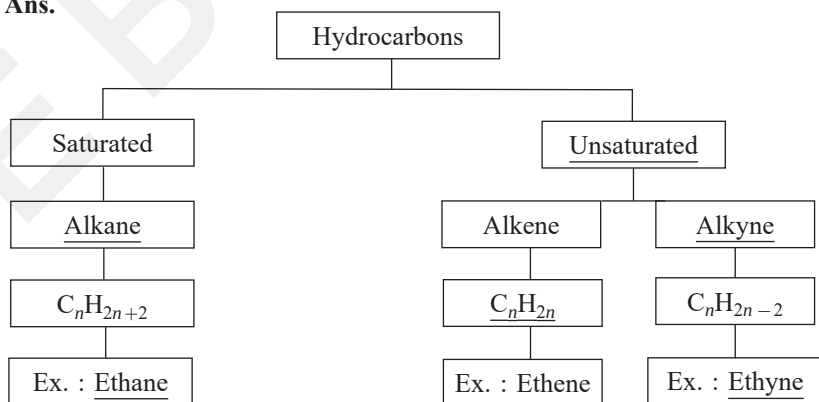
Sr. No.	Common Name	Structural Formula	IUPAC Name
1	Ethylene	$\text{CH}_2 = \text{CH}_2$	<u>Ethene</u>
2	Acetylene	<u>$\text{CH} \equiv \text{CH}$</u>	Ethyne
3	Acetic acid	$\text{CH}_3 - \text{COOH}$	<u>Ethanoic acid</u>
4	Methyl alcohol	<u>$\text{CH}_3 - \text{OH}$</u>	Methanol
5	<u>Acetone</u>	$\text{CH}_3 - \text{CO} - \text{CH}_3$	Propane-2-one

(3) Complete the following flow chart :

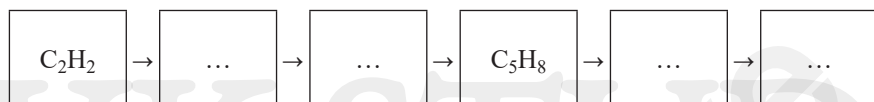
(3 marks) (March '23)



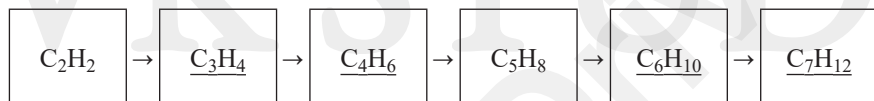
Ans.



(4) Complete the flow chart : Homologous series of alkynes : (2 marks)



Ans.



(5) Complete the following table :

Ans. (Answers are directly given and underlined.)

Hetero atom	Name	Structural formula	Condensed structural formula	Example
Halogen, [Chlorine, Bromine, Iodine]	Halo	$\text{-- X(} \underline{\text{-- Cl, -- Br, -- I}} \text{)}$	$\text{-- X(} \underline{\text{-- Cl, -- Br, -- I}} \text{)}$	<u>CH₃Cl Chloromethane</u>
Oxygen	Alcohol	-- O -- H	<u>-- OH</u>	<u>CH₃ -- CH₂ -- OH Ethanol</u>
Nitrogen	Amine	$\begin{array}{c} \text{-- N -- H} \\ \\ \text{H} \end{array}$	<u>-- NH₂</u>	<u>CH₃ -- CH₂ -- NH₂</u> <u>Ethylamine</u>

(6) Complete the following table :

(March '20) (3 marks)

Ans. (Answers are directly given and underlined.)

Straight chain of carbon compounds	Structural formula	Molecular formula	Name
C	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$	CH ₄	Methane
C – C	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	C ₂ H ₆	Ethane
C – C – C	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$	C ₃ H ₈	<u>Propane</u>
C – C – C – C	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$	C ₄ H ₁₀	<u>Butane</u>

Chapter 10 : Space Missions

• Complete the following table :

(March '25) (3 marks)

	Type of the satellite	Function of the satellite	The names of the Indian satellite series and their launch vehicles
(a)		Fix the location of any place on the earth's surface	
(b)	Weather satellite		
(c)			IRS Launcher : PSLV

Ans.

	Type of the satellite	Function of the satellite	The names of the Indian satellite series and their launch vehicles
(a)	<u>Navigational satellite</u>	Fix the location of any place on the earth's surface	<u>IRNSS</u> <u>Launcher : PSLV</u>
(b)	Weather satellite	<u>Study and prediction of weather</u>	<u>INSAT and GSAT</u> <u>Launcher : GSLV</u>
(c)	<u>Earth Observation Satellite</u>	<u>Study of forests, deserts, oceans, polar ice on the earth's surface, exploration and management of natural resources, observation and guidance in case of natural calamities like flood and earthquake.</u>	IRS Launcher : PSLV

Question Set 17

MISCELLANEOUS QUESTIONS

Remember :

- ☛ This set contains various types of important questions. These questions may form part of any question.

Chapter 2 : Periodic Classification of Elements

(1) Describe the merits of Mendeleev's periodic table.

(March '19) (3 marks)

Ans. (1) To give the proper place in the periodic table, atomic masses of some elements were revised in accordance with their properties. For example, the previously determined atomic mass of beryllium, 14.09, was changed to the correct value 9.4, and beryllium was placed before boron.

(2) Mendeleev had kept some vacant places in the periodic table for elements that were yet to be discovered. Three of these unknown elements were given the names eka-boron, eka-aluminium and eka-silicon from the known neighbours and their atomic masses were indicated as 44, 68 and 72, respectively. Their properties were also predicted. Later on these elements were discovered subsequently and were named as scandium (Sc), gallium (Ga) and germanium (Ge) respectively. The properties of these elements matched well with those predicted by Mendeleev. This brought out the importance of Mendeleev's periodic table.

(3) There was no place reserved for noble gases in Mendeleev's original periodic table. When noble gases such as helium, neon and argon were discovered, Mendeleev created the 'zero group' without disturbing the original periodic table. The noble gases were placed very well in the new table.

(2) What are the demerits of Mendeleev's periodic table?

(Sept. '21; March '22) (3 marks)

Ans. (1) The elements cobalt (Co) and nickel (Ni) have the same whole number atomic mass. As a result there was an ambiguity regarding their sequence in Mendeleev's periodic table.

(2) Isotopes were discovered long time after Mendeleev put forth the periodic table. A challenge was posed in placing isotopes in Mendeleev's periodic table as isotopes have the same chemical properties but different atomic masses.

(3) The increase in atomic mass does not appear to be uniform when elements are arranged in an increasing order of atomic masses. It was not possible, therefore, to predict how many elements could be discovered between two heavy elements.

(4) Position of hydrogen : Hydrogen shows similarity with halogens (group VII). For example, the molecular formula of hydrogen is H_2 while the molecular formulae of fluorine and chlorine are F_2 and Cl_2 , respectively. In the same way, there is a similarity in the chemical properties of hydrogen and alkali metals (group I). There is a similarity in the molecular formulae of the compounds of hydrogen alkali metals (Na, K, etc.) formed with chlorine and oxygen. On considering the above properties it is difficult to decide the correct position of hydrogen whether it is in the group of alkali metals (group I) or in the group of halogens (group VII).

Compounds of H	Compounds of Na
HCl	NaCl
H_2O	Na_2O
H_2S	Na_2S

Similarity in hydrogen and alkali metals

Element (Molecular formula)	Compounds with metals	Compounds with nonmetals
H_2	NaH	CH_4
Cl_2	NaCl	CCl_4

Similarity in hydrogen and halogens

(3) An element has its electronic configuration as 2, 8, 1. Now answer the following questions :

- What is the atomic number of this element?
- What is the group of this element?
- To which period does the element belong?

(March '23)

Ans. (a) The atomic number of this element : 11

(b) The group of this element : 1

(c) The period of this element : 3

(4) An element has its electronic configuration as 2, 8, 2 and answer the following questions given below : (March '20; July '22) (3 marks)

- (a) What is the atomic number of this element?
- (b) What is the group of this element?
- (c) To which period does this element belong?

Ans. (a) The atomic number of this element : 12

(b) The group of this element : 2

(c) The period of this element : 3

(5) Describe the structure of the modern periodic table. (5 marks)

Ans. (1) In the modern periodic table, the elements are arranged in the order of their increasing atomic number. In the modern periodic table, there are seven horizontal rows called periods and eighteen vertical columns (1 to 18) called groups. The arrangement of the periods and groups results into formation of boxes. Atomic numbers are serially indicated in the upper part of these boxes.

(2) Each box represents the place for one element. Apart from these seven rows, there are two rows of elements placed separately at the bottom of the periodic table. They are lanthanides and actinides series. There are 118 boxes in the periodic table including the two series that means there are 118 places for elements in the modern periodic table. The formation of a few elements was established experimentally very recently and thereby the modern periodic table is now completely filled with 118 elements.

(3) On the basis of the electronic configuration, the elements in the modern periodic table are divided into four blocks, viz. *s*-block, *p*-block, *d*-block and *f*-block. The *s*-block contains groups 1 and 2. The groups 13 to 18 constitute the *p*-block. Groups 3 to 12 constitute the *d*-block, while the lanthanide and actinide series at the bottom form the *f*-block. The *d*-block elements are called transition elements. A zigzag line is shown in the *p*-block of the periodic table. This zig-zag line shows the three traditional types of elements, i.e. metals, nonmetals and metalloids. The metalloid elements lie along the border of zigzag line. All the metals lie on the left side of the zigzag line while all the nonmetals lie on the right side.

(6) Define atomic size. How does it vary in a period and a group?

(3 marks)

Ans. (1) The distance between the centre of the atom and the outermost shell of the atom is called the atomic radius. The size of an atom is indicated by

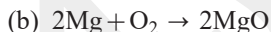
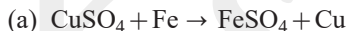
its radius. Atomic radius is expressed in unit picometre (pm). The size of atom depends on the number of shells; the more the number of shells the larger is the atomic size.

(2) In a group, while going down a group the atomic size goes on increasing, because newer shells are successively added. This increases the distance between the outermost electron and the nucleus. Hence, the nuclear attraction on these electrons goes on decreasing. Thus, in a group atomic size increases.

(3) While going from left to right within a period, atomic radius goes on decreasing and the atomic number goes on increasing one by one. The positive charge on the nucleus increases by one unit at a time. However, the additional electron gets added to the same outermost shell. Due to the increased nuclear charge, the electrons are pulled towards the nucleus to a greater extent; as a result, the size of the atom decreases.

Chapter 3 : Chemical Reactions and Equations

(1) Identify the type of chemical reaction given below :



Ans. (a) Displacement reaction

(b) Combination reaction

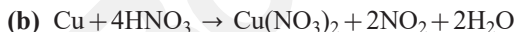
(c) Decomposition reaction

(2) (a) What happens when copper reacts with concentrated nitric acid? What is the colour of the gas released during the reaction?

(b) Write its balanced chemical equation.

(c) Write the names of reactants and products. *(March '25) (3 marks)*

Ans. (a) Copper on reaction with concentrated nitric acid gives reddish coloured poisonous nitrogen dioxide gas.



(conc.)

(c) Reactants : Copper, conc. nitric acid

Products : Copper nitrate, nitrogen dioxide, water

(3) Write the balanced equations for the following reactions :

(3 marks each)



Ans. Step 1 : Rewrite the given equation as it is



Step 2 : Write the number of atoms of each element in the unbalanced equation on both sides of equations.

Element	Number of atoms in reactants (left side)	Number of atoms in products (right side)
H	4	2
S	2	1
O	8	4

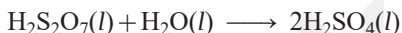
Step 3 : To equalise the number of hydrogen atoms, sulphur atoms and oxygen atoms we use 2 as the coefficient or factor in the product.

Element	Number of atoms in reactants (left side)	Number of atoms in products (right side)
H	4	2×2
S	2	1×2
O	8	4×2
Total	14	14

Now the equation becomes $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \longrightarrow 2\text{H}_2\text{SO}_4$

Now, count the atoms of each element on both sides of the equation. The number of atoms on both sides are equal. Hence, the balanced equation is $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \longrightarrow 2\text{H}_2\text{SO}_4$.

Now indicate the physical states of the reactants and products.



Ans. Step 1 : Rewrite the given equation as it is



Step 2 : Write the number of atoms of each element in the unbalanced equation on both sides of equations.

Element	Number of atoms in reactants	Number of atoms in products
Na	1	2
S	1	1
O	5	5
H	3	2

The number of oxygen atoms involved in different compounds on both sides (reactants and products) are equal. Therefore, balance the number of atoms of the second element, sodium.

Step 3 : To balance the number of sodium atoms :

Number of atoms of Sodium	In reactants	In products
To begin with	1 (in NaOH)	2 (in Na ₂ SO ₄)
To balance	1 × 2	2

To equalise the number of sodium atoms, we use 2 as the factor of NaOH in the reactants. Now, the partly balanced equation becomes as follows :



Step 4 : Now, balance the number of hydrogen atoms :

Number of atoms of hydrogen	In reactants	In products
To begin with	(in H ₂ SO ₄) 2 (in NaOH)	2 (in H ₂ O)
To balance	4	2 × 2

To equalise the number of hydrogen atoms, we use 2 as the factor of H₂O in the products. The equation then becomes



Now, count the atoms of each element on both sides of the equation. The number of atoms on both sides are equal. Hence, the balanced equation is



Now indicate the physical states of the reactants and the products.



(4) How does the rate of reaction depend on the nature of the reactants?

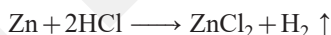
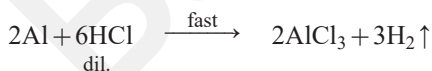
Illustrate with suitable example.

(3 marks)

Ans. (1) When the reactant combines with two or more other reactants, the rate of a chemical reaction depends on the nature of the reactants.

(2) Both Al and Zn react with dilute hydrochloric acid, H₂ gas is liberated and water soluble salts of these metals are formed. However, aluminium reacts faster with dil. HCl as compared to zinc.

(3) Al is more reactive than Zn. Therefore, the rate of reaction of Al with hydrochloric acid is higher than that of Zn. Hence, the nature of the reactant affects the rate of a reaction.



(5) How does the rate of a reaction depend upon the concentration of the reactants? Give a suitable example. (3 marks)

Ans. (1) A chemical reaction takes place due to collisions of the reactant molecules. The higher the concentrations of the reactants the more is the frequency of collisions and higher is the rate of the reaction.

(2) In the reaction of dil. HCl and CaCO_3 , CaCO_3 disappears slowly and CO_2 also liberates slowly. On the other hand, the reaction with concentrated HCl takes place rapidly and CaCO_3 disappears fast.

(3) Concentrated acid reacts faster than dilute acid; that means the rate of a reaction is proportional to the concentration of reactants.

Slow reaction :



Fast reaction :

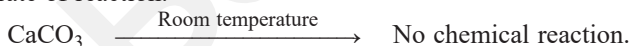


(6) How does the rate of a reaction depend upon the temperature of reactants? Give a suitable example. (3 marks)

Ans. (i) (1) When the temperature of the reactants is increased, the reactant molecules start moving with more velocity and their kinetic energy increases. As a result, the number collisions in a given time interval increases. Hence, the rate of chemical reaction increases.

(2) Lime stone on heating decomposes to give CO_2 , which turns lime water milky. On the other hand, the lime water does not turn milky before heating the lime stone; because of the zero rate of reaction. The above observation indicates that the rate of a reaction increases on increasing the temperature.

(ii) Solid CaCO_3 does not decompose at room temperature. When heated, it decomposes to give CaO and CO_2 ; that means the rise in temperature increases the rate of reaction.



(7) How does the rate of a reaction depend upon the catalyst? Give a suitable example. (2 marks)

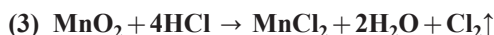
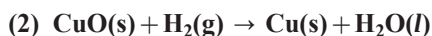
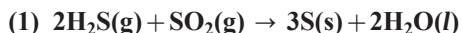
Ans. (1) The substance in whose presence the rate of a chemical reaction changes, without causing any chemical change to it, is called a catalyst.

(2) On heating potassium chlorate (KClO_3) decomposes into potassium chloride and oxygen slowly.



The rate of the above reaction neither increases by reducing the particle size nor by increasing the reaction temperature. However in the presence of manganese dioxide, KClO_3 decomposes at a comparatively lower temperature and oxygen is produced more briskly. No chemical change takes place in MnO_2 in this reaction. It acts as a catalyst.

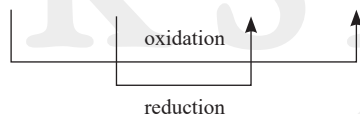
(8) Identify the substances that are oxidized and the substances that are reduced in the following reactions : **(1 mark each)**



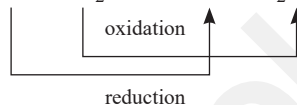
Manganese
dioxide

Manganese
chloride

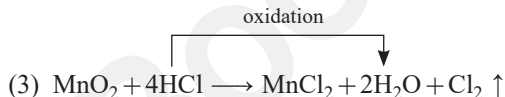
Ans. When oxidation and reduction take place simultaneously in a given chemical reaction, it is known as a redox reaction.



H_2S is oxidized and SO_2 is reduced.



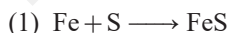
CuO is reduced and H_2 is oxidized.

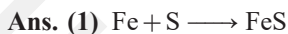


(1) Oxidation : $\rightarrow \text{H}_2\text{S}$, H_2O and HCl .

(2) Reduction : $\rightarrow \text{SO}_2$, CuO and MnO_2 .

(9) Identify the following reactions the reactants that undergo oxidation and reduction. **(1 mark each)**

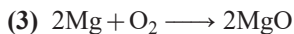




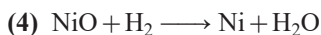
In this reaction, oxidation of Fe takes place and reduction of sulphur takes place.



In this reaction, reduction of Ag_2O takes place.



In this reaction, oxidation of Mg takes place.



In this reaction, reduction of NiO takes place and oxidation of H_2 takes place.

(10) How can corrosion be prevented?

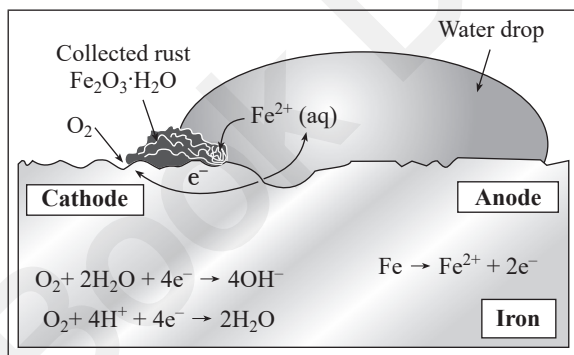
(2 marks)

Ans. (1) Corrosion damages buildings, bridges, automobiles, ships, iron railings and other articles made of iron.

(2) It can be prevented by using an anti-rust solution, coating the surface by a paint, processes like galvanizing and electroplating with other metals.

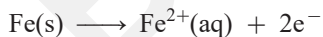
(11) Observe the following picture and write down the chemical reaction with explanation.

(5 marks)

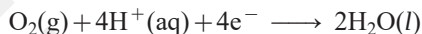


Ans. The rusting of iron is an oxidation process. The rust on iron does not form by a simple reaction between oxygen and iron surface. The rust is formed by an electrochemical reaction. Fe oxidizes to $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$ on one part of iron surface while oxygen gets reduced to H_2O on another part of surface. Different regions on the surface of iron become anode and cathode.

(1) Fe is oxidized to Fe^{2+} in the anode region.

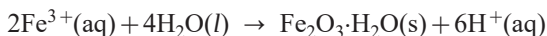


(2) O_2 is reduced to form water in the cathode region.



When Fe^{2+} ions migrate from the anode region they react with water and further get oxidized to form Fe^{3+} ions.

A reddish coloured hydrated oxide is formed from Fe^{3+} ions. It is called rust. It collects on the surface.



Because of various components in the atmosphere, oxidation of metals takes place, consequently resulting in their damage. This is called 'corrosion'. Iron rusts and a reddish coloured layer is formed on it. This is corrosion of iron.

(12) Complete the process of iron rusting by filling in the blanks.

Suggest a way to prohibit the process.

(3 marks)

The iron rust is formed due to reaction. Different regions on iron surface become anode and cathode.

● Reaction on anode region :



● Reaction on cathode region :



When Fe^{2+} ions migrate from anode region they react with to form Fe^{3+} ions.

A reddish coloured hydrated oxide is formed from ions. It is called rust.



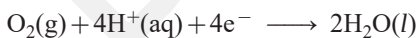
A way to prevent rusting

Ans. The iron rust is formed due to electrochemical reaction. Different regions on iron surface become anode and cathode.

● Reaction on anode region :

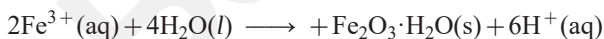


● Reaction on cathode region :



When Fe^{2+} ions migrate from anode region they react with water to form Fe^{3+} ions.

A reddish coloured hydrated oxide is formed from Fe^{3+} ions. It is called rust.



A way to prevent rusting by colouring with acrylic paints, Zn plating, galvanizing, anodizing, alloying, etc.

Chapter 4 : Effects of Electric Current

(1) What is heating effect of electric current? What is its origin?

(2 marks)

Ans. The production of heat in a resistor due to the electric current flowing through it when it is connected in an electrical circuit, is called the heating effect of electric current.

When a potential difference is applied across a metallic conductor, free electrons in the conductor move from the end at the lower potential to the end at the higher potential giving rise to electric current. These electrons collide with the atoms and positive ions and transfer some kinetic energy to them. This energy is converted into heat. Hence, the temperature of the conductor begins to rise, i.e. the conductor becomes hot. This is the origin of the heating effect of electric current.

(2) Statement 1 : Electric current (flow of electrons) creates heat in a resistor.

Statement 2 : Heat in the resistor is created according to the law of energy conservation.

Explain Statement 1 with the help of Statement 2. (2 marks)

Ans. (1) When electrons flow through a resistor (during flow of electric current) electrons possess kinetic energy.

(2) During the flow of electrons there is a decrease in the kinetic energy of the electrons due to collisions with atoms, ions and molecules.

(3) According to the law of conservation of energy, this decrease in the kinetic energy of the electrons gets converted into heat.

(3) Explain the term short circuiting. What does a short circuit lead to?

OR

How does a short circuit form? What is its effect? (2 marks)

Ans. If a bare live wire (phase wire) and a bare neutral wire touch each other (come in direct contact) or come very close to each other, the resistance of the circuit becomes very small and hence huge (very high) electric current flows through it. This condition is called a short circuit or short circuiting.

In this case, a large amount of heat is produced and the temperature of the components involved becomes very high. Hence, the circuit catches fire.

(4) What is the effect on the magnetic needle in Oersted's experiment, when (1) a current is passed through the wire (2) the current through the wire is increased (3) the current through the wire is stopped (4) the current through the wire is reversed (5) the distance between the magnetic needle and the wire is increased, keeping the current through the wire constant?

(3 marks)

Ans. In Oersted's experiment, when there is no current in the wire, the magnetic needle is at rest along the north-south direction.

(1) When a current is passed through the wire, the needle is deflected.

(2) When the current through the wire is increased, the deflection of the needle increases.

(3) When the current through the wire is stopped, the needle comes to rest in its original position along the north-south direction.

(4) When the current through the wire is reversed, the needle is deflected in the direction opposite to that in the first case.

(5) When the distance between the magnetic needle and the wire is increased, keeping the current through the wire constant, the deflection of the needle becomes less.

(5) State the conclusions that can be drawn from Oersted's experiment.

(2 marks)

Ans. Conclusions that can be drawn from Oersted's experiment :

(1) An electric current produces a magnetic field around it. The moving charge in the conducting wire is a source of magnetic field.

(2) The direction of the magnetic field produced by the current is the direction in which the north pole of the magnetic needle is deflected. Hence, from the experimental observations we can conclude that at any point near the current-carrying conductor, the magnetic field is perpendicular to (i) the length of the conductor and (ii) the line joining the conductor and the given point.

Chapter 5 : Heat

(1) Name the behaviour of water between its temperature from 0°C to 4°C.

(March '25) (1 mark)

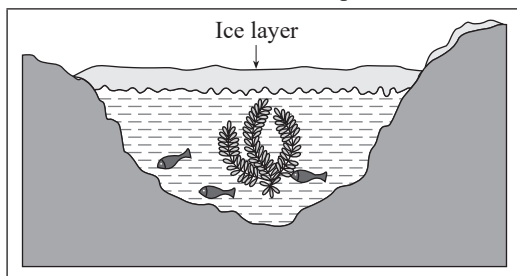
Ans. The behaviour of water between its temperature from 0°C to 4°C is called anomalous behaviour of water.

(2) Explain the role of anomalous behaviour of water in preserving aquatic life in regions of cold climate.

(3 marks)

Ans. In cold regions, during winter, the temperature of the atmosphere falls

well below 0°C . As the temperature decreases, the water at the surfaces of lakes and ponds starts contracting. Hence, its density increases and it sinks to the bottom. This process continues till the temperature of all the water in a lake falls to 4°C . As the water at the surface cools further, i.e. its temperature falls below 4°C , it starts



Aquatic plants and animals in cold regions

expanding instead of contracting. Therefore, its density decreases and it remains at the surface. The temperature of the water at the surface continues to fall to 0°C . Finally, the water at the surface is converted into ice, but the water below the layer of ice is at 4°C . Ice is a bad conductor of heat. Hence, the layer of the ice at the surface does not allow transfer of heat from the water to the atmosphere. As the water below the layer of ice remains at 4°C , fish and other aquatic animals and plants can survive in it. (See the figure.)

(3) Explain the following : In cold regions in winter, the rocks crack due to anomalous expansion of water. (2 marks)

Ans. Sometimes water enters into crevices of the rocks. When the temperature of the atmosphere falls below 4°C , water expands. Even when water freezes to form ice, there is increase in its volume. As there is no room for expansion, it exerts a tremendous pressure on the rocks which crack and break up into small pieces.

(4) During winter, sometimes we see a white trail at the back of a flying aeroplane in a clear sky. Explain why. (2 marks)

Ans. In winter, air temperature is low. Hence, when an aeroplane flies, the vapour released by its engine condenses and forms white clouds. If the relative humidity of the air surrounding the plane is high, we see this white trail at the back of the plane for a long time before it disappears. If the relative humidity is low, the white trail is short and disappears quickly. If the relative humidity is very low, there is no formation of the white trail.

(5) Explain how the specific heat capacity of a solid can be determined (measured) by the method of mixture. (3 marks)

Ans. A hot solid is put in water in a calorimeter. The mixture is stirred continuously and the maximum temperature of the mixture is measured with a thermometer. Heat exchange between the hot solid, water and calorimeter

results in all bodies attaining the same temperature after some time. Hence, according to the principle of heat exchange, heat lost by the solid = heat gained by the water in the calorimeter + heat gained by the calorimeter. Now, heat lost by the solid (Q) = mass of the solid $\times$ its specific heat capacity $\times$ decrease in its temperature, heat gained by the water (Q_1) = mass of the water $\times$ its specific heat capacity $\times$ increase in its temperature and heat gained by the calorimeter (Q_2) = mass of the calorimeter $\times$ its specific heat capacity $\times$ increase in its temperature.

Heat lost by the hot object = heat gained by the calorimeter + heat gained by the water. $Q = Q_2 + Q_1$

Using this equation, the specific heat capacity of the solid can be determined (measured) when the other quantities are known.

(6) Explain the following : How can you relate the formation of water droplets on the outer surface of a bottle taken out of a refrigerator with formation of dew? (2 marks)

Ans. At a given temperature, there is a limit on how much water vapour the given volume of air can hold. The lower the temperature, the lower is the capacity of air to hold water vapour.

The temperature of a bottle kept in a refrigerator is lower than room temperature. Hence, when the bottle is taken out of the refrigerator, the temperature of the air surrounding the bottle is lowered. Therefore, the capacity of the air to hold water vapour becomes less. Hence, the excess water vapour condenses to form water droplets (like dew) on the outer surface of the bottle.

Chapter 6 : Refraction of Light

(1) What is the reason for twinkling of stars? (March '25) (1 mark)

Ans. Stars twinkles due to changing refractive index of atmospheric gases.

(2) Why does the apparent position of stars keep changing a bit?

(July '23) (2 marks)

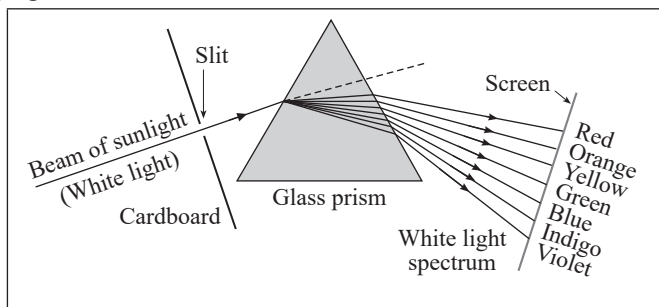
Ans. (1) As a star is far away from the earth, it appears as a point source of light. Air is always in motion. The density and temperature of air change continuously.

(2) This results in the change in the refractive index of air in a given region of space. Hence, there is a continuous change in the apparent position and brightness of a star.

(3) With a neat labelled diagram, describe in brief the experiment to demonstrate dispersion of sunlight (white light) by a prism. (3 marks)

Ans. Experiment :

(i) **Procedure :** Keep a glass prism on a table in a dark room. Hold a plane mirror outside the room so that it reflects a beam of sunlight into the room. Allow this beam to pass through a narrow slit made in a cardboard and then fall on the prism. Place a white screen on the other side of the prism as shown in the following figure.



Dispersion of sunlight (white light) by a glass prism

(ii) Observations :

- (1) A pattern of various colours is observed on the screen. This pattern is called the spectrum.
- (2) It is found that in dispersion, the ray corresponding to violet colour deviates the most.
- (3) The ray corresponding to red colour deviates the least.
- (4) The deviation of rays corresponding to other colours is intermediate.

(iii) **Conclusion :** When sunlight (white light) is incident on a prism, dispersion of light takes place, forming a spectrum.

(4) What is a spectrum? Why do we get a spectrum of seven colours when white light is dispersed by a prism? (2 marks)

Ans. A band of coloured components of a light beam is called a spectrum.

When white light is incident on a prism, the rays corresponding to different colours bend through different angles on refraction.

Of the various colours in the visible region, red light bends the least and violet light bends the most. Each colour emerges through the prism along a different path and becomes distinct. Hence, we get a spectrum of seven colours.

Chapter 7 : Lenses

(1) An object is kept in front of a lens of focal length + 10 cm. Describe the nature of the image in the following cases : (1) The object distance is 25 cm. (2) The object distance is 5 cm. (2 marks)

Ans. Since the focal length of the lens (+ 10 cm) is positive, it is a convex lens.

(1) If an object is kept at 25 cm from the lens, the image will be real, inverted and smaller than the object.

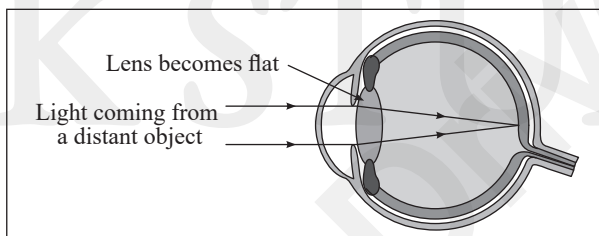
(2) If an object is kept at 5 cm from the lens, the image will be virtual, erect and larger than the object.

(2) Explain the function of the iris and the muscles connected to the lens in the human eye. (5 marks)

Ans. When the incident light is very bright, the muscles of the iris stretch to reduce the size of the pupil. When the incident light is dim, the muscles of the iris relax to increase the size of the pupil. Thus, the iris controls the size of the pupil and thereby regulates the amount of light entering the eye.

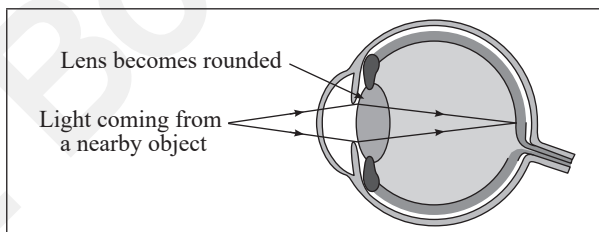
When a distant object is to be observed, the ciliary muscles relax so that the eye lens becomes flat. This increases the focal length of the lens. Therefore, a sharp image of the distant object is formed on the retina.

Thus, we can see a distant object clearly.



Formation of an image of a distant object

When an object closer to the eye is to be observed, the ciliary muscles contract increasing the curvature of the eye lens. The eye lens, therefore, becomes rounded. This decreases the focal length of the lens. Therefore, a sharp image of the nearby object is formed on the retina. Thus, we can see a nearby object clearly.



Formation of an image of a nearby object

(3) What is presbyopia? State the reason for this defect. How is presbyopia corrected? (2 marks)

Ans. Presbyopia is the defect of vision in which aged people find it difficult to see nearby objects comfortably and clearly without spectacles.

Reason of presbyopia : The power of accommodation of eye usually decreases with ageing. The muscles near the eye lens lose their ability to change the focal length of the lens. Therefore, the near point of the eye lens shifts farther from the eye.

This defect is corrected using a convex lens of appropriate power. The lens converges light rays before they fall on the eye lens such that the action of the eye lens forms the image on the retina.

(4) What is a bifocal lens? (1 mark)

Ans. A bifocal lens is a lens of which the upper part is a concave lens to correct myopia and the lower part is a convex lens to correct hypermetropia.

(5) How is the phenomenon of persistence of vision used in motion pictures? (2 marks)

Ans. In motion pictures, photographs of a moving object are taken at the rate of more than sixteen pictures per second. These photographs are projected on the screen at the same rate.

Each picture is slightly different from the other. As a result of persistence of vision, we get the impression of observing the object in continuous motion.

(6) What is colour-blindness? (1 mark)

Ans. In the retina of some persons, cone-shaped cells responding to certain specific colours are absent. Hence, these persons are unable to distinguish between certain colours, i.e. they are colour-blind. This defect is known as colour-blindness.

(7) Explain the perception of colour in the human eye. OR

Explain in short perception of colour. OR

Write a note on perception of colour. (2 marks)

Ans. (1) In nature we find objects of various colours. Perception of colour means to be able to respond to colour.

(2) We can distinguish between various colours due to perception of colour.

(3) The cone-shaped cells on the retina of the eye respond to the various colours when light is bright and communicate to the brain about the colours of the image formed on the retina. This gives us the proper idea about the colours of the object.

(4) If, in the retina of a person, the cone-shaped cells responding to certain specific colours are absent, the person is unable to distinguish between the colours. As a result, he lacks perception of colour.

Chapter 8 : Metallurgy

(1) Name the main ore of aluminium. What are the main impurities found in aluminium ore? (2 marks) (March '22)

Ans. The main ore of aluminium is bauxite ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$). It is a hydrated aluminium oxide. It contains 30% to 70% Al_2O_3 . The main impurities present in aluminium ore are iron oxide (Fe_2O_3), sand (SiO_2) and titanium oxide (TiO_2).

(2) What is meant by aqua regia? (2 marks)

Ans. Aqua regia is a highly corrosive and fuming liquid. It is a freshly prepared mixture of conc. HCl and conc. HNO_3 in the ratio of 3 : 1 by volume. Most of the substances dissolve in it. Aqua regia is a reagent which dissolves gold and platinum.

(3) How does a metal react with a nonmetal? (2 marks)

Ans. By oxidation of a metal, cations are formed; on the other hand by reduction of a nonmetal, anions are formed. The ionic compound is formed due to the metal losing electrons while the nonmetal accepts the electrons. The ionic compound of sodium chloride is formed as sodium loses one electron while chlorine accepts one electron.



Similarly, Mg and K form ionic compounds MgCl_2 and KCl .

(4) From which ore is aluminium extracted? What are the stages in its extraction (give only names)? (2 marks)

Ans. Aluminium is extracted from bauxite (Al_2O_3).

Stages in the extraction : (i) Concentration of ore, i.e. conversion of bauxite into alumina. (ii) Electrolytic reduction of alumina.

(5) Explain the term corrosion with a suitable example. (2 marks)

Ans. The process in which a metal is destroyed gradually by the action of air, moisture or a chemical (like an acid) on its surface is called corrosion. OR

Corrosion is degradation of a material due to reaction with its environment.

The major problem of corrosion occurs with iron, as it is used as a structural material in construction, bridges, ship building.

Iron gets covered by reddish brown flakes when exposed to atmosphere. This is an example of corrosion.

(6) Explain the different methods to prevent corrosion of metals.

(1 mark each)

Ans. (1) Corrosion of a metal can be prevented if the contact between metal and air is cut off.

(2) Corrosion of a metal is prevented by coating with something which does not allow moisture and oxygen to react with it.

(3) A layer of oil or paint or grease is applied on the surface of a metal to prevent corrosion. The rusting or corrosion of iron can be prevented by this method.

(4) Corrosion is also prevented by coating a corrosive metal with a non-corrosive metal. Galvanizing, tinning, electroplating, anodizing and alloying are the different methods in which a metal is coated with a non-corrosive metal to prevent corrosion.

(7) Write three methods of preventing rusting of iron. (3 marks)

Ans. (1) The rusting of iron can be prevented by painting, oiling, greasing or varnishing its surface.

(2) Galvanization is another method of protecting iron from rusting by coating iron with a thin layer of zinc.

(3) Corrosion of iron is prevented by coating iron with non-corrosive substance like carbon. This process is termed as alloying.

Chapter 9 : Carbon Compounds

Q. Answer the following questions :

(a) Define hydrocarbons.

(b) Name the types of Hydrocarbons.

(c) Name two carbon compounds used in day-to-day life.

(March '20) (3 marks)

Ans. (a) The compounds which contain carbon and hydrogen as the only two elements are called hydrocarbons.

(b) Types of Hydrocarbons : (i) Saturated hydrocarbon (ii) Unsaturated hydrocarbon.

(c) Food items, threads, papers, medicines, fuel.

Chapter 10 : Space Missions

(1) The 'rocket' is a type of fire cracker used in Diwali.

(March '25) (2 marks)

(a) Name the launcher.

(b) Name the law on which its working is based.

Ans. (a) The launcher is a rocket.

(b) The working of the rocket is based on Newton's third law of motion.

(2) What is an artificial satellite? Name any two types of artificial satellite and state their functions.

(March '20) (3 marks)

Ans. A man-made object that revolves around the earth or any other planet in a fixed orbit is called an artificial satellite.

(i) Weather satellite : It is used for the study and prediction of weather.

(ii) Communication satellite : It is used for establishing communication between different locations in the world by using specific electromagnetic waves.

(iii) Broadcast satellite : It is used for telecasting television programmes.

(iv) Navigational satellite : It is used to fix the location of any place on the earth in terms of the latitude and longitude of the place.

(v) Military satellite : It is used to collect information for security aspects in order to protect the country.

(vi) Earth observation satellite : It is used for the study of forests, deserts, oceans, polar ice on the earth's surface, exploration and management of natural sources, observation and guidance in the event of natural calamities like flood, earthquake and storm.

(3) On what basis and how are the orbits of artificial satellites classified?

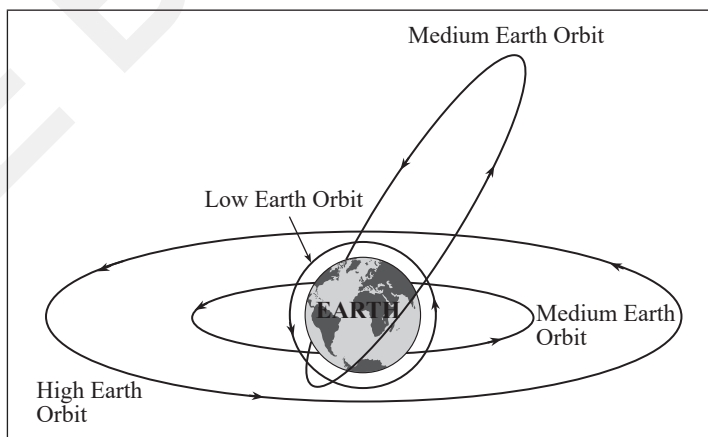
(3 marks)

Ans. Orbits of artificial satellites can be classified on the basis of

(1) the angle of the orbital plane : Orbital plane of a satellite can be the equatorial plane of the earth or it can be at an angle to it.

(2) the nature of the orbit : Orbital plane can be circular or elliptical in shape.

(3) the height of the satellite : from the earth's surface.



Orbits of satellites

A satellite orbiting at a height equal to or greater than 35780 km above the earth's surface is called a High Earth Orbit satellite. A satellite orbiting at a height between 2000 km and 35780 km above the earth's surface is called a Medium Earth Orbit satellite. A satellite orbiting at a height between 180 km and 2000 km above the earth's surface is called a Low Earth Orbit satellite.

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LONG ANSWER QUESTIONS

[Q. 4] (5 marks each)

Remember :

☛ Many questions in this set contain diagrams.

Chapter 1 : Gravitation

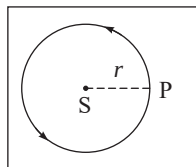
(1) With the help of Kepler's third law, show that gravitational force, $F \propto \frac{1}{r^2}$ (in the usual notation).

Ans. Kepler's third law : The square of the period of revolution of a planet around the Sun is directly proportional to the cube of the mean distance of the planet from the Sun.

Thus, if r is the average distance of the planet from the Sun and T is its period of revolution, then,

$$T^2 \propto r^3, \text{ i.e. } \frac{T^2}{r^3} = \text{constant} = K.$$

For simplicity we shall assume the orbit to be a circle. In the adjoining figure, S denotes the position of the Sun, P denotes the position of a planet at a given instant and r denotes the radius of the orbit ($\equiv$ the distance of the planet from the Sun). Here, the speed of the planet is uniform. It is



Circular motion of a planet around the Sun

$$v = \frac{\text{circumference of the circle}}{\text{period of revolution of the planet}} = \frac{2\pi r}{T}.$$

If m is the mass of the planet, the centripetal force exerted on the planet by the Sun ($\equiv$ gravitational force), $F = \frac{mv^2}{r}$

$$\therefore F = \frac{m(2\pi r/T)^2}{r} = \frac{4\pi^2 mr^2}{T^2 r} = \frac{4\pi^2 mr}{T^2}$$

According to Kepler's third law,

$$T^2 = Kr^3$$

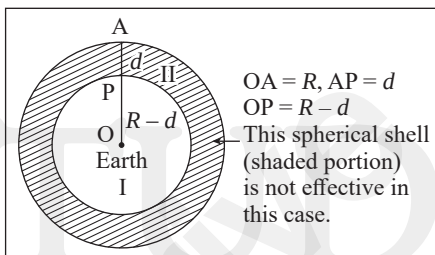
$$\therefore F = \frac{4\pi^2 mr}{Kr^3} = \frac{4\pi^2 m}{K} \left(\frac{1}{r^2} \right)$$

Thus, $F \propto \frac{1}{r^2}$ as $\frac{4\pi^2 m}{K}$ is constant in a particular case.

(2) With a neat labelled diagram, explain how the value of g changes while going deep inside the earth.

Ans. The value of g changes while going deep inside the earth. It goes on decreasing as we go from the earth's surface towards the earth's centre.

We shall treat the earth as a sphere of uniform density. If we consider a particle of mass m at point P at a distance $(R-d)$ from the earth's centre, where R is the radius of the earth and d is the depth below the earth's surface, the gravitational force on the particle due to the earth is



g inside the earth

$$F = \frac{GmM'}{(R-d)^2}, \text{ where } M' \text{ is the mass of the sphere of radius } (R-d).$$

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

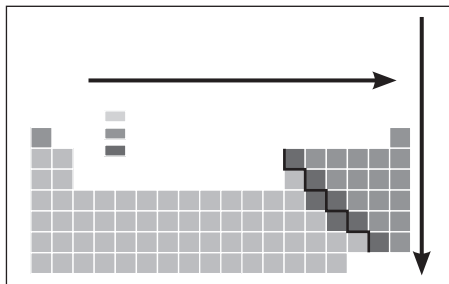
$$\therefore M' = \text{volume} \times \text{density} = \frac{4}{3}\pi (R-d)^3 \times \frac{M}{\frac{4}{3}\pi R^3} = \frac{M(R-d)^3}{R^3}. \text{ This is because}$$

the outer spherical shell is not effective. In this case, the acceleration due to gravity is

$$g = \frac{F}{m} = \frac{G}{(R-d)^2} \times \frac{M(R-d)^3}{R^3} = \frac{GM(R-d)}{R^3}, \text{ where } M \text{ is the mass of the earth.}$$

Thus, g decreases as d increases. It is less than that at the earth's surface $\left(\frac{GM}{R^2} \right)$. At the earth's centre, $d = R$. $\therefore g = 0$.

(1) Write the answers to the questions with reference to the structural of the periodic table.



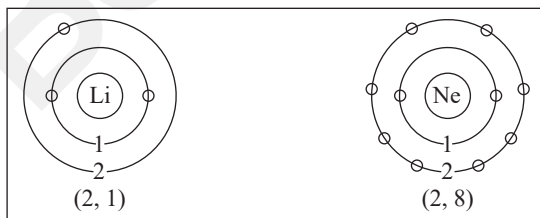
- (a) Which points are considered for the modern periodic table?
- (b) How are blocks indicated?
- (c) Which elements are present near the zigzag line?
- (d) Draw the electronic configuration of the second row elements of first group in the periodic table.
- (e) In a periodic table, while going from left to right atomic radius decreases. Explain.

Ans. (a) In the modern periodic table, the elements are arranged in the order of their increasing atomic number. In the modern periodic table, there are seven horizontal rows called periods and eighteen vertical columns (1 to 18) called groups. The arrangement of the periods and groups results into formation of boxes. Atomic numbers are serially indicated in the upper part of these boxes.

(b) On the basis of the electronic configuration, the elements in the modern periodic table are divided into four blocks, viz. the *s*-block, *p*-block, *d*-block and *f*-block. The *s*-block constitutes the groups 1 and 2. Groups 13 to 18 constitute the *p*-block. Groups 3 to 12 constitute the *d*-block, while the lanthanide and actinide series at the bottom form the *f*-block. The *d*-block elements are called transition elements. A zig-zag line is shown in the *p*-block of the periodic table.

(c) The zigzag line shows the three traditional types of elements, i.e. metals, nonmetals and metalloids. The metalloid elements lie along the border of the zigzag line. All the metals lie on the left side of the zigzag line while all the nonmetals lie on the right side.

(d) The electronic configuration of the second row elements of the first group in the periodic table is shown below :



(e) (1) In a period while going from left to right, atomic radius goes on decreasing and the atomic number goes on increasing one by one. It means the positive charge on the nucleus increases by one unit at a time.

(2) However, the additional electron gets added to the same outermost shell. Due to the increased nuclear charge the electrons are pulled towards the nucleus to a greater extent. As a result, the size of atom decreases, i.e. the atomic radius decreases.

(2) Discuss the trends in the variation of metallic and nonmetallic properties in a period and in a group.

Ans. (1) Metals have a tendency to lose the valence electrons to form cations having a stable noble gas configuration. This tendency of an element, called electropositivity, is the metallic character of that element.

(2) Nonmetals have a tendency to accept the valence electrons to form anions having a stable noble gas configuration. This tendency of an element, called electronegativity, is the nonmetallic character of that element.

(3) In a group, while going down a group a new shell gets added, resulting in an increase in the distance between the nucleus and the valence electrons. This results in lowering the effective nuclear charge and thereby lowering the attractive force on the valence electrons. As a result of this the tendency of the atom to lose electrons increases. Also the penultimate shell becomes the outermost shell on losing valence electrons. The penultimate shell is a complete octet. Therefore, the resulting cation attains special stability. The metallic character of an atom is its tendency to lose electrons. Therefore, the following trend is observed : The metallic character of elements increases while going down the group.

(4) While going from left to right within a period the outermost shell remains the same. However, the positive charge on the nucleus goes on increasing while the atomic radius goes on decreasing and thus the effective nuclear charge goes on increasing. Therefore, the valence electrons are held with greater attractive force. This is called electronegativity. As a result of this, the tendency of atom to lose valence electrons decreases within a period from left to right, i.e. electronegativity increases. Thus, nonmetallic character of elements increases within a period from left to right.

(3) Taking into consideration the period of the elements given below, answer the following questions : (March '19)

Element	O	B	C	N	Be	Li
Atomic radius (pm)	66	88	77	74	111	152

- Arrange the above elements in a decreasing order of their atomic radii.
- State the period to which the above elements belong.
- Why this arrangement of elements is similar to the above period of modern periodic table?
- Which of the above elements have the biggest and the smallest atom?
- What is the periodic trend observed in the variation of atomic radius while going from left to right within a period?

Ans. (a) The above elements are arranged in a decreasing order of their atomic radii :

Li	Be	B	C	N	O
152	111	88	77	74	66

- The above elements belong to period 2.
- As we move from left to right within a period, the atomic number increases one by one means the positive charge on the nucleus increases by one unit at a time, but the electrons are added to the same orbit thereby increasing the pull towards the nucleus which decreases the size of the atom.
- The biggest atom : Lithium (Li)
The smallest atom : Oxygen (O)
- While going from left to right in a period, atomic number increases, atomic radius decreases. Therefore, atomic size gradually decreases.

Chapter 4 : Effects of Electric Current

(1) Observe the following diagram (refer to the diagram of Q. 2) and answer the questions given below : (March '22)

- Identify the above diagram.
- Write the principle on which the above appliance works.
- Write the working of the above appliance.
- Write the use of the above appliance.

Ans. (a) AC electric generator.

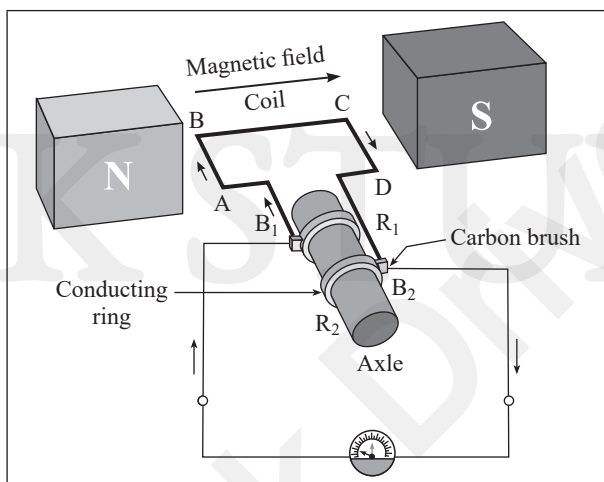
(b) Electromagnetic induction.

(c) See the answer to Q. (2).

(d) It is used to generate alternating current.

(2) With a neat labelled diagram, explain the construction and working of an electric generator (AC).

Ans. Figure shows the construction of an AC electric generator. Here, a coil ABCD of copper wire is kept between the pole pieces (N and S) of a strong magnet. The ends of the coil are connected to the conducting rings R_1 and R_2 via carbon brushes B_1 and B_2 . The rings are fixed to the axle and there is a resistive coating in between the rings and the axle. The stationary brushes are connected to a galvanometer used to show the direction of the current in the circuit.

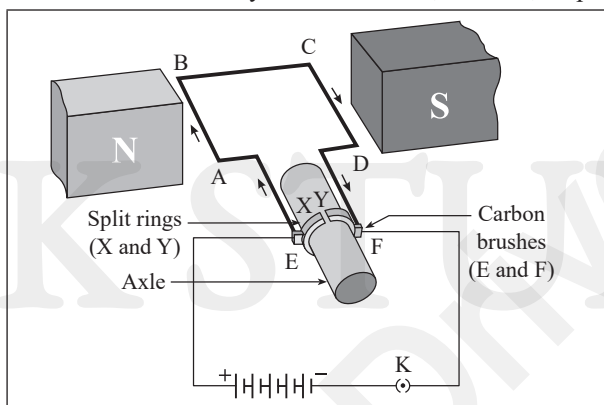


Working : When the axle is rotated with a machine from outside, the coil ABCD starts rotating. Suppose the coil rotates in clockwise direction, as observed from the side AD. Then as the branch AB moves upward, the branch CD moves downward. By Fleming's right hand rule, the induced current flows in the direction $A \rightarrow B \rightarrow C \rightarrow D$ and in the external circuit, it flows from B_2 to B_1 through the galvanometer. The induced current is proportional to the number of turns of the copper wire in the coil. After half a rotation, AB and CD interchange their places. Hence, the induced current flows in the direction $D \rightarrow C \rightarrow B \rightarrow A$. As AB is always in contact with B_1 and CD is in contact with B_2 , the current in the external circuit flows from B_1 to B_2 through the galvanometer. Thus, the direction of the current in the external circuit is opposite to that in the previous half rotation. The process goes on repeating and alternating current is generated.

(3) Explain the construction and working of an electric motor. Draw a neat diagram and label it. OR

With a neat labelled diagram, explain the construction and working of an electric motor.

Ans. Construction : The following figure shows the construction of an electric motor. Here, a rectangular loop ABCD of copper wire with resistive coating is placed between the north pole and south pole of a strong magnet, such as a horseshoe magnet, such that the branches AB and CD are perpendicular to the direction of the magnetic field. The ends of the loop are connected to the two halves, X and Y, of split rings – X and Y have resistive coating on their inner surfaces and are tightly fitted on the axle. The outer conducting surfaces of X and Y are in contact with two stationary carbon brushes, E and F, respectively.



Electric motor : Construction and working

Working : (1) When the circuit is completed with a plug key or switch, the current flows in the direction $E \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow F$. As the magnetic field is directed from the north pole to the south pole, the force on AB is downward and that on CD is upward by Fleming's left hand rule. Hence, AB moves downward and CD upward. These forces are equal in magnitude and opposite in direction. Therefore, as observed from the side AD, the loop ABCD and the axle start rotating in anticlockwise direction.

(2) After half a rotation, X and Y come in contact with brushes F and E respectively and the current flows in the direction EDCBAF. Hence the force on CD is downward and that on AB is upward. Therefore, the loop and the axle continue to rotate in the anticlockwise direction.

(3) After every half rotation, the current in the loop is reversed and the loop and the axle continue to rotate in anticlockwise direction.

When the current is switched off, the loop stops rotating after some time.

Chapter 6 : Refraction of Light

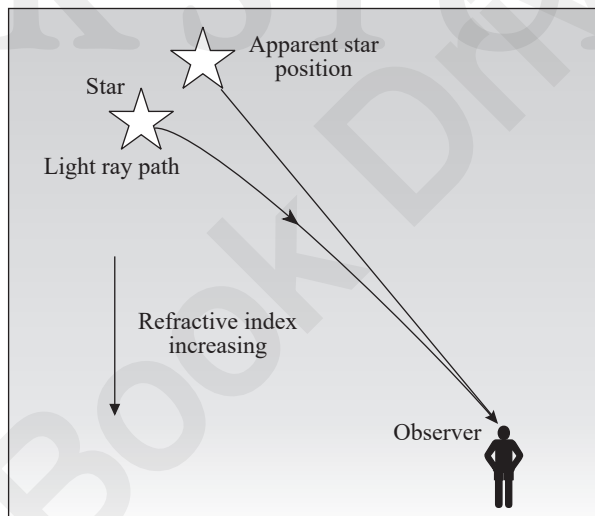
(1) With a neat labelled diagram, explain twinkling of a star. Also explain why a planet does not twinkle.

Ans. (1) As a star is far away from the earth, it appears as a point source of light. The density of air decreases with height above the earth's surface. Hence, the refractive index of air also decreases with height. When starlight enters the earth's atmosphere, it undergoes refraction continuously in the medium with gradually changing refractive index. The bending of starlight occurs towards the normal as it passes from the optically rarer part of the medium to the optically denser part.

(2) Hence, when a star is observed near the horizon, its apparent position is slightly higher than the actual position as shown in the figure.

(3) Further, the apparent position varies with time as the medium is not stationary due to mobility of air and change in temperature. When more light is refracted towards the observer the star appears bright. When less light is refracted towards the observer, the star appears dim.

(4) Thus, there is fluctuation in the brightness of a star when observed from the earth. This is called twinkling of a star.



Actual position and apparent position of a star
(Schematic diagram)

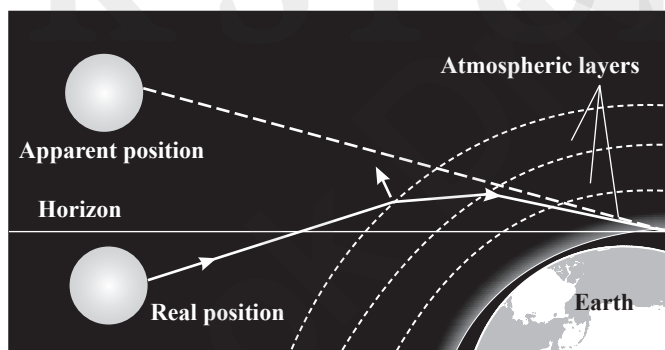
(5) Compared to stars, planets are relatively closer to the earth. Hence, a planet appears as a collection of a large number of point sources. Due to the changes in the refractive index of air, there is a change in the position and

brightness of these point sources. There is an increase in intensity of light coming from some point sources while there is a decrease in intensity of light coming from equal number of other point sources, on an average. The average brightness of a planet remain the same. Also, there is no change in the average position of a star. Hence, a planet does not twinkle.

(2) Explain advanced sunrise and delayed sunset.

Ans. (1) The sunrise (the appearance of the sun above the horizon) is advanced due to atmospheric refraction of sunlight. An observer on the earth sees the sun two minutes before the sun reaches the horizon. A ray of sunlight entering the earth's atmosphere follows a curved path due to atmospheric refraction before reaching the earth. This happens due to a gradual variation in the refractive index of the atmosphere. For the observer on the earth, the apparent position of the sun is slightly higher than the actual position. Hence, the sun is seen before the sun reaches the horizon.

(2) Increased atmospheric refraction of sunlight occurs also at the sunset (the sun disappearing below the horizon). In this case, the observer on the earth continues to see the setting sun for two minutes after the sun has dipped below the horizon, thus delaying the sunset.



Effect of atmospheric refraction

The advanced sunrise and delayed sunset increases the duration of day by four minutes.

Chapter 7 : Lenses

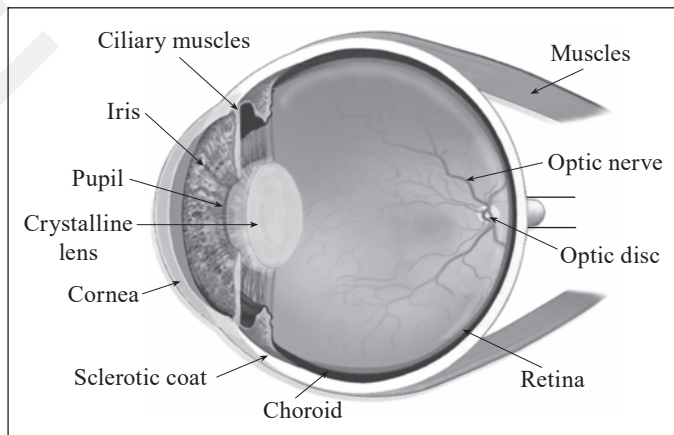
(1) Answer the following questions :

(March '25)

- Draw a neat labelled diagram of human eye.
- What is the minimum distance of distinct vision for a normal human eye?
- Name the capacity of the eye lens to change its focal length as per need.

(d) Name the defect of eye vision in which the focussing power of eye lens decreases with age.

Ans. (a)



(b) The minimum distance of distinct vision for a normal human eye is 25 cm.

(c) The capacity of the eye lens to change its focal length as per need is called Power of accommodation of the eye.

(d) The defect of eye vision in which the focusing power of eye lens decreases with age is called presbyopia.

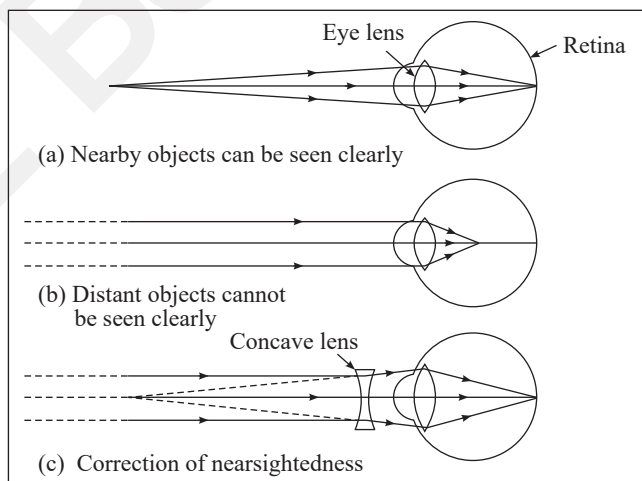
(2) What is myopia or nearsightedness? What are the possible reasons of myopia? How is myopia corrected? Explain with diagrams.

Ans. Myopia or nearsightedness is the defect of vision in which a human eye can see nearby objects distinctly but is unable to see distant objects clearly as they appear indistinct.

In this case the image of a distant object is formed in front of the retina instead of on the retina. [Figs. (a), (b)]

Possible reasons of myopia : (1) The curvature of the cornea and the eye lens increases. The muscles near the lens cannot relax so that the converging power of the lens remains large. (2) The distance between the eye lens and the retina increases as the eyeball elongates.

Myopia is corrected using a suitable concave lens. Light rays are diverged by the concave lens before they strike the eye lens. A concave lens of proper focal length is chosen to produce the required divergence. Hence, after the converging action of the eye lens, the image is formed on the retina. [Fig. (c)]



(3) What is hypermetropia or farsightedness? What are the possible reasons of hypermetropia? How is hypermetropia corrected? Explain with diagrams.

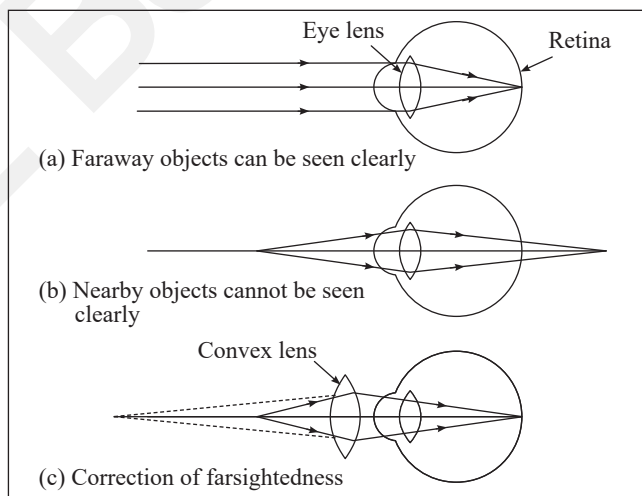
Ans. Hypermetropia or farsightedness is the defect of vision in which a human eye can see distant objects clearly but is unable to see nearby objects clearly.

In this case the image of a nearby object would fall behind the retina instead of on the retina.

[Figs. (a), (b)]

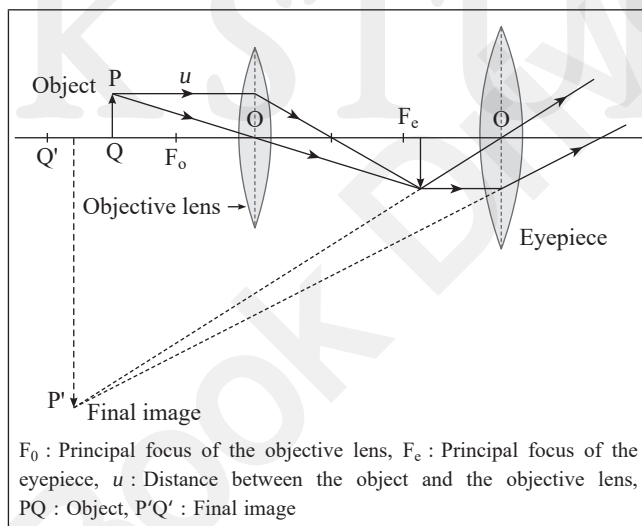
Possible reasons of hypermetropia : (1) Curvature of the cornea and the eye lens decreases. Hence, the converging power of the eye lens becomes less. (2) The distance between the eye lens and retina decreases (relative to the normal eye) and the focal length of the eye lens becomes very large due to the flattening of the eyeball.

Hypermetropia is corrected using a suitable convex lens. Light rays are converged by the convex lens before they strike the eye lens. A convex lens of proper focal length is chosen to produce the required convergence. Hence, after the converging action of the eye lens, the image is formed on the retina. Fig. (c)



(4) With a neat labelled diagram, explain the construction and working of a compound microscope.

Ans.



A compound microscope

Construction of a compound microscope : (1) A compound microscope consists of a metal tube fitted with two convex lenses at the two ends. These lenses are called the objective lens (the lens directed towards the object) and the eyepiece (the lens directed towards the eye). Both the lenses are small in size, but the cross section of the objective lens is less than that of the eyepiece. The

objective lens has a short focal length. The focal length of the eyepiece is more than that of the objective lens.

(2) The metal tube is mounted on a stand. The principal axes of the objective lens and the eyepiece are along the same line. The distance between the object and the objective lens can be changed with a screw. It is possible to change the distance between the objective lens and the eyepiece.

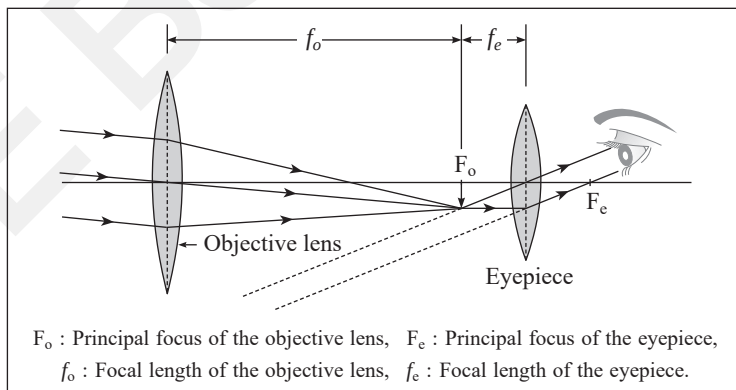
Working : (1) The object to be observed is illuminated and placed in front of the objective lens, slightly beyond the focal length of the objective lens. Its real, inverted and enlarged image is formed by the objective lens on the other side.

(2) This intermediate image lies within the focal length of the eyepiece. It serves as an object for the eyepiece. The eyepiece works as a simple microscope. The final image is virtual, highly enlarged and inverted with respect to the original object. It can be formed at the minimum distance of distinct vision from the eyepiece. The final image is observed by keeping the eye close to the eyepiece.

(5) With a neat labelled diagram, explain the working of a refracting telescope. OR

Explain the working of an astronomical telescope using refraction of light.

Ans. Construction of a refracting telescope : It consists of two convex lenses called the objective lens (directed towards the object) and the eyepiece (directed towards the eye). The focal length and diameter of the objective lens are respectively greater than the focal length and diameter of the eyepiece. The objective lens is fitted at one end of a long metal tube. A metal tube of smaller diameter is fitted in this metal tube and the eyepiece is fitted at the outer end of the smaller tube. With the help of a screw it is possible to change the distance between the eyepiece and the objective lens by sliding the tube fitted with the eyepiece. The principal axes of the objective lens and the eyepiece are along the same line. A telescope is usually mounted on a stand.



Refracting telescope

Working : When the objective lens is pointed towards the distant object to be observed, the rays of light from the distant object, which are almost parallel to each other, pass through the objective lens. The objective lens collects maximum amount of light as it is large in size. It forms a real, inverted and diminished image in the focal plane of the objective lens. Now, the position of the eyepiece is adjusted such that this image falls just within the focal length of the eyepiece and serves as the object for the eyepiece which works as a simple microscope. The final image is highly magnified, virtual, on the same side as that of the object and inverted with respect to the original object. The final image can be observed by keeping the eye close to the eyepiece. If the image formed by the objective lens lies in the focal plane of the eyepiece, the final image is formed at infinity.

Chapter 8 : Metallurgy

(1) Explain the reactivity series of the metals.

Ans. The arrangement of metals in decreasing order of their reactivity in the form of a series is called the reactivity or activity series of the metals.

The most reactive metal is placed at the top of the list and the least reactive metal is placed at the bottom of the list.

On the basis of reactivity, we can classify metals into the following categories :

(1) Highly reactive metals (2) Moderately reactive metals (3) Less reactive metals.

(1) Extraction of highly reactive metals : The metals which are placed at the top of the reactivity series are very reactive. They are never found in nature as free elements, e.g. sodium, potassium, calcium and aluminium. These metals are obtained by electrolytic reduction.

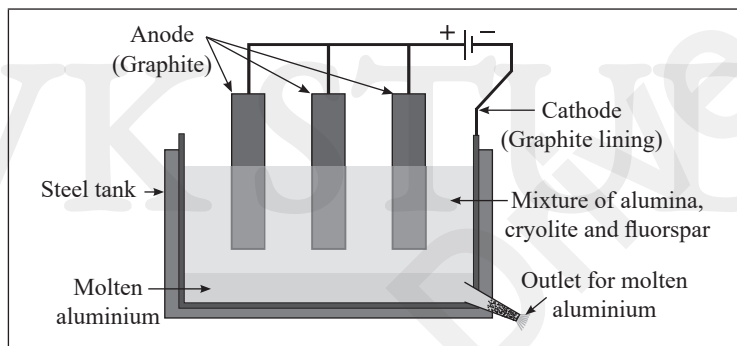
(2) **Extraction of moderately reactive metals** : The metals in the middle of reactivity series such as iron, zinc, lead, copper are moderately reactive. These elements are present as sulphides or carbonates in nature. Generally metals are obtained from their oxide as compared to their sulphides and carbonates.

(3) **Extraction of less reactive metals** : The metals which are placed at the bottom of the reactivity series are least reactive. They occur in free state, e.g. gold, silver and copper. Copper and silver are also found in the combined state as sulphide and oxide ores. These metals are obtained from their ores by just heating the ores in air.

(2) **Describe the process of preparation of aluminium by the electrolysis of alumina.** OR **Explain the electrolytic reduction of alumina.**

Ans. Electrolytic reduction of alumina :

(1) The electrolytic cell consists of a rectangular steel tank lined from inside with graphite.



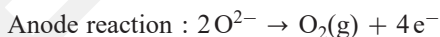
(2) The carbon lining (graphite) acts as a cathode. The anode consists of graphite rods suspended in the molten electrolyte.

(3) Alumina has very high melting point ($> 2000^{\circ}\text{C}$). The electrolysis of alumina is carried out at a low temperature by dissolving it in molten cryolite (Na_3AlF_6). The solution of alumina in cryolite and small amount of fluorspar (CaF_2) is added in the mixture to lower its melting point up to 1000°C .

(4) On passing an electric current, alumina is electrolysed.

(5) Molten aluminium is collected at the cathode, while oxygen gas is evolved at the anode.

The electrode reactions are shown below :



The molten aluminium is heavier than the electrolyte. Therefore, it sinks to the bottom of the electrolyte and is removed from time to time. About 99% of pure aluminium is obtained by this process.

The oxygen gas liberated reacts with carbon anode and forms carbon dioxide. As the anode gets oxidised during the electrolysis of alumina, it has to be replaced from time to time.

Chapter 9 : Carbon Compounds

Q. Atomic number of chlorine is 17.

(March '25)

- (a) Write the electronic configuration of chlorine 2, 8, 9.
- (b) What is the number of electrons in the valence shell of chlorine?
- (c) Write the molecular formula of chlorine.
- (d) Name the type of bond in the formation of chlorine molecule.
- (e) Draw the electron dot structure of a chlorine molecule.

Ans. (a) 2,8,7.

(b) 7 electrons.

(c) Cl_2 .

(d) Covalent bond.

(e) $\text{:}\ddot{\text{Cl}}\text{:}\ddot{\text{Cl}}\text{:}$

**Question
Set
19**

PARAGRAPH-BASED QUESTIONS

[Q. 3 or 4]

Remember :

- ☛ This set contains **paragraph-based questions**.
- ☛ Read the paragraph and answer the questions.

Chapter 4 : Effects of Electric Current

Q. Read the given passage and answer the following questions :

(March '20) (3 marks)

The home electrical connection consists of 'live', 'neutral' and 'earth' wires. The 'live' and the 'neutral' wires have potential difference of 220 V. The 'earth' wire is connected to ground. Due to a fault in the equipment or if the plastic coating on the 'live' and the 'neutral' wires gives away the two wires come in contact with each other and a large current flows through it producing heat. If any inflammable material (such as wood, cloth, plastic, etc.) exists around that place it can catch fire. Therefore a fuse wire is used as a precautionary measure.

- (a) Name the two wires having potential difference of 220 V.
- (b) What is short circuit?
- (c) Write the function of a fuse.

Ans. (a) The potential difference between the live wire and the neutral wire is 220 V.

(b) If a bare live wire (phase wire) and a bare neutral wire touch each other (come in direct contact) or come very close to each other, the resistance of the circuit becomes very small and hence huge (very high) electric current flows through it. This condition is called a short circuit or short circuiting.

In this case, a large amount of heat is produced and the temperature of the components involved becomes very high. Hence, the circuit catches fire.

(c) A fuse protects electrical circuits and appliances by stopping the flow of electric current when it exceeds a specified value. For this, it is connected in series with the appliance (or circuit) to be protected. A fuse is a piece of wire

made of an alloy of low melting point (e.g. an alloy of lead and tin). If a current larger than the specified value flows through the fuse, its temperature increases enough to melt it. Hence, the circuit breaks and the appliance is protected from damage.

Chapter 5 : Heat

(1) Read the following paragraph and answer the questions. (5 marks)

If heat is exchanged between a hot object and a cold object, the temperature of the cold object goes on increasing due to gain of energy and the temperature of the hot object goes on decreasing due to loss of energy. The change in temperature continues till the temperatures of both the objects attain the same value. In this process, the cold object gains heat energy and the hot object loses heat energy. If the system of both the objects is isolated from the environment by keeping it inside a heat resistant box (meaning that the energy exchange takes place between the two objects only), then no energy can flow from inside the box or come into the box.

Questions :

- (1) Heat is transferred from where to where?
- (2) Which principle do we learn about from this process?
- (3) How will you state the principle briefly?
- (4) Which property of the substance is measured using this principle?

Ans. (1) Heat is transferred from a hot object to a cold object.

(2) This process shows the principle of heat exchange.

(3) In this process, the cold object gains heat energy and the hot object loses energy. If a system of two objects is isolated from the surroundings, heat energy lost by the hot object = heat energy gained by the cold object.

(4) This principle is used to measure the specific heat capacity of a substance.

[Note : This question carried 3 marks in the Board examination conducted in March 2022, but it did not contain the subquestion (4).]

(2) Study the following procedure and answer the questions below :

(3 marks)

- (1) Take 3 spheres of iron, copper and lead of equal mass.
- (2) Put all the 3 spheres in boiling water in a beaker for some time.
- (3) Take 3 spheres out of the water. Put them immediately on a thick slab of wax.
- (4) Note the depth that each sphere goes into the wax.
 - (i) Which property of a substance can be studied with this procedure?
 - (ii) Describe that property in minimum words.
 - (iii) Explain the rule of heat exchange with this property.

Ans. (i) Specific heat.

(ii) Specific heat : The amount of heat energy required to raise the temperature of a unit mass of an object by 1°C .

(iii) According to the rule/principle of heat exchange, heat energy lost by the hot object = heat energy gained by the cold object.

In this activity, heat absorbed by the iron sphere is transmitted more in the wax, hence the sphere goes deepest into the wax, while the lead sphere absorbs less heat, resulting in less transmission of heat in the wax, hence the sphere goes the least depth into the wax.

Chapter 7 : Lenses

(1) Read the following paragraph and answer the questions given below it :

(5 marks)

Compound microscope : Construction :

(1) A compound microscope consists of a metal tube fitted with two convex lenses at the two ends. These lenses are called the objective lens (the lens directed towards the object) and the eyepiece (the lens directed towards the eye). Both the lenses are small in size, but the cross section of the objective lens is less than that of the eyepiece. The objective lens has a short focal length. The focal length of the eyepiece is more than that of the objective lens.

(2) The metal tube is mounted on a stand. The principal axes of the objective lens and the eyepiece are along the same line. The distance between the object and the objective lens can be changed with a screw. It is possible to change the distance between the objective lens and the eyepiece.

Working :

(1) The object to be observed is illuminated and placed in front of the objective lens, slightly beyond the focal length of the objective lens. Its real, inverted and enlarged image is formed by the objective lens on the other side.

(2) This intermediate image lies within the focal length of the eyepiece. It serves as an object for the eyepiece. The eyepiece works as a simple microscope. The final image is virtual, highly enlarged and inverted with respect to the original object. It can be formed at the minimum distance of distinct vision from the eyepiece. The final image is observed by keeping the eye close to the eyepiece.

Use : This microscope is used to observe blood cells, microorganisms, etc.

Questions :

- (1) In a compound microscope, which lens has greater focal length?
- (2) Where do you place the object to be observed with a compound microscope?
- (3) State which distance is adjusted to observe the object with a compound microscope.
- (4) State the nature of the final image in a compound microscope relative to the object.
- (5) State the use of a compound microscope.

Answers :

- (1) In a compound microscope, the eyepiece has greater focal length.
 - (2) In a compound microscope, the object to be observed is placed in front of the objective lens, slightly beyond the focus of the objective lens.
 - (3) To observe the object with a compound microscope, the distance between the object and objective lens is adjusted.
 - (4) In a compound microscope, the final image is virtual, highly enlarged and inverted relative to the object.
 - (5) A compound microscope is used to observe blood cells, microorganisms, etc.
-

Section 3**ASSIGNMENT**
(Additional Important Questions for
Self-Practice)**Q. 1 (A) : 5 Marks**

Q. Choose the correct alternative and write the correct option :

- (1) The gravitational force between two particles separated by a distance r varies as
(A) $\frac{1}{r}$ (B) r (C) r^2 (D) $\frac{1}{r^2}$ [Ch. 1]
- (2) The law of octaves was given by
(A) Dobereiner (B) Newlands
(C) Mendeleev (D) Moseley [Ch. 2]
- (3) Noble gases belong to group in the modern periodic table.
(A) 15 (B) 16 (C) 17 (D) 18 [Ch. 2]
- (4) The reaction $\text{KNO}_3(\text{s}) + \text{H}_2\text{O}(\text{l}) + \text{Heat} \longrightarrow \text{KNO}_3(\text{aq})$ is a/an reaction.
(A) exothermic (B) endothermic
(C) oxidation (D) reduction [Ch. 3]
- (5) The Δ sign written above the arrow indicates of the reaction.
(A) the reactant (B) the product
(C) the heat (D) the direction of the reaction [Ch. 3]
- (6) Electromagnetic induction means
(A) charging of an electric conductor.
(B) production of magnetic field due to a current flowing through a coil.
(C) generation of a current in a coil due to relative motion between the coil and the magnet.
(D) motion of the coil around the axle in an electric motor. [Ch. 4]
- (7) heat is needed to raise the temperature of 1 kg of water from 14.5°C to 15.5°C .
(A) 4180 J (B) 10^3 J (C) 1 cal (D) 4180 cal [Ch. 5]
- (8) has the highest refractive index.
(A) Air (B) Water
(C) Glass (D) Diamond [Ch. 6] (July '23)

- (9) Inside water, an air bubble behaves
(A) like a flat plate (B) like a concave lens
(C) like a convex lens (D) like a concave mirror [Ch. 7]
- (10) is a metal which is in liquid form at ordinary temperature and pressure.
(A) Magnesium (B) Sodium
(C) Scandium (D) Mercury [Ch. 8]
- (11) is an allotropic form of a nonmetal which conducts electricity.
(A) Sulphur (B) Graphite
(C) Chlorine (D) Iodine [Ch. 8]
- (12) Silver gets corroded due to in air.
(A) oxygen (B) hydrogen sulphide
(C) carbon dioxide (D) nitrogen [Ch. 8]
- (13) What type of reaction is shown below?
$$\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{Sunlight}} \text{CH}_3\text{Cl} + \text{HCl}$$

(A) Addition (B) Substitution
(C) Decomposition (D) Reduction [Ch. 9]
- (14) When acetic acid reacts with sodium metal gas is formed.
(A) oxygen (B) hydrogen
(C) chlorine (D) nitrogen [Ch. 9]
- (15) Which of the following satellite launchers is developed by India?
(A) INSAT (B) IRNSS
(C) EDUSAT (D) PSLV [Ch. 10]
-

Q. 1 (B) : 5 Marks

Q. Attempt the following questions :

- (1) Give one example of centripetal force. [Ch. 1]
- (2) State any one characteristic of gravitational force. [Ch. 1]
- (3) Name an alkaline earth metal in the period 3. [Ch. 2]
- (4) State the number of elements in the modern periodic table. [Ch. 2]

- (5) How many valence electrons are there in the outermost shell of silicon. [Ch. 2]
- (6) Write the electronic configuration : $_{13}\text{Al}$. [Ch. 2]
- (7) State True or False : In a group, the metallic character decreases from top to bottom. [Ch. 2]
- (8) What do you mean by fast speed reaction? [Ch. 3]
- (9) What is corrosion? [Ch. 3]
- (10) State the use of antioxidants in food containing oils. [Ch. 3]
- (11) Complete the correlation :
 $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$: Displacement reaction : :
 $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$: [Ch. 3]
- (12) State True or False :
(a) The chemical reactions in which the heat is liberated are called endothermic reactions. [Ch. 3]
(b) Electric power = I^2R . [Ch. 4]
- (13) What is one kilowatt-hour? [Ch. 4]
- (14) What is relative humidity? [Ch. 5]
- (15) State two effects of humidity present in atmosphere? [Ch. 5]
- (16) What is mirage? [Ch. 6]
- (17) Name the device in which total internal reflection of light is used. [Ch. 6]
- (18) What is a lens formula? [Ch. 7]
- (19) What is the shape and the size of the human eyeball? [Ch. 7]
- (20) Name a common ore of aluminium. [Ch. 8]
- (21) Name two highly reactive metals. [Ch. 8]
- (22) Complete the correlation :
Sulphide ores : Froth floatation method : : Cassiterite ore : [Ch. 8]
- (23) Find the odd one out : C_2H_4 , C_4H_{10} , C_3H_8 , CH_4 . [Ch. 9]
- (24) Define : Addition reaction. [Ch. 9]
- (25) Write IUPAC name of sodium acetate. [Ch. 9]
- (26) How many covalent bonds are there in a molecule of cyclohexane? [Ch. 9]
- (27) State True or False : The escape velocity on the moon is less than that on the earth. [Ch. 10]
- (28) What is a Medium Earth Orbit (MEO) satellite? [Ch. 10]
-

Q. 2 (A) : 4 Marks

Q. Give scientific reasons : (Any two)

- (1) The weight of a body is different on different planets. [Ch. 1]
- (2) Zero group elements (inert gases) are called noble gases. [Ch. 2]
- (3) While going down the second group, the reactivity of the alkaline earth metal increases. [Ch. 2]
- (4) While preparing dilute sulphuric acid from concentrated sulphuric acid in the laboratory, the concentrated sulphuric acid is added slowly to water with constant stirring. [Ch. 3]
- (5) In practice the unit kW·h, rather than the joule, is used for the measurement of electric energy. [Ch. 4]
- (6) Water droplets are seen on the outer surface of a cold drink bottle. [Ch. 5]
- (7) The sun is seen on the horizon for sometime even after it has gone below the horizon. [Ch. 6]
- (8) One can sense colours only in bright light. [Ch. 7]
- (9) Pine oil is used in froth floatation process. [Ch. 8]
- (10) Copper vessels turn greenish and silver articles turn blackish when kept open in air for a long time. [Ch. 8] (Nov. '20)
- (11) The hydrogenation of vegetable oil in the presence of nickel catalyst forms vanaspati ghee. [Ch. 9]
- (12) Geostationary satellites are not useful for studies of polar regions. [Ch. 10]

Q. 2 (B) : 6 Marks

Q. Answer the following questions : (Any three)

- (1) State the importance of Newton's universal law of gravitation. [Ch. 1]
- (2) Two small bodies are separated by a distance of 10 metres from each other. Their masses are 1 kg and 0.1 kg respectively. What is the gravitational force between them? $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$. [Ch. 1]

- (3) Rearrange the columns 2 and 3 so as to match with the column 1 : [Ch. 2]

Column 1	Column 2	Column 3
i. Nucleus	a. Positive charge on the nucleus	(a) Dobereiner
ii. Electron	b. Sequential change in molecular formulae	(b) Rutherford (c) Moseley

- (4) State the electronic configuration of nitrogen and phosphorus. [Ch. 2]
- (5) Write down what do you understand from the following chemical reaction :
 $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}\downarrow + \text{NaNO}_3(\text{aq})$ [Ch. 3]
- (6) How does the rate of a reaction depend on the size of the particles of reactants? [Ch. 3]
- (7) What is electromagnetic induction? Who discovered it? [Ch. 4]
- (8) A bulb of 40W is connected across a source of 220 V. Find the resistance of the bulb. [Ch. 4]
- (9) What is absolute humidity? State its unit. [Ch. 5]
- (10) How much heat energy is necessary to raise the temperature of 10 kg of water from 30°C to 100°C?
 [Specific heat capacity of water (c)=1 kcal/kg.°C] [Ch. 5] (July '22)
- (11) Explain why the bottom of a pond appears raised. [Ch. 6]
- (12) Light travels with a velocity 1.5×10^8 m/s in a medium. On entering second medium its velocity becomes 1.25×10^8 m/s. What is the refractive index of the second medium with respect to the first medium? [Ch. 6]
- (13) In general, when a ray of light passes through a lens, there occurs a change in its direction of propagation. Why? [Ch. 7]
- (14) An object is kept at 60 cm in front of a convex lens. Its real image is formed at 20 cm from the lens. Find the focal length of the lens. [Ch. 7]
- (15) What happens when aluminium ore is heated with caustic soda? Write the balanced chemical equation for the same. [Ch. 8]
- (16) Explain the term galvanising with example. [Ch. 8]
- (17) (i) Define functional group.
 (ii) Write functional group and formula for ethyl alcohol. [Ch. 9]
- (18) Define : (i) Alkene (ii) Esterification. [Ch. 9]
- (19) What are the objectives of the space mission? [Ch. 10]
- (20) Name any two types of artificial satellite and state their functions. [Ch. 10]

Q. 3 : 15 Marks

Q. Answer the following : (Any five)

- (1) Explain why the value of g is zero at the centre of the earth. [Ch. 1]
- (2) Using the law of conservation of energy, obtain the expression for the escape velocity. [Ch. 1]
- (3) Considering the elements of period 3 in the modern periodic table, answer the following questions : [Ch. 2]
 - (a) Name the 'element' in which all the shells are completely filled with electrons.
 - (b) Name the element which has one electron in the outermost shell.
 - (c) State the most electronegative element in this period.
- (4) Classify the following elements into group 1, 16 and 17 :
Hydrogen, Oxygen and Bromine. [Ch. 2]
- (5) What is corrosion? Do gold ornaments corrode? Justify. [Ch. 3]
- (6) $\text{Ag(s)} + \text{HCl(aq)} \longrightarrow \text{AgCl}\downarrow + \text{H}_2\uparrow$ [Ch. 3]
- (7) What is meant by a double displacement reaction? Give two examples. [Ch. 3]
- (8) What is overloading? When does it occur? What does it cause? [Ch. 4]
- (9) With a neat labelled diagram, explain the construction of an electric motor. [Ch. 4]
- (10) What is meant by specific heat capacity? Determine its SI unit. [Ch. 5]
- (11) Explain how the specific heat capacity of a solid can be determined by the method of mixture. [Ch. 5]
- (12) If the angle of incidence and angle of emergence of a light ray falling on a glass slab are i and e respectively, prove that $i = e$. [Ch. 6]
- (13) With a neat labelled diagram, explain twinkling of a star. Also explain why a planet does not twinkle. [Ch. 6]
- (14) State the rules used for drawing ray diagrams for the formation of an image by a convex lens. [Ch. 7]
- (15) State the characteristics of an image formed by a concave lens. [Ch. 7]
- (16) Explain the following terms :
 - (i) Roasting
 - (ii) Calcination
 - (iii) Minerals. [Ch. 8]

- (17) Complete the table, if a metal reacts with the reagent then mark ✓ and if not then mark ✗.

Metal	Ferrous sulphate	Silver nitrate	Zinc sulphate
Cu			
Al			

[Ch. 8]

- (18) Explain the following reactions with the help of balanced equation.
- (a) Ethanol is reacted with sodium?
 - (b) Ethanol is heated at 170°C with excess of conc. H_2SO_4 .
 - (c) Ethanol is treated with sodium hydroxide. [Ch. 9]
- (19) Explain the structure of benzene. [Ch. 9]
- (20) Write in brief on significant space missions carried out by man. [Ch. 10]
- _____

Q. 4 : 5 Marks

Q. Answer any one of the following questions :

- (1) Explain the construction and working of an electric generator (AC). Draw a neat diagram and explain it. [Ch. 4]
- (2) Describe the structure of the modern periodic table. [Ch. 2]
- (3) With a neat labelled diagram, explain the construction and working of a compound microscope. [Ch. 7]
- (4) Describe Bayer's process for concentration of bauxite. [Ch. 8]
- (5) What causes the existence of very large number of carbon compound?

[Ch. 9]

Note : Refer to 'Navneet's Science & Technology Digest–Std. X'
for Answers of Assignment Section

Section 4**BOARD'S ACTIVITY SHEET : JULY 2025
(FOR PRACTICE)****Time : 2 Hours]****(Answer on QR Code)****[Max. Marks : 40****Note :** (i) All questions are compulsory.

(ii) Use of a calculator is not allowed.

(iii) The numbers to the right of the questions indicate full marks.

(iv) In case of MCQs [Q. No. 1 (A)], only the first attempt will be evaluated and will be given credit.

(v) Scientifically correct, labelled diagrams should be drawn wherever necessary.

(vi) Each new question should be started on the new page.

Q. 1. (A) Choose the correct alternative :**5**

(i) Alkaline earth metals have valency 2. This means that their position in the modern periodic table is in

(A) Group 2 (B) Group 16 (C) Period 2 (D) d-block

(ii) When the positive charge on an atom or an ion increases or the negative charge on them decreases it is called

(A) reduction (B) oxidation
(C) displacement (D) decomposition

(iii) device is used to produce electricity.

(A) Electric motor (B) Galvanometer
(C) Electric generator (D) Voltmeter

(iv) Light changes its direction when going from one transparent medium to another transparent medium. This is called the of light.

(A) reflection (B) dispersion
(C) refraction (D) absorption

(v) The height of High earth orbit from the surface of the earth is

(A) = 20,000 km (B) = 27,000 km
(C) < 30,780 km (D) > 35,780 km

Q. 1. (B) Answer the following questions :**5**

- (i) Find the odd one out :
Fuse wire, bad conductor, rubber gloves, generator.
- (ii) Complete the co-relation.
Red light wavelength close to : 700 nm : : Violet light wavelength close to :
- (iii) Match the pair :

Group 'A'	Group 'B'
Farsightedness	(a) Faraway object can be seen clearly (b) Nearby object can be seen clearly (c) Faraway object cannot be seen clearly

- (iv) Write the electronic configuration of Aluminium.
- (v) State true or false :
The escape velocity on the moon is less than that on the earth.

Q. 2. (A) Give scientific reasons : (Any two)**4**

- (i) Sodium is always kept in kerosene.
- (ii) We cannot clearly see an object kept at a distance less than 25 cm from the eye.
- (iii) Tungsten metal is used to make a solenoid type coil in an electric bulb.

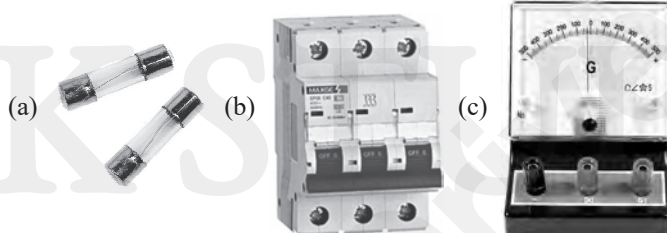
Q. 2. (B) Answer the following questions : (Any three)**6**

- (i) An object takes 5 seconds to reach the ground from a height of 5 m on a planet. What is the value of g on the planet?
- (ii) An element has its electronic configuration as 2, 8, 2. Now answer the following questions :
(a) What is the atomic number of this element?
(b) To which period does this element belong?
- (iii) Complete the following reaction and give the names of products :
 $\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \longrightarrow \dots + \dots$
- (iv) Define specific heat of an object. Write its SI and CGS units.
- (v) Draw the electron-dot structure of methane (CH_4) and ethane (C_2H_6).

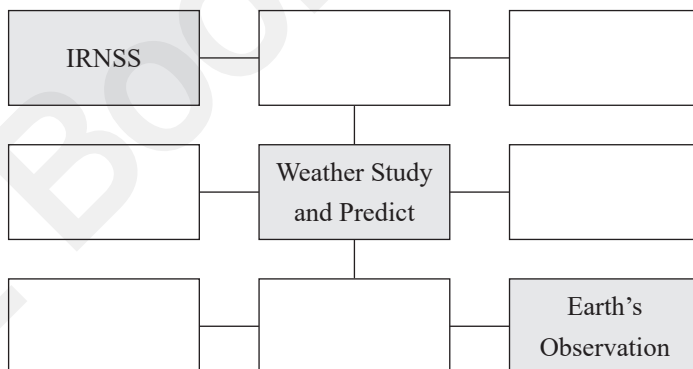
Q. 3. Answer the following questions : (Any five)

15

- (i) Explain the concept escape velocity in detail.
- (ii) Answer the following sub-questions regarding the modern periodic table :
 - (a) Write the modern periodic law.
 - (b) Write the number of periods and groups.
 - (c) What is the valency of halogens?
- (iii) State whether the following reactions are endothermic or exothermic :
 - (a) $\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O} + \text{Heat}$
 - (b) $2\text{KClO}_{3(s)} \xrightarrow{\Delta} 2\text{KCl}_{(s)} + 3\text{O}_{2(g)} \uparrow$
 - (c) $\text{CaO} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{Heat}$
- (iv) Identify the figure and write one use of it :



- (v) How much heat energy is necessary to raise the temperature of 5 kg of water from 20°C to 100°C ?
 $[c = 1 \text{ kcal/kg}\cdot^\circ\text{C}]$
- (vi) Write the Cartesian sign convention for lenses.
- (vii) Complete the following chart :



- (viii) Give the chemical equations for the following reactions :
- Thermit reaction
 - Burning of magnesium in air
 - Iron nail/dust is dropped in aqueous solution of copper sulphate.

Q. 4. Answer the following questions : (Any one)

5

- “A rainbow is the combined effect of the refraction, dispersion and total internal reflection of light.” Prove this statement with the help of a diagram and explain it.
- Complete the following table :

Sr. No.	Straight chain of carbon atoms	Molecular formula	Name
1.	C	CH ₄	
2.	C—C		Ethane
3.	C—C—C	C ₃ H ₈	
4.	C—C—C—C		Butane
5.	C—C—C—C—C	C ₅ H ₁₂	
6.	C—C—C—C—C—C		Hexane
7.	C—C—C—C—C—C—C		Heptane
8.	C—C—C—C—C—C—C—C	C ₈ H ₁₈	
9.	C—C—C—C—C—C—C—C—C		Nonane
10.	C—C—C—C—C—C—C—C—C—C		Decane

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